

SingleRAN

PKI Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....	1
1.1 SRAN22.1 Draft B (2026-01-30).....	1
1.2 SRAN22.1 Draft A (2025-12-31).....	1
2 About This Document.....	5
2.1 General Statements.....	5
2.2 Applicable RAT.....	5
2.3 Features in This Document.....	6
3 Overview.....	8
4 PKI.....	10
4.1 Principles.....	10
4.1.1 PKI System Logical Architecture.....	10
4.1.1.1 Overview.....	10
4.1.1.2 EE.....	11
4.1.1.3 CA.....	11
4.1.1.4 RA.....	14
4.1.1.5 Digital Certificate and CRL Database.....	15
4.1.1.6 CMP.....	15
4.1.2 Introduction to Digital Certificates.....	16
4.1.2.1 Device Certificate.....	16
4.1.2.2 Root Certificate, Certificate Chain, Trust Certificate, and Trust Group.....	19
4.1.2.3 Cross-Certificate.....	23
4.1.2.4 CRL.....	25
4.1.2.5 OCSP.....	26
4.1.3 Certificate Management and Application Scenarios.....	27
4.1.3.1 Certificate Preconfiguration Phase.....	27
4.1.3.2 Deployment Phase.....	29
4.1.3.2.1 Base Station Deployment Phase.....	29
4.1.3.2.2 Base Station Controller Deployment Phase (GSM/UMTS).....	32
4.1.3.3 Operation Phase.....	34
4.1.3.3.1 Applying for an Operator's Certificate.....	34
4.1.3.3.2 Replacing Certificates.....	40
4.1.3.3.3 Certificate Sharing.....	43

4.1.3.3.4 Checking Certificate Validity.....	43
4.1.3.3.5 Updating Certificates.....	46
4.1.3.3.6 Revoking Certificates.....	51
4.1.3.3.7 Offline Certificate Monitoring.....	56
4.1.3.4 Application of Digital Certificates.....	57
4.1.3.4.1 Certificate Usage in UMPT+UMPT Cold Backup Mode.....	57
4.1.3.4.2 Estimation of Certificate Reconfiguration Impact.....	58
4.1.3.4.3 Certificate Application of eCPRI Connections.....	59
4.1.3.4.4 Certificate Application When Transmission Link Backup of Main Control Boards (LTE/NR) Is Deployed.....	62
4.2 Network Analysis.....	63
4.2.1 Benefits.....	63
4.2.2 Impacts.....	63
4.3 Requirements.....	64
4.3.1 Licenses.....	64
4.3.2 Functions.....	64
4.3.2.1 GBFD-113526 BTS Supporting PKI.....	64
4.3.2.1.1 Prerequisite Functions.....	64
4.3.2.1.2 Mutually Exclusive Functions.....	64
4.3.2.1.3 Function Impacts.....	64
4.3.2.2 WRFD-140210 NodeB PKI Support.....	64
4.3.2.2.1 Prerequisite Functions.....	65
4.3.2.2.2 Mutually Exclusive Functions.....	65
4.3.2.2.3 Function Impacts.....	65
4.3.2.3 GBFD-160211 BSC Supporting PKI.....	65
4.3.2.3.1 Prerequisite Functions.....	65
4.3.2.3.2 Mutually Exclusive Functions.....	65
4.3.2.3.3 Function Impacts.....	65
4.3.2.4 WRFD-160276 RNC Supporting PKI.....	65
4.3.2.4.1 Prerequisite Functions.....	65
4.3.2.4.2 Mutually Exclusive Functions.....	65
4.3.2.4.3 Function Impacts.....	66
4.3.2.5 LBFD-003010 Public Key Infrastructure (PKI).....	66
4.3.2.5.1 Prerequisite Functions.....	66
4.3.2.5.2 Mutually Exclusive Functions.....	66
4.3.2.5.3 Function Impacts.....	66
4.3.2.6 MLBFD-12000312 Public Key Infrastructure (PKI).....	66
4.3.2.6.1 Prerequisite Functions.....	66
4.3.2.6.2 Mutually Exclusive Functions.....	66
4.3.2.6.3 Function Impacts.....	66
4.3.2.7 TDLBFD-003010 Public Key Infrastructure (PKI).....	66
4.3.2.7.1 Prerequisite Functions.....	66
4.3.2.7.2 Mutually Exclusive Functions.....	66

4.3.2.7.3 Function Impacts.....	66
4.3.2.8 FBFD-010023 Security Mechanism (PKI).....	67
4.3.2.8.1 Prerequisite Functions.....	67
4.3.2.8.2 Mutually Exclusive Functions.....	67
4.3.2.8.3 Function Impacts.....	67
4.3.3 Hardware.....	67
4.3.4 Others.....	68
4.4 Operation and Maintenance.....	68
4.4.1 When to Use.....	68
4.4.2 Precautions.....	69
4.4.3 Deployment of PKI on the GBTS/eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station.....	70
4.4.3.1 Data Preparation.....	70
4.4.3.2 Using MML Commands.....	87
4.4.3.3 Using the MAE-Deployment.....	89
4.4.3.4 Activation Verification.....	92
4.4.4 Deployment of PKI on the BSC6900/BSC6910 (GSM/UMTS).....	93
4.4.4.1 Data Preparation.....	93
4.4.4.2 Using MML Commands.....	102
4.4.4.3 Activation Verification.....	108
4.4.5 Deployment of PKI on the BSC6960 (GSM/UMTS).....	110
4.4.5.1 Data Preparation.....	110
4.4.5.2 Using MML Commands.....	121
4.4.5.3 Activation Verification.....	127
4.4.6 Deployment of PKI on a Board with eCPRI Ports.....	127
4.4.6.1 Data Preparation.....	127
4.4.6.2 Using MML Commands.....	127
4.4.6.3 Using the MAE-Deployment.....	128
4.4.6.4 Activation Verification.....	128
4.4.7 Deployment of PKI for Transmission Link Backup of Main Control Boards (LTE/NR).....	128
4.4.7.1 Data Preparation.....	128
4.4.7.2 Using MML Commands.....	141
4.4.7.3 Activation Verification.....	143
4.4.8 Replacing the Initial Certificate.....	143
4.4.9 Reconfiguration.....	144
4.4.10 Network Monitoring.....	145
5 NE Supporting PKI Redundancy.....	146
5.1 Principles.....	146
5.2 Network Analysis.....	147
5.2.1 Benefits.....	147
5.2.2 Impacts.....	147
5.3 Requirements.....	147
5.3.1 Licenses.....	147

5.3.2 Functions.....	147
5.3.2.1 GBFD-160210 BTS Supporting PKI Redundancy.....	147
5.3.2.1.1 Prerequisite Functions.....	148
5.3.2.1.2 Mutually Exclusive Functions.....	148
5.3.2.1.3 Function Impacts.....	148
5.3.2.2 GBFD-160208 BSC Supporting PKI Redundancy.....	148
5.3.2.2.1 Prerequisite Functions.....	148
5.3.2.2.2 Mutually Exclusive Functions.....	148
5.3.2.2.3 Function Impacts.....	148
5.3.2.3 WRFD-160275 NodeB Supporting PKI Redundancy.....	148
5.3.2.3.1 Prerequisite Functions.....	148
5.3.2.3.2 Mutually Exclusive Functions.....	149
5.3.2.3.3 Function Impacts.....	149
5.3.2.4 WRFD-160277 RNC Supporting PKI Redundancy.....	149
5.3.2.4.1 Prerequisite Functions.....	149
5.3.2.4.2 Mutually Exclusive Functions.....	149
5.3.2.4.3 Function Impacts.....	149
5.3.2.5 LOFD-070212 eNodeB Supporting PKI Redundancy.....	149
5.3.2.5.1 Prerequisite Functions.....	149
5.3.2.5.2 Mutually Exclusive Functions.....	149
5.3.2.5.3 Function Impacts.....	149
5.3.2.6 MLOFD-070212 eNodeB Supporting PKI Redundancy.....	149
5.3.2.6.1 Prerequisite Functions.....	150
5.3.2.6.2 Mutually Exclusive Functions.....	150
5.3.2.6.3 Function Impacts.....	150
5.3.2.7 TDLOFD-070212 eNodeB Supporting PKI Redundancy.....	150
5.3.2.7.1 Prerequisite Functions.....	150
5.3.2.7.2 Mutually Exclusive Functions.....	150
5.3.2.7.3 Function Impacts.....	150
5.3.2.8 FBFD-010023 Security Mechanism (gNodeB Supporting PKI Redundancy).....	150
5.3.2.8.1 Prerequisite Functions.....	150
5.3.2.8.2 Mutually Exclusive Functions.....	150
5.3.2.8.3 Function Impacts.....	150
5.3.3 Hardware.....	151
5.3.4 Networking.....	151
5.3.5 Others.....	151
5.4 Operation and Maintenance.....	151
5.4.1 Precautions.....	151
5.4.2 Deployment of PKI Redundancy on the eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station.....	151
5.4.2.1 Data Preparation.....	152
5.4.2.2 Using MML Commands.....	153
5.4.2.3 Activation Verification.....	154

5.4.3 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the eGBTS/ NodeB/eNodeB/gNodeB/Multimode Base Station.....	154
5.4.3.1 Data Preparation.....	156
5.4.3.2 Data Configuration.....	157
5.4.3.3 Activation Verification.....	158
5.4.4 Deployment of PKI Redundancy on the BSC6900/BSC6910 (GSM/UMTS).....	158
5.4.4.1 Data Preparation.....	158
5.4.4.2 Using MML Commands.....	159
5.4.4.3 Activation Verification.....	159
5.4.5 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the BSC6900/ BSC6910 (GSM/UMTS).....	160
5.4.5.1 Data Preparation.....	162
5.4.5.2 Data Configuration.....	163
5.4.5.3 Activation Verification.....	164
5.4.6 Deployment of PKI Redundancy on the BSC6960 (GSM/UMTS).....	164
5.4.6.1 Data Preparation.....	164
5.4.6.2 Using MML Commands.....	164
5.4.6.3 Activation Verification.....	165
5.4.7 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the BSC6960 (GSM/UMTS).....	166
5.4.7.1 Data Preparation.....	168
5.4.7.2 Data Configuration.....	169
5.4.7.3 Activation Verification.....	169
5.4.8 Network Monitoring.....	170
6 NE Supporting Digital Certificate Whitelist Management.....	171
6.1 Principles.....	171
6.2 Network Analysis.....	172
6.2.1 Benefits.....	172
6.2.2 Impacts.....	172
6.3 Requirements.....	172
6.3.1 Licenses.....	172
6.3.2 Functions.....	173
6.3.2.1 GBFD-181202 BTS Supporting Digital Certificate Whitelist Management.....	173
6.3.2.1.1 Prerequisite Functions.....	173
6.3.2.1.2 Mutually Exclusive Functions.....	174
6.3.2.1.3 Function Impacts.....	174
6.3.2.2 WRFD-181220 NodeB Supporting Digital Certificate Whitelist Management.....	174
6.3.2.2.1 Prerequisite Functions.....	174
6.3.2.2.2 Mutually Exclusive Functions.....	174
6.3.2.2.3 Function Impacts.....	174
6.3.2.3 LOFD-111203 eNodeB Supporting Digital Certificate Whitelist Management.....	174
6.3.2.3.1 Prerequisite Functions.....	174
6.3.2.3.2 Mutually Exclusive Functions.....	174

6.3.2.3.3 Function Impacts.....	174
6.3.2.4 MLOFD-111203 eNodeB Supporting Digital Certificate Whitelist Management.....	174
6.3.2.4.1 Prerequisite Functions.....	175
6.3.2.4.2 Mutually Exclusive Functions.....	175
6.3.2.4.3 Function Impacts.....	175
6.3.2.5 FOFD-010080 IPsec (gNodeB Supporting Digital Certificate Whitelist Management).....	175
6.3.2.5.1 Prerequisite Functions.....	175
6.3.2.5.2 Mutually Exclusive Functions.....	175
6.3.2.5.3 Function Impacts.....	175
6.3.3 Hardware.....	175
6.3.4 Others.....	176
6.4 Operation and Maintenance.....	176
6.4.1 When to Use.....	176
6.4.2 Data Configuration.....	176
6.4.2.1 Data Preparation.....	176
6.4.2.2 Using MML Commands.....	178
6.4.2.3 Using the MAE-Deployment.....	179
6.4.3 Activation Verification.....	179
6.4.4 Network Monitoring.....	179
7 Parameters.....	180
8 Counters.....	182
9 Glossary.....	183
10 Reference Documents.....	184

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised the descriptions of certificate usage in UMPT+UMPT cold backup mode. For details, see [4.1.3.4.1 Certificate Usage in UMPT+UMPT Cold Backup Mode](#).

Revised the descriptions of automatic certificate re-application triggered by base stations. For details, see [Support for Automatic Re-Application When the Certificate Issuer Changes](#).

Revised related descriptions in [4.1.3.4.2 Estimation of Certificate Reconfiguration Impact](#).

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 08 (2025-12-14).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added the switch control for initial certificate verification on the BSC6910. For details, see: • 4.1.3.3.2 Replacing Certificates • 4.4.4.1 Data Preparation • 4.4.4.2 Using MML Commands	Added the CERTCHKTSK.VEND ORCERTCHKSW parameter.	BSC6910
Changed the default and recommended values of the certificate validity check period for the BSC6910. For details, see 4.4.4.2 Using MML Commands.	Changed the default and recommended values of the CERTCHKTSK.PERIOD parameter.	BSC6910
Changed the default and recommended values of the certificate validity check period for the BSC6960. For details, see 4.4.5.2 Using MML Commands.	Changed the default and recommended values of the CERTCHKTSK.PERIOD parameter.	BSC6960
Added the support for preset RSA initial certificates and ECDSA initial certificates by the UMPThb series main control boards. For details, see: • 4.1.2.1 Device Certificate • 4.1.2.2 Root Certificate, Certificate Chain, Trust Certificate, and Trust Group • 4.1.3.1 Certificate Preconfiguration Phase	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	Base Station Model
Added the support for automatic retrieval of an operator-issued device certificate for the standby UMPT. For details, see: • 4.1.3.3.1 Applying for an Operator's Certificate • 4.1.3.3.4 Checking Certificate Validity • 4.1.3.3.5 Updating Certificates • 4.1.3.4.1 Certificate Usage in UMPT +UMPT Cold Backup Mode • 4.1.3.4.4 Certificate Application When Transmission Link Backup of Main Control Boards (LTE/NR) Is Deployed	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added enhanced automatic certificate management on the MAE. For details, see Replacing Certificates in Batches.	None	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the support for obtaining CRLs over HTTP. For details, see: • 4.1.1.5 Digital Certificate and CRL Database • 4.1.3.3.6 Revoking Certificates • 4.3.4 Others • 4.4.3.1 Data Preparation • 4.4.7.1 Data Preparation	Added the enumerated value HTTP to the CRLTSK.CRLGETMET HOD (LTE eNodeB, 5G gNodeB) parameter.	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised configuration descriptions related to the **RMV CRLTSK** command. For details, see [4.4.3.2 Using MML Commands](#), [4.4.4.2 Using MML Commands](#), [4.4.5.2 Using MML Commands](#), and [4.4.7.2 Using MML Commands](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and New Radio (NR).

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RA T	Feature ID	Feature Name	Chapter/Section
LTE FD D	LBFD-003010	Public Key Infrastructure (PKI)	4 PKI
LTE TD D	TDLBFD-003010	Public Key Infrastructure (PKI)	
NB- IoT	MLBFD-12000312	Public Key Infrastructure (PKI)	
NR	FBFD-010023	Security Mechanism (PKI)	
GS M	GBFD-113526	BTS Supporting PKI	
GS M	GBFD-160211	BSC Supporting PKI	
UM TS	WRFD-140210	NodeB PKI Support	
UM TS	WRFD-160276	RNC Supporting PKI	
GS M	GBFD-160210	BTS Supporting PKI Redundancy	5 NE Supporting PKI Redundancy
GS M	GBFD-160208	BSC Supporting PKI Redundancy	
UM TS	WRFD-160275	NodeB Supporting PKI Redundancy	
UM TS	WRFD-160277	RNC Supporting PKI Redundancy	
LTE FD D	LOFD-070212	eNodeB Supporting PKI Redundancy	
LTE TD D	TDLOFD-070212	eNodeB Supporting PKI Redundancy	
NB- IoT	MLOFD-070212	eNodeB Supporting PKI Redundancy	

RA T	Feature ID	Feature Name	Chapter/Section
NR	FBFD-010023	Security Mechanism (gNodeB Supporting PKI Redundancy)	6 NE Supporting Digital Certificate Whitelist Management
GS M	GBFD-181202	BTS Supporting Digital Certificate Whitelist Management	
UM TS	WRFD-181220	NodeB Supporting Digital Certificate Whitelist Management	
LTE FD D	LOFD-111203	eNodeB Supporting Digital Certificate Whitelist Management	
NB- IoT	MLOFD-111203	eNodeB Supporting Digital Certificate Whitelist Management	
NR	FOFD-010080	IPsec (gNodeB Supporting Digital Certificate Whitelist Management)	

3 Overview

The public key infrastructure (PKI) system provides information security services using asymmetric key algorithms. It is used for digital certificate application and management on networks and is the basis and core of network security construction.

A digital certificate is an electronic identity credential issued by a certification authority (CA) that binds the identity information of a certificate holder with its public and private key pair. This digital certificate is also called a public key certificate or an identity certificate. This document mainly describes digital certificates used by devices. Therefore, digital certificates are also called device certificates or certificates for short.

A device certificate includes the serial number, holder name, issuer, validity period, holder's public key, and extension fields. The SubjectAltName extension field in a digital certificate contains the base station/base station controller's identity information, such as the electronic serial number (ESN) of the main control board on the base station.

Digital certificates are used for identity authentication, information encryption, and digital signature between devices. Digital certificate-based authentication uses asymmetric keys to authenticate device identities. During the course of communication, the sender uses its private key to digitally sign randomly generated data and sends its certificate and signature data to the receiver over the network. The receiver uses the public key in the certificate to verify the signature data and sender identity. With digital certificates, both the receiver and the sender confirm each other's identity to protect against spoofing and eavesdropping.

Huawei base stations and base station controllers use PKI-based end-to-end certificate management solutions. This facilitates the deployment, application, and management of digital certificates on NEs.

Before delivery, Huawei base stations or base station controllers are preconfigured with Huawei initial certificates on boards using the Huawei PKI system. On a live network, a Huawei initial certificate can be used by a device to securely access an operator's network. After the device is connected to the operator's network, the certificate should be replaced with the certificate issued by the PKI system on the operator's live network in a timely manner. For details about the certification practices statement (CPS) of Huawei PKI system, see "RootCA CPS" at <https://support.huawei.com/additionalres/pki>.

NOTE

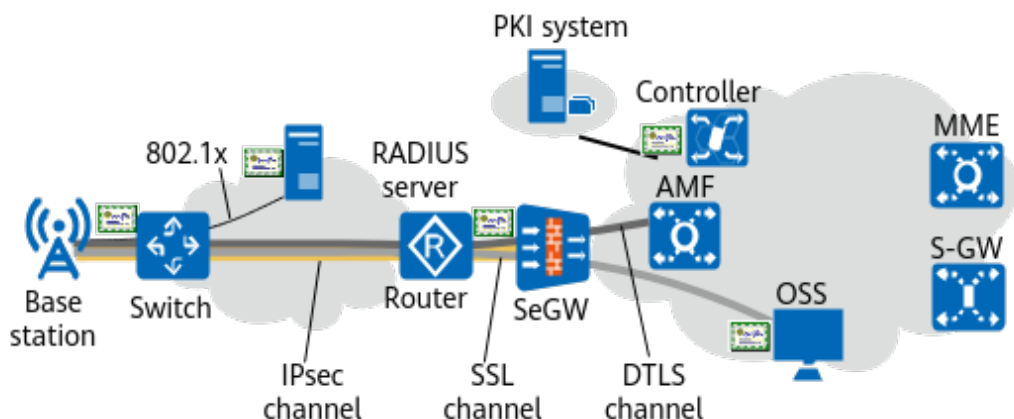
A digital certificate contains configuration data of applications such as IPsec and Secure Sockets Layer (SSL) on a live network. A Huawei initial certificate preconfigured on a board is used only for accessing an operator's network securely during site deployment. During the normal operation phase, an operator-issued device certificate is required. Therefore, the operator needs to deploy its own PKI system.

For Huawei products, digital certificates apply to the following scenarios:

- Authentication during the setup of an IPsec tunnel between a base station and a security gateway (SeGW) on a radio bearer network. For details, see *IPsec*.
- Authentication during the setup of an SSL connection between the eGBTS/NodeB/eNodeB/gNodeB/RNC/BSC and the MAE to secure data transmission on the management plane. For details, see *SSL* and *OM Security*.
- Identity authentication between the base station and the 5GC (AMF) when Datagram Transport Layer Security (DTLS) is used for transmission protection between the gNodeB and the AMF. For details, see *DTLS*.
- 802.1x-based identity authentication when 802.1x-based access control is required for the eGBTS/NodeB/eNodeB/gNodeB. For details, see *Access Control based on 802.1x*.
- Setup of separate IPsec tunnels for each operator, thereby implementing secure service isolation in RAN sharing scenarios when multiple operators share a base station and each operator deploys a separate PKI server. For details, see *Base Station Supporting Multi-operator PKI*.

Figure 3-1 shows the application of digital certificates on a network. For details about digital certificates, see **4 PKI**. The MACSec, eCPRI, WebLMT, and WDT in the base station also use digital certificates for authentication.

Figure 3-1 Example of networking that uses digital certificates



PKI redundancy is introduced to improve the reliability of PKI-based networking. For details, see **5 NE Supporting PKI Redundancy**.

If no PKI system is deployed on an operator's network, the base station can access the operator's network by using the digital certificate whitelist management function. For details, see **6 NE Supporting Digital Certificate Whitelist Management**.

4 PKI

4.1 Principles

Attacks on and threats to the infrastructure deployed on a network are inevitable. To protect the infrastructure, the PKI system must be deployed to implement security control measures and reduce network threats. A PKI system manages digital certificates for network equipment. This helps operators establish trusted security domains and trust relationships between equipment of different vendors to establish a secure network environment.

This chapter describes the logical architecture of the PKI system, basic concepts of digital certificates, and certificate management and application.

The logical architecture of the PKI system consists of PKI system components, entities using certificates, and certificate management protocols.

Digital certificates include device certificates and CA certificates. Different types of CAs have different CA certificate names, such as root certificates, subordinate CA certificates, and cross-certificates.

Digital certificate management involves the factory phase, deployment phase, and operation phase.

Digital certificate application includes certificate application in UMPT+UMPT cold backup mode, estimation of certificate reconfiguration impacts, and certificate application of eCPRI connections.

4.1.1 PKI System Logical Architecture

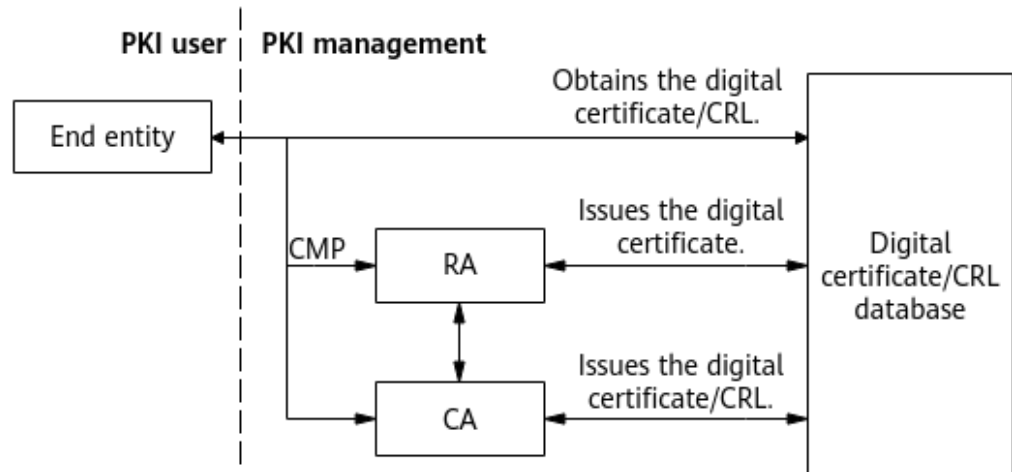
4.1.1.1 Overview

The PKI system of a base station supports both IPv4 networking and IPv6 networking. The PKI system of a base station controller supports only IPv4 networking.

As shown in [Figure 4-1](#), a PKI system on a wireless network generally consists of the following network elements (NEs):

- End entities (EEs) that use certificates, such as base stations, base station controllers, SeGWs, and MAEs. These EEs are PKI users and holders of digital certificates.
- PKI management nodes, including the Certificate Authority (CA), registration authority (RA), and digital certificate and certificate revocation list (CRL) database. These components jointly provide certificate issuance and lifecycle management for users, devices, and servers on networks.

Figure 4-1 PKI system logical architecture



NOTE

For details about PKI, see RFC 5280. Digital certificates described in this document comply with X.509v3 specifications. CRLs comply with X.509v2 specifications. Digital certificates and CRLs of earlier versions are not supported.

The CA integrated in the MAE serves only the NEs managed by the MAE, that is, Huawei base stations, base station controllers, and MAE itself. If the SeGW is deployed and certificate authentication is used between the base station and the SeGW, the operator needs to build an independent PKI system provided by a third party to meet the end-to-end certificate management requirements of the entire network.

4.1.1.2 EE

An EE is a PKI system user who holds and uses a digital certificate.

An EE may be a person, a device, a software program, or the like, for example, an NE, an OSS, an SeGW, or an FTP server. It is located at a leaf node of the PKI system architecture and is the end user of digital certificates.

4.1.1.3 CA

A CA is the central management node in a PKI system. It is usually established by an enterprise to issue and manage certificates. As shown in [Figure 4-1](#), a CA manages certificates as follows:

- Approves or rejects certificate applications and issues certificates for approved applications.
- Handles requests for certificate updates, verifications, and queries.
- Handles certificate revocation and CRL publishing issues.

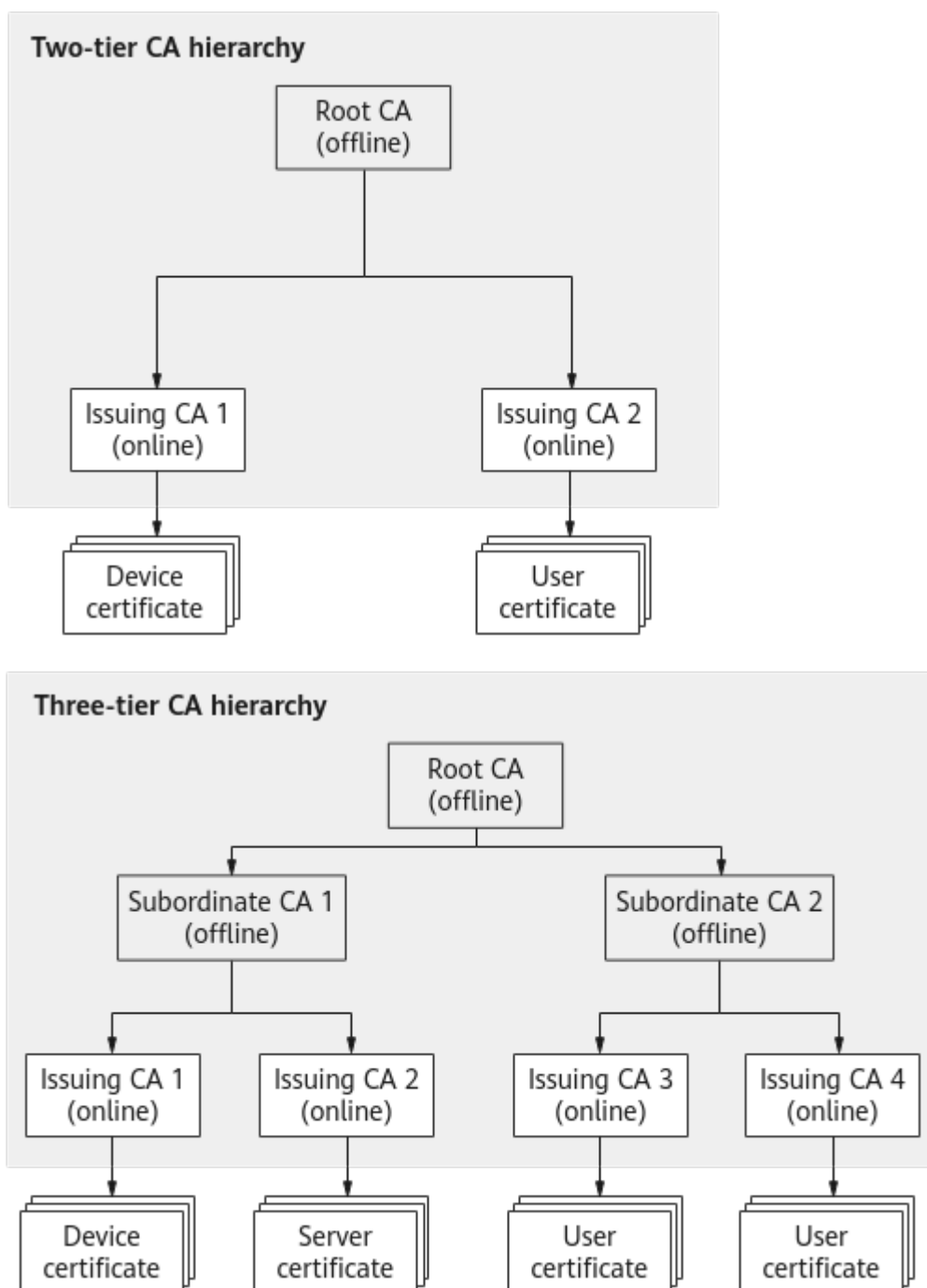
- Generates certificates and CRLs and publishes them in the certificate and CRL database.

Common CAs involved in wireless network solutions are as follows:

- Huawei CA: CA deployed by Huawei. It is used to preconfigure digital certificates for devices during device manufacturing.
- Operator CA: self-signed CA deployed by an operator. It is used to issue digital certificates to devices on the operator's live network.
- LiteCA: CA service integrated into Huawei MAE. It is used to issue digital certificates to network devices managed by the MAE. LiteCA is a type of operator CA. It can be controlled and managed by operators. For details about the LiteCA, see *iMaster MAE Product Documentation*.

On a live network, a CA system can use a layered structure to meet the requirements for CA deployment across different areas, departments, and vendors to meet certificate issuing requirements. This relieves the management pressure on a single CA. [Figure 4-2](#) shows an example of the CA system architecture.

Figure 4-2 Example of the CA system architecture



When building a PKI system, an operator needs to determine the root CA domain based on the operator's business scale and global network distribution.

- **Root CA:** The root CA is at the top of the CA hierarchy. Each CA hierarchy starts from the root CA and can contain multiple subordinate CAs. All subordinate CAs must be authenticated by their upper-level CAs until the root CA. The root CA has the highest security and trustworthiness and must be stored in a secure domain. Generally, the root CA is an independent offline CA.

- Subordinate CA: An operator uses an upper-level CA to issue certificates to lower-level CAs. CAs at each level can directly sign and issue certificates to end users or lower-level CAs. All certificates from end users to the root CA form a certificate chain. Users can authenticate the certificates in a certificate chain after obtaining the peer's root CA certificate and subordinate CA certificates at each level. Certificate-chain-based authorization facilitates certificate deployment because the root CA does not need to sign and issue certificates for all end users.
- Issuing CA: An issuing CA is also a subordinate CA and is located at a leaf node in a CA hierarchy. It is used to issue digital certificates to devices, servers, applications, and users on an operator network. There can be multiple issuing CAs, one for generating device certificates, one for generating server certificates, and one for issuing certificates to client users.

There is no strict limitation on the number of tiers in a CA system. Operators can divide the system into a number of tiers as required. Generally, a three-tier CA system can meet the requirements of most operators. However, a two-tier CA system is recommended to reduce management costs and complexity.

CA hierarchy planning is important because it affects the application, verification, and OM of digital certificates on a network. Operators can plan CA deployment based on the service scale, involved functional departments, device vendors, and network infrastructure types, geographical distribution, security domains, costs, and OM management.

- If the digital certificate application involves external users (for example, Internet web service users), an external commercial CA system can be used to deploy digital certificates on a live network.
- If digital certificates are used only inside an enterprise and a large number of digital certificates are required, you are advised to purchase a CA system from a commercial CA vendor and deploy a self-managed PKI system in the organization, including the root CA and subordinate CAs required by each department.

NOTE

Currently, Huawei base stations support a CA system with a maximum of four tiers.

A logical or physical entity can be both an EE entity and a CA entity. For example, a CA system that integrates all PKI management functions is deployed on a network. When NEs on the network apply for digital certificates from the CA system in online mode, the CA system plays two roles for the NEs. It is a CA entity that issues digital certificates to the NEs. It is also a communication entity, such as the CMP server. In this case, the CA system is an EE entity and requires an identity certificate.

4.1.1.4 RA

An RA is a certificate registration and approval authority. As shown in [Figure 4-1](#), an RA interacts with EEs, registers certificate applicant information, and verifies applicant qualifications. The RA then determines whether to issue a certificate to an applicant based on the verification result. If the application is approved, the RA sends the application information to the CA, which eventually issues the certificate.

An RA is an optional component of a CA. An RA is not required in a small-sized PKI system because the CA itself can handle interactions with EEs. In a large-sized PKI system, the CA focuses on certificate management and an RA takes over the functions of exchanging information between the CA and the EE.

4.1.1.5 Digital Certificate and CRL Database

As shown in [Figure 4-1](#), the digital certificate/CRL database stores CA-issued digital certificates and CRLs, which can be obtained by relying parties such as base stations and base station controllers. Digital certificates are approved and signed files, while CRLs contain information about certificates revoked by CAs.

On a live network, the digital certificate/CRL database can be deployed as an independent entity on a server in the demilitarized zone (DMZ). This enables users on the network to obtain digital certificates/CRLs online, without posing a security risk to the CA system.

This database is generally deployed on a File Transfer Protocol (FTP), Lightweight Directory Access Protocol (LDAP), or Hypertext Transfer Protocol (HTTP) server.

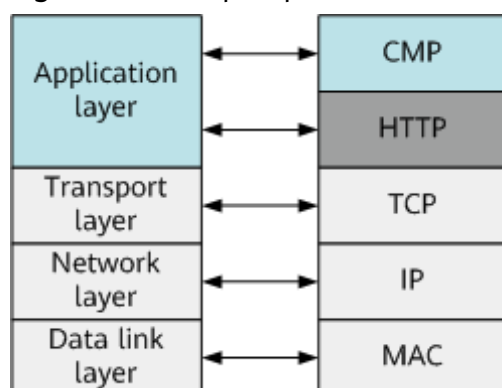
4.1.1.6 CMP

The PKI system uses the Certificate Management Protocol version 2 (CMPv2) to automatically manage digital certificates. 3GPP TS 33.310 specifies the use of this protocol in wireless networks. CMP-based interworking between NEs and the RA/CA must comply with this requirement.

CMPv2 is defined in RFC 4210, RFC 4211, and RFC 6712. On a secure network, a base station/base station controller can automatically apply for an operator-issued certificate using the initial certificate to access the operator's network or apply for a new operator-issued certificate using the old operator-issued certificate to update the operator-issued certificate through CMPv2. The base station/base station controller/MAE uses HTTP/HTTPS as the bearer protocol for CMPv2. HTTP/HTTPS is defined in RFC 6712.

[Figure 4-3](#) shows the transport protocol stack for CMPv2.

Figure 4-3 Transport protocol stack for CMPv2



NOTICE

Automatic digital certificate management for NEs through CMPv2 requires that the NEs always have valid device certificates.

4.1.2 Introduction to Digital Certificates

4.1.2.1 Device Certificate

A device certificate is used to identify an NE. During network protocol authentication, the device certificate is used to verify the validity of the NE identity. Each device certificate has a unique private-public key pair. The key pair is used for digital signature and verification during authentication between a base station/base station controller and the SeGW/MAE.

Device certificates used by base stations/base station controllers are Huawei initial certificates and operator-issued device certificates.

Huawei Initial Certificates

A Huawei initial certificate is issued by a Huawei-signed CA. In compliance with 3GPP TS 33.310, Huawei equipment is preconfigured with an initial certificate issued by Huawei CA during the manufacturing phase. The certificate is used to identify Huawei equipment and is used by the equipment to securely access an operator's network during initial deployment on a live network.

Huawei base stations adopt two PKI systems. The issued initial certificates are as follows:

- The Huawei ECDSA initial certificate that uses the ECDSA algorithm, with ECC curve set to P384.
- The Huawei RSA initial certificate that uses the RSA algorithm, with a key length of 2048 bits.

The Huawei initial certificate, preconfigured on Huawei devices during the production phase, is an essential factory identity credential for Huawei devices. The following statements are provided regarding its use:

1. A Huawei initial certificate is only used in the deployment phase to establish a secure channel between a device and a customer's network. The Huawei ECDSA initial certificate is used only to apply for an operator-issued device certificate for connecting a device to the customer network during deployment. That is, the Huawei ECDSA initial certificate is used for HTTPS authentication, CMP-based certificate request message signature, and eCPRI interface authentication during CMP-based certificate procedure.
2. A Huawei RSA initial certificate is valid for 15 years since the manufacturing date or until 2041, whichever comes sooner. A Huawei ECDSA initial certificate is valid until 2073.
3. Huawei recommends that customers replace the initial certificates on network devices with customer-managed certificates, and use short-validity certificates and regularly rotate certificates in alignment with industry best practices to

minimize the risk of private key exposure and prevent service disruption due to certificate expiration.

The following devices are preconfigured with initial certificates:

- Each Huawei base station is preconfigured with device certificates issued by Huawei CA (Huawei initial certificates or initial certificates for short) before delivery. The certificates are stored on the main control board (UMPT/UMDU/GTMUc/MDUC), UTRPc, and baseband processing unit (BBP). Each certificate is bound with the ESN of the board.

A board with eCPRI ports is also preconfigured with a Huawei initial certificate before delivery. The certificate can be replaced with an operator-issued device certificate.

NOTE

- The Huawei initial certificate preconfigured on the GTMUc in a GBTS can only be used for SSL connections between the GBTS and the site maintenance terminal (SMT). It cannot be used for PKI or IPsec authentication.
- Before delivery, the following boards of Huawei base stations are preconfigured with both RSA initial certificates and ECDSA initial certificates: (1) the UMPTha/UMPTThb/UBBPh series boards; (2) the AAUs complying with the eCPRI protocol whose **Start Base Station Version That Supports Board Software Mapping** is V100R020C10SPCxxx or later. For the **Start Base Station Version That Supports Board Software Mapping** of an AAU, see *3900 & 5900 Series Base Station Software and Hardware Compatibility Description* in *3900 & 5900 Series Base Station Product Documentation*. For details about AAU models that support the eCPRI protocol, see CPRI port specifications in AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- In a Huawei base station, the RSA initial certificate file is named **appcert.pem**, and the ECDSA initial certificate file is named **vendorcert.pem**.
- Each Huawei base station controller is preconfigured with an initial certificate issued by Huawei CA before delivery. This certificate uses the RSA algorithm with a 2048-bit key and is automatically activated before delivery. The BSC6900/BSC6910 certificate is bound with the ESN of the OMU board and is named **hwusercert.pem**. The BSC6960 certificate is bound with the ESN of the CMPU board and is named **appcert.pem**.

The application scenarios of Huawei initial certificates are as follows:

- If a PKI system is deployed in an operator's network, Huawei initial certificates are used for authentication during the operator-issued device certificate application process. When a base station/base station controller accesses the operator's network, it applies for an operator-issued device certificate from the operator's CA by sending a CMPv2 message. The operator-issued device certificate is then used for authentication during subsequent communication processes.
- Huawei MAE integrates the CA service function to issue digital certificates to network devices managed by the MAE. This facilitates device certificate management on an operator's live network where no PKI system is deployed. In this case, the preconfigured Huawei initial certificate can be used to apply for a digital certificate from the LiteCA. It is recommended that the LiteCA be enabled and the Huawei initial certificate be replaced with a certificate issued by the LiteCA in a timely manner.

If a preconfigured Huawei initial certificate of a base station (using a UMPT/UMDU/GTMUc/MDUC as the main control board or using the BBP or transmission

board of UTRPc) or a base station controller becomes invalid (for example, the certificate is lost, damaged, or revoked, or has expired), the base station will report ALM-26841 Certificate Invalid, the BSC6900/BSC6910 will report ALM-20851 Digital Certificate Loss, Expiry, or Damage, and the BSC6960 will report ALM-21031 Certificate Invalid. You can determine whether the invalid certificate is the preconfigured Huawei initial certificate based on the alarm parameter **Certificate Name** of the preceding alarms at the base station, BSC6900/BSC6910, or BSC6960.

- If a preconfigured Huawei initial certificate of the base station is lost or damaged, and the certificate is required for identity authentication, contact Huawei technical support engineers to replace the involved board.
- If a preconfigured Huawei initial certificate **hwusercert.pem/appcert.pem** of a base station controller is lost and this certificate is required for identity authentication, contact Huawei technical support engineers to replace the involved board. The initial certificate **usercert.pem** is preconfigured by software and is not bound with the ESN of the OMU board. If such a certificate is lost, contact Huawei technical support engineers to reinstall software.

The initial certificate on the board for certificate deployment can be queried by running the **DSP CERTMK** command on the base station controller.

The initial certificate on the board for certificate deployment can be queried by running the **LST VENDORCERT** command on the base station. The initial certificates on all boards can be queried by running the **DSP BRDCERTMK** command on the base station with the **CERTFILETYPE** parameter set to **DEVCERT**.

NOTICE

Huawei CA does not issue digital certificates for devices on an operator's network or provide online certificate management services. Therefore, Huawei initial certificates do not support online certificate update, application, or revocation.

Operator-Issued Device Certificate

An operator-issued device certificate is a digital certificate issued by an operator's CA to a device on a live network. The CA system deployed by an operator on a live network is controlled and managed by the operator. The service security domain of the CA system is limited to the internal network of the operator or even the network infrastructure served by the CA system, improving network security.

If a PKI system is deployed on the operator's network, a Huawei device uses a Huawei initial certificate for identity authentication to apply for an operator-issued device certificate. The operator-issued device certificates are then used for base station/base station controller authentication during subsequent communication processes.

The validity period of an operator-issued device certificate is configured by the operator on the CA as required. Operator-issued device certificates support certificate application, update, and revocation. Compared with Huawei initial certificates, operator-issued device certificates feature flexible management and various risk control methods. It is recommended that Huawei initial certificates be

replaced with operator-issued device certificates immediately after the base station/base station controller connects to the operator's network.

The base station and BSC6960 can apply for RSA and ECDSA certificates as operator-issued device certificates through the **CA.KEYTYPE (5G gNodeB, LTE eNodeB)** and **CERTREQ.KEYTYPE (5G gNodeB, LTE eNodeB)** parameters. The BSC6900/BSC6910 can only apply for RSA certificates as operator-issued device certificates.

The base station/base station controller allows users to run the **DSP CERTMK** command to query the operator-issued device certificate on a board. The base station also allows users to run the **DSP BRDCERTMK** command with the **CERTFILETYPE** parameter set to **DEVCERT** to query the operator-issued device certificates on all boards.

NOTICE

Except when the active and standby transmission boards work in cold backup mode, operator-issued certificates of base stations should be deployed on the main control board in most cases and **Certification Deploy Position Type** in the **CERTDEPLOY** MO must be set to **DEFAULT**. The base station does not allow a BBP to be configured as the certificate deployment position.

4.1.2.2 Root Certificate, Certificate Chain, Trust Certificate, and Trust Group

Root Certificate

A root certificate is the certificate of the root CA and is used to verify the validity of device certificates issued by the root CA.

Huawei base stations adopt two PKI systems, and the base stations use two PKI root certificates as trust certificates:

- The preset Huawei-issued ECDSA root certificate that uses the ECDSA algorithm, with ECC curve set to P521.
- The preset Huawei-issued RSA root certificate that uses the RSA algorithm, with a key length of 4096 bits.

A Huawei base station is preconfigured with Huawei root certificates on its main control board (UMPT/UMDU/GTMUc/MDUC), UTRPc, and BBP as trust certificates before delivery. The root certificates are used to verify Huawei initial certificates.

A Huawei GSM/UMTS base station controller is preconfigured with a Huawei root certificate as the trust certificate before delivery. The root certificate is used to verify Huawei initial certificates. The preconfigured root certificate of the BSC6900/BSC6910 is named **rootca.pem**, and that of the BSC6960 is named **caroot.pem**.

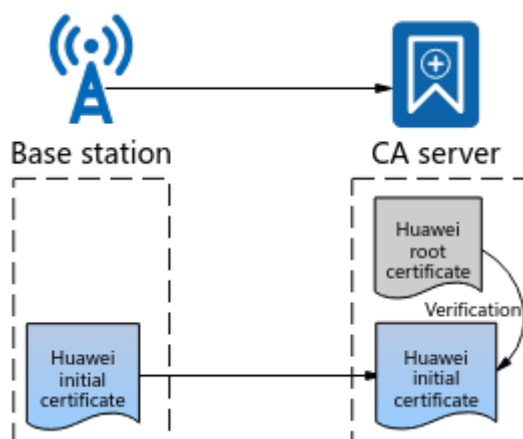
NOTE

- Huawei wireless-network CA system is a two-tier CA system. **caroot.pem**, **vendorroot.pem**, and **rootca.pem** are files in the two-tier certificate chain.
- Before delivery, the following boards of Huawei base stations are preconfigured with both RSA trust certificates and ECDSA trust certificates: (1) the UMPTha/UMPTb/UBBPh series boards; (2) the AAUs complying with the eCPRI protocol whose **Start Base Station Version That Supports Board Software Mapping** is V100R020C10SPCxxx or later. For the **Start Base Station Version That Supports Board Software Mapping** of an AAU, see *3900 & 5900 Series Base Station Software and Hardware Compatibility Description* in *3900 & 5900 Series Base Station Product Documentation*. For details about AAU models that support the eCPRI protocol, see CPRI port specifications in AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.
- In a Huawei base station, the RSA trust certificate file is named **caroot.pem**, and the ECDSA trust certificate file is named **vendorroot.pem**.

If a Huawei base station/base station controller uses an operator-issued device certificate to connect to an operator's network, the base station/base station controller must be preconfigured with and trust the operator's root certificate or certificate chain to authenticate the operator's device, such as the SeGW or a third-party FTP server. Similarly, the operator's device must be preconfigured with and trust the Huawei root certificate or certificate chain to authenticate the base station/base station controller. During authentication, the communicating parties use their respective trust certificates to verify the validity of the peer's device certificate.

Figure 4-4 shows an example of authenticating the Huawei initial certificate by using the root certificate or certificate chain. A Huawei certificate chain is preconfigured on the CA server. During identity authentication, the base station sends a Huawei initial certificate to the CA server. The CA server uses the preconfigured Huawei root certificate and intermediate CA certificate to verify the validity of the Huawei initial certificate sent from the base station. Since Huawei wireless-network CA system has two root CAs and three subordinate CAs, the certificates of these CAs should be preconfigured and trusted on the CA server.

Figure 4-4 Base station authentication by a CA server



To query the status of the preconfigured root certificate on the main control board (UMPT/UMDU/GTMUc/MDUC)/BBP/UTRPc, you can run the **DSP BRDCERTMK** command with **CERTFILETYPE** set to **TRUSTCERT**.

Certificate Chain

If there are multiple levels of CAs in a PKI system, the PKI system searches for a trusted root CA from the upper-level CAs based on the issuer information of a device certificate to verify the validity of the device certificate. A CA certificate chain consists of a series of certificates in sequence. The CA of the device certificate, its upper-level CAs, until the trusted root CA can be traced one by one.

The base station supports the parsing and authentication of the following three types of certificate chains. The specific method depends on the signature algorithm of each CA certificate in the CA hierarchy.

- RSA certificate chain: All certificates on the chain are RSA certificates.
- ECDSA certificate chain: All certificates on the chain are ECDSA certificates.
- Hybrid certificate chain: The certificates on the chain consist of both RSA and ECDSA certificates.

NOTICE

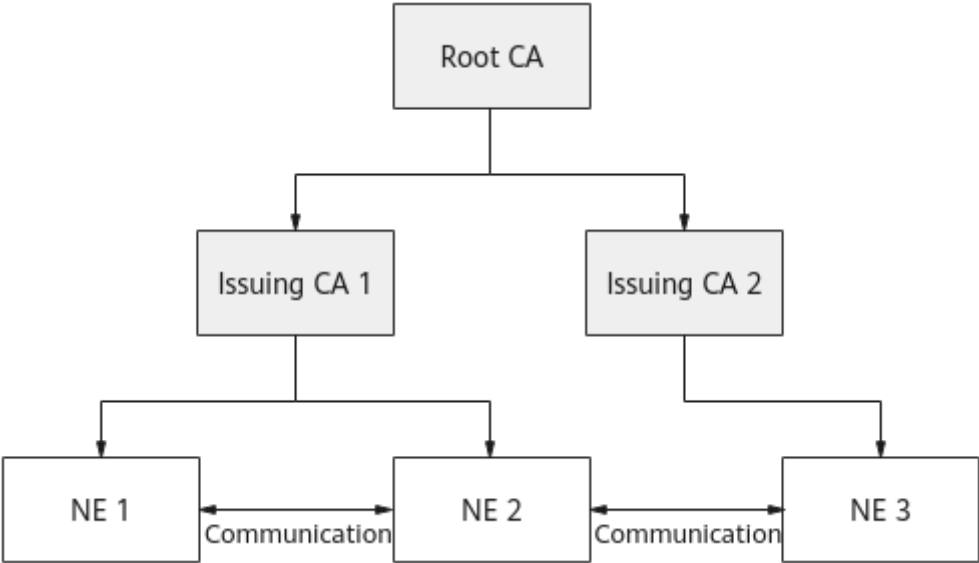
If the operator's certificate chain is a hybrid one but the peer device adopting protocols such as IPsec or SSL cannot parse a hybrid certificate chain, a security channel based on such protocols may fail to be established.

Trust Certificate

A trust certificate is a root certificate, subordinate CA certificate, or CA certificate chain that is trusted by an NE and loaded onto an NE device. Among these trusted CA certificates, only the root certificate is used as the trust anchor during certificate authentication. Therefore, when a Huawei device is connected to another device, if the peer device and the Huawei device are not in the same root CA domain, you need to configure the peer root CA certificate and CA certificate chain on the Huawei device. There are two common trust models:

1. Hierarchical trust model, as shown in [Figure 4-5](#)

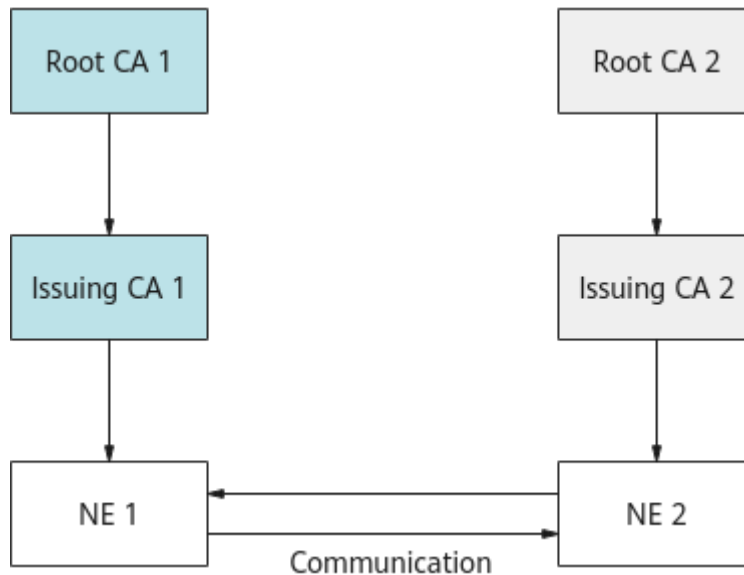
Figure 4-5 Hierarchical trust model



In the single-root CA hierarchical deployment mode, the identity certificates used for communication between all NEs on the network belong to the same root CA trust domain. Each NE needs to trust only one root CA and is configured with an entire CA certificate chain. This is the simplest CA trust model.

2. Peer-to-peer trust model, as shown in [Figure 4-6](#)

Figure 4-6 Peer-to-peer trust model



In the multi-root CA hierarchical deployment mode, the identity certificates used for communication between NEs on the network belong to different root CA trust domains. In addition to being configured with and trusting the root CA certificate and CA certificate chain of the local end, each NE must be configured with and trust the root CA certificate and CA certificate chain of the peer end, or cross-certificates issued by both local and peer ends. This is a complex CA trust model.

Typical scenarios are as follows:

- A Huawei device uses the initial certificate issued by Huawei CA, and the peer device uses the certificate issued by the operator's CA. For example, a Huawei device accesses an operator's network for the first time and applies for a digital certificate from the operator's CA system.
- In RAN sharing scenarios, RAN devices access operator core networks using IPsec and each operator has deployed an independent CA to issue IPsec digital certificates.
- The internal network domains of the operator are isolated from each other. Different network domains use independent root CA policies. When devices in two network domains need to communicate with each other securely, mutual trust between CAs is involved.
- The LiteCA is used, each OSS is deployed with an independent root CA, and NE devices are migrated from OSS A to OSS B or NE devices under two OSSs communicate with each other.

The **DSP BRDCERTMK** command can be executed to query the trust certificate information of all boards in a base station. In this case, the **CERTFILETYPE** parameter must be set to **TRUSTCERT**.

Trust Group

A combination of any existing trust certificates in a base station constitutes a trust group. A trust group can provide a minimum set of trust certificates for each link. This ensures that only the device certificates planned by the customer can be authenticated on the service link. Compared with the original method of directly applying all trust certificates of the base station to the service link, the trust group improves the link security.

A trust certificate can be added to the **TRUSTGROUP** MO by running the **ADD TRUSTCERT2GROUP** command on the base station.

If a service link is not associated with a trust group, the Huawei base station must be preconfigured with the root CA of the peer device certificate so that the peer device certificate can be successfully authenticated. If **Trust Group ID** is **65535**, it indicates that no trust group has been associated.

If a service link is associated with a trust group, the Huawei base station uses the trust certificates in the trust group or cross-certificates to verify the device certificate of the peer end. A service link cannot be associated with an empty trust group.

Customers are advised to plan trust groups to ensure that only the peer end with a valid identity certificate can pass the authentication on each service link.

If a trust certificate is added to or deleted from a trust group, the referenced trust certificates in the group take effect during the next service link setup or negotiation. The established service links are not affected.

When configuring a trust group, comply with the following rules:

- If a trust group contains **caroot.pem**, only the non-root CA trust certificates of **caroot.pem** can be added to the trust group.
- If the base station uses operator-issued certificates for authenticating NEs in the networking, **caroot.pem** does not need to be added to any trust group of the base station.
- If the base station uses a Huawei-issued certificate for authenticating NEs in the networking and a link is associated with a trust group, the customer needs to add the subordinate CA certificates corresponding to the device certificate of the authorized peer end to the trust group of the base station.

4.1.2.3 Cross-Certificate

A cross-certificate is issued by one CA to another in order to establish a trust relationship between them. Cross-certification is a process in which a CA in a PKI system issues a cross-certificate to a CA in another PKI system to establish a trust relationship between the entities under the two CAs. The trust relationship can be unidirectional or bidirectional. Unidirectional trust means that only one CA issues a cross-certificate to another CA, and bidirectional trust means that both CAs issue cross-certificates to each other.

Cross-authentication usually occurs in the peer-to-peer trust model. In this model, two communication devices trust only the root CA of the local end and do not trust the root CA of the peer end. In this scenario, if a secure transmission channel based on certificate authentication needs to be established between the two devices, the certificate management organizations of the two devices need to negotiate, issue a cross-certificate to each other, and deploy the cross-certificates on the two devices. Otherwise, the trust relationship cannot be established.

As shown in **Figure 4-7**, cross-certification involves two processes:

1. Two CA entities reach an agreement, issue cross-certificates to each other, and establish a trust relationship between the two CAs.
2. When authenticating digital certificates, the EEs in the two CA domains use cross-certificates to establish a trust path and verify the trust relationship.

Figure 4-7 Cross-certificate issuing and authentication

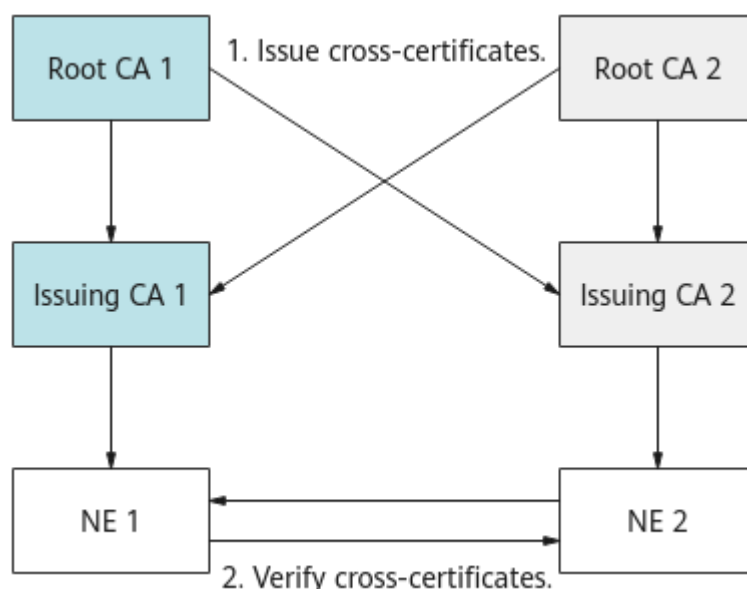
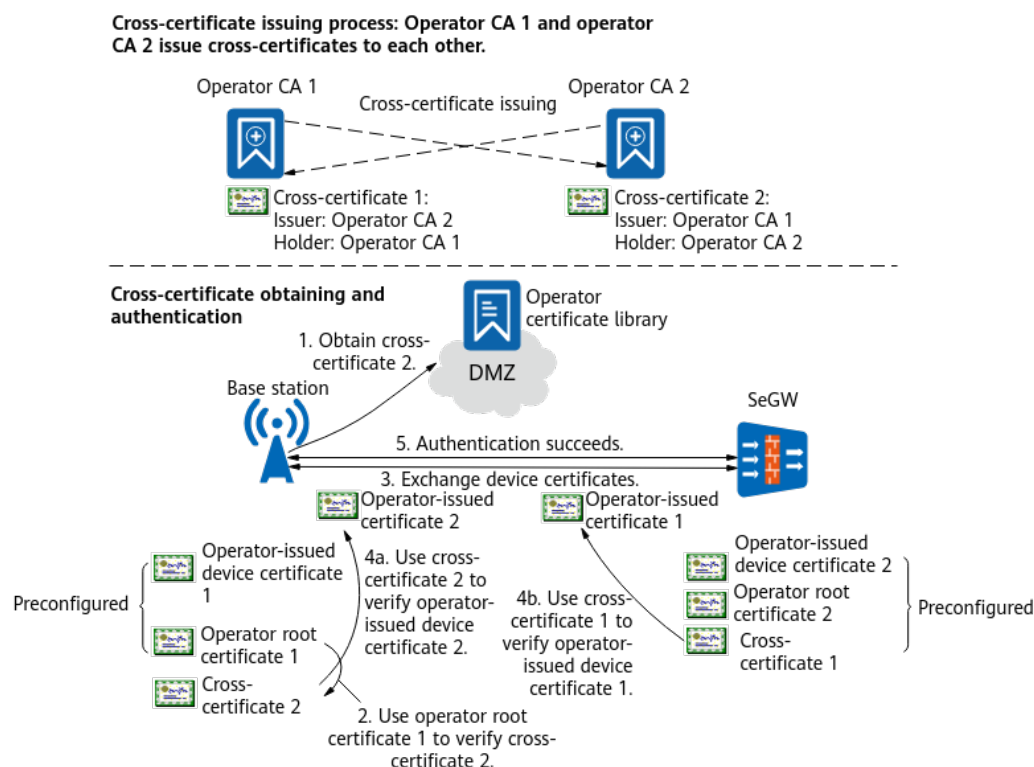


Figure 4-8 shows the cross-certification process between operator CA 1 and operator CA 2.

Figure 4-8 Process of issuing cross-certificates

To use cross-certificates for authentication, ensure that operator CA 1 and operator CA 2 can issue cross-certificates to each other. This process requires cooperation between operator 1 and operator 2 and is complex. Therefore, this process is not recommended.

NOTICE

A cross-certificate must be added by running the **ADD CROSSCERT** command. If a cross-certificate is added by running the **ADD TRUSTCERT** command, IKE authentication may fail.

4.1.2.4 CRL

CRL is used to verify the validity of the peer certificate. The peer device is not trustworthy if its certificate is recorded in a CRL.

A certificate needs to be revoked when the private key of the certificate is disclosed or the device that uses the certificate is changed or discarded. Revoked certificates are recorded in CRLs. After an NE obtains a CRL, during identity authentication, the NE uses the CRL to check whether the device certificate sent by a peer device has been revoked.

- The CRL usage policy for the GBTS is set by running the **SET BTSCRLPOLICY** command.
- The CRL usage policy for the eGBTS/NodeB/eNodeB/gNodeB/base station controller is set by running the **SET CRLPOLICY** command.

The setting of the **CRLPOLICY.CRLPOLICY (5G gNodeB, LTE eNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **BTSCRLPOLICY.CRLPOLICY** parameter (for the GBTS) is as follows:

- When the parameter is set to **NOVERIFY**, the base station/base station controller does not perform CRL-based certificate validity checks.
- When the parameter is set to **ALARM**, the base station reports ALM-26834 Peer Certificate Invalid, the BSC6900/BSC6910 reports ALM-20854 Peer Certificate Invalid, and the BSC6960 reports ALM-21029 Peer Certificate Invalid if a revoked device certificate of the peer end is detected against the CRL.
- When the parameter is set to **DISCONNECT**, the base station/base station controller reports the preceding alarms and disconnects from the peer end if a revoked device certificate of the peer end is detected against the CRL.

NOTE

If this parameter is set to **NOVERIFY**, the revocation of peer certificates cannot be detected in a timely manner. As a result, the revoked certificates may be used by unauthorized users. Therefore, it is recommended that this parameter not be set to **NOVERIFY**. The value **ALARM** or **DISCONNECT** is recommended.

A cross-certificate cannot be revoked using the CRL.

4.1.2.5 OCSP

Generally, the certification authority (CA) issues a new certificate revocation list (CRL) at a specified interval. Therefore, the CRL cannot reflect the actual certificate status in a timely manner. The Online Certificate Status Protocol (OCSP) can meet the requirement of real-time online query of certificate status. Using OCSP to check the validity of digital certificates is more convenient than the traditional method of downloading and processing CRLs. [Figure 4-9](#) shows how a base station obtains the OCSP status of local device certificates.

Figure 4-9 Process of obtaining the OCSP status of local device certificates



An OCSP server can be any of the following forms:

- The CA itself
- A server whose public key is trusted by the OCSP requester
- A proxy server authorized by the CA

 NOTE

The base station can obtain the OCSP status only from the OCSP server in the first or second form. If the OCSP server is in the second form, the response from the OCSP server must contain the certificate chain. Otherwise, the base station's check on the OCSP status fails and it reports ALM-26842 Automatic Certificate Update Failed.

Currently, the base station does not support signing OCSP requests. The base station will fail to obtain the OCSP status of the local device certificate if the OCSP server requires to verify the OCSP request's signature during the obtaining. In this case, ALM-26842 Automatic Certificate Update Failed is reported.

According to RFC 8954, the base station's OCSP request supports a 32-byte nonce extension to prevent replay attacks. The peer OCSP server must be able to process the 32-byte nonce. Otherwise, the base station will fail to obtain the OCSP status and report ALM-26842 Automatic Certificate Update Failed.

A base station can check whether the device certificate sent by a peer NE is valid based on OCSP status information during identity authentication with this NE.

For the eGBTS, NodeB, eNodeB, and gNodeB, run the **SET CRLPOLICY** command to set the policy of using OCSP.

The **CRLPOLICY.CRLPOLICY (5G gNodeB, LTE eNodeB)** (eGBTS/NodeB/eNodeB/gNodeB) parameter values and the corresponding actions are as follows:

- **NOVERIFY:** Under this setting, the base station does not perform CRL-based certificate validity checks on the peer NE but performs OCSP-based certificate validity checks.
- **ALARM:** Under this setting, the base station performs CRL- and OCSP-based certificate validity checks on the peer NE, and reports ALM-26834 Peer Certificate Invalid if the peer certificate is invalid.
- **DISCONNECT:** Under this setting, the base station performs CRL- and OCSP-based certificate validity checks on the peer NE, and reports ALM-26834 Peer Certificate Invalid and disconnects from the NE if the peer certificate is invalid.

For details about OCSP, see RFC 6960.

 NOTE

In the case that the base station needs to determine whether the peer certificate is invalid based on CRL and OCSP, it determines that the peer certificate is invalid and reports ALM-26834 Peer Certificate Invalid if any of the following conditions is met:

- The peer certificate's OCSP status is revoked or unknown.
- Verification of the peer certificate's OCSP status fails.
- Acquisition of the peer certificate's OCSP status fails.
- The CRL contains the revocation record of the certificate in question.
- The CRL file expires or the signature verification fails.

4.1.3 Certificate Management and Application Scenarios

4.1.3.1 Certificate Preconfiguration Phase

The certificate preconfiguration phase includes the following activities:

- The Huawei CA server issues a Huawei root certificate and the Huawei initial certificates signed by Huawei CA to each of the following boards:

- Main control board (UMPT/UMDU/GTMUc/MDUC), transmission board (UTRPc), and BBP of the base station
- OMU board of the BSC6900/BSC6910 and CPU board of the BSC6960
- AAU supporting the eCPRI protocol

 NOTE

Before delivery, the following boards of Huawei base stations are preconfigured with both RSA initial certificates and ECDSA initial certificates: (1) the UMPTha/UMPTb/UBBPh series boards; (2) the AAUs complying with the eCPRI protocol whose **Start Base Station Version That Supports Board Software Mapping** is V100R020C10SPCxxx or later. For the **Start Base Station Version That Supports Board Software Mapping** of an AAU, see *3900 & 5900 Series Base Station Software and Hardware Compatibility Description* in *3900 & 5900 Series Base Station Product Documentation*. For details about AAU models that support the eCPRI protocol, see CPRI port specifications in AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

- The Huawei root certificates and CRLs are released.
Huawei root certificates and CRLs are released to operators through a web server. Huawei has two root CAs. If the certificate verification function is enabled, the root CA certificates, subordinate CA certificates, and CRLs of the old and new CAs need to be downloaded. They are published at <https://support.huawei.com/additionalres/pki>.
- Huawei wireless-network RSA system has an old and a new subordinate CA. When preconfiguring a Huawei certificate chain, you need to preconfigure the certificates of the two RSA subordinate CAs and the certificates of Huawei root CA. If the CRL verification function is enabled, the CRLs of both the old and new RSA subordinate CAs must be downloaded. The certificates and CRLs of Huawei subordinate CAs are released to operators through a web server.

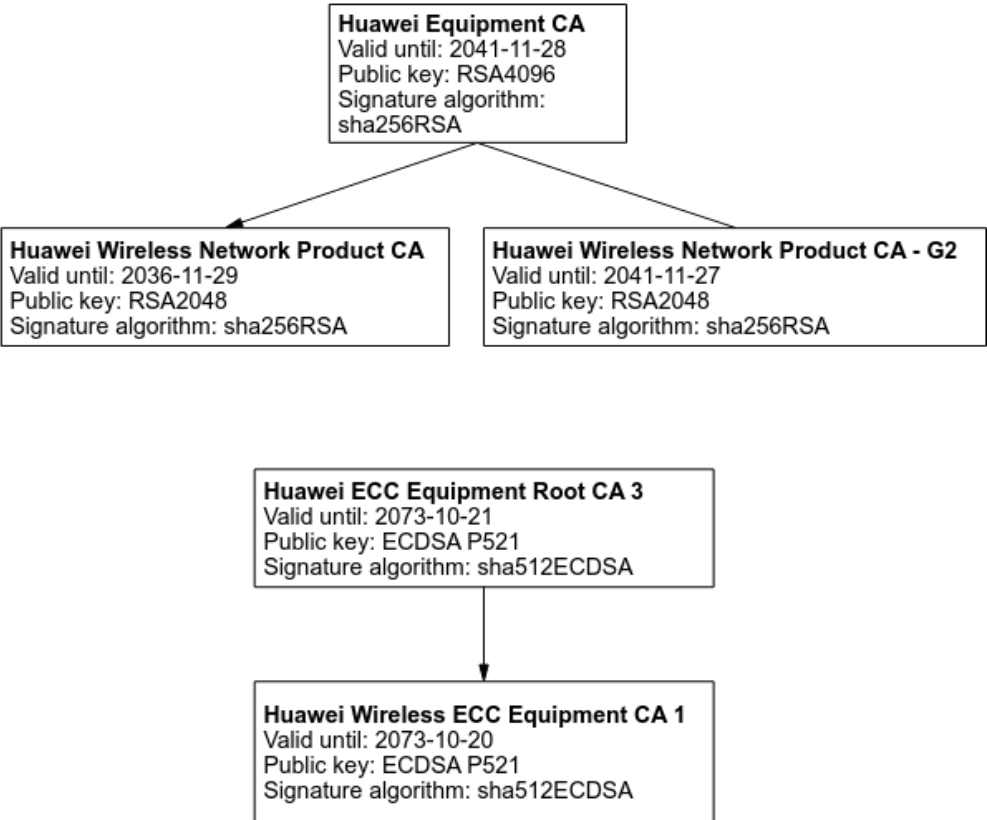
 NOTE

Huawei CA-related files are released at <https://support.huawei.com/additionalres/pki>. Huawei CA-related files are as follows:

- RSA root certificates (common name: Huawei Equipment CA)
- ECDSA root certificate (common name: Huawei ECC Equipment Root CA 3)
- ECDSA subordinate CA certificate (common name: Huawei Wireless ECC Equipment CA 1)
- Certificates of the new wireless RSA subordinate CA (common name: Wireless Network Product CA - G2)
- Certificates of the old wireless RSA subordinate CA (common name: Wireless Network Product CA)
- CRLs issued by the preceding CAs

Figure 4-10 shows the PKI hierarchy.

Figure 4-10 PKI hierarchy



4.1.3.2 Deployment Phase

4.1.3.2.1 Base Station Deployment Phase

Before delivery, Huawei base stations are preconfigured with Huawei initial certificates on their boards. The operator's network must be deployed with a PKI system. During base station deployment, the base stations must apply for device certificates issued by the operator's CA to access the operator's network.

This section describes a scenario where IPsec is used and digital certificates are used for authentication. [Figure 4-11](#) shows such an example.

Figure 4-11 IPsec networking

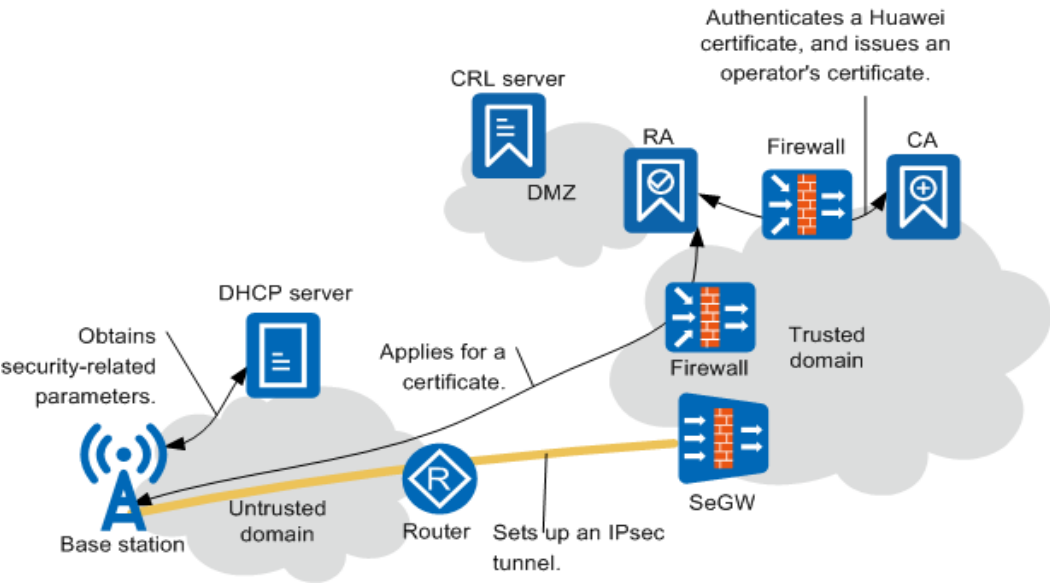
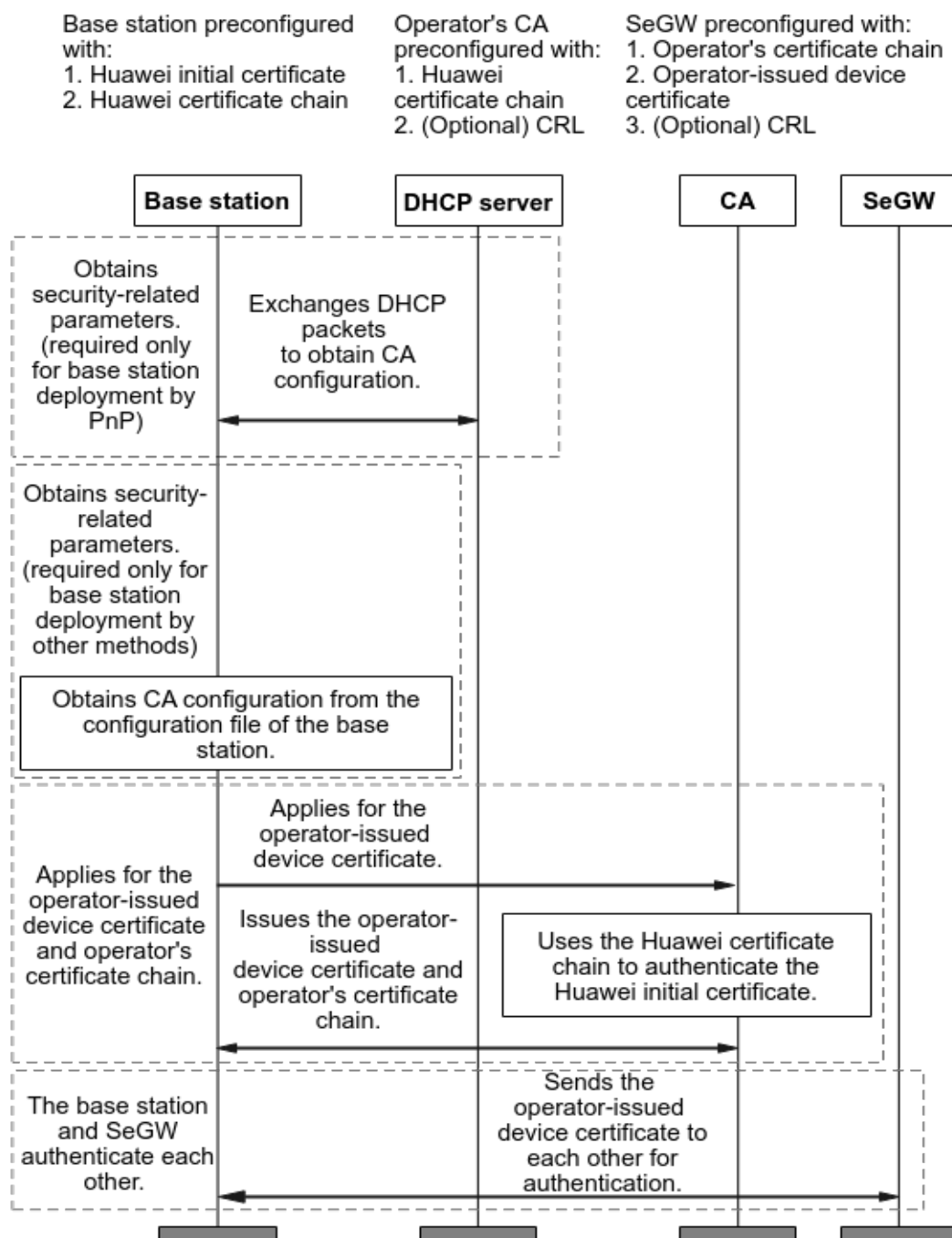


Figure 4-12 shows the certificate application procedure during automatic base station deployment.

Figure 4-12 Certificate application procedure during automatic base station deployment



During automatic base station deployment, the Huawei initial certificate preconfigured on the base station is used as follows:

- If the base station has obtained CA configuration from the DHCP server, the operator requires the base station to use an operator-issued device certificate for authentication. The CA information is used to apply for certificates and includes the certificate key type, IP address of the CA, and the name of the initial certificate file used for HTTPS authentication and certificate request message signing to obtain operator certificate by the base station.

- If the base station has a valid operator-issued device certificate, the base station directly uses this certificate and does not use the preconfigured Huawei initial certificate.
- If the base station has an operator-issued device certificate but information about the issuer is inconsistent with the CA information, this certificate is considered invalid and cannot be used. If the **CERTCHKTSK.AUTOREAPPLYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**, the base station uses the Huawei initial certificate specified by the DHCP server to apply for a new operator-issued device certificate.
- If the base station fails to obtain the operator-issued device certificate or if the request for the device certificate times out, the base station uses the preconfigured Huawei initial certificate. If the base station cannot be automatically deployed by using the Huawei initial certificate, it restarts and attempts to obtain the operator-issued device certificate again. The ECDSA initial certificate is not used for OMCH establishment. Therefore, only the RSA initial certificate is used.
- If CA information fails to be obtained during base station deployment, the base station uses the preconfigured Huawei RSA initial certificate.

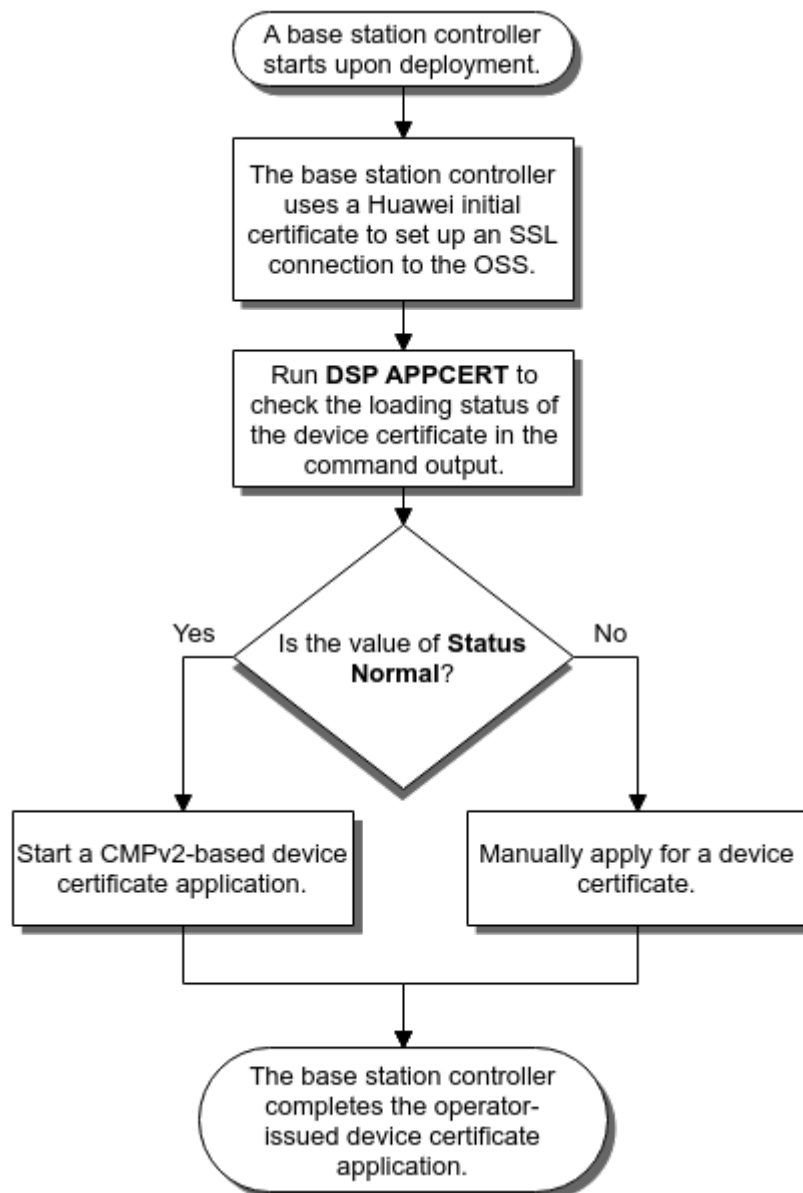
 NOTE

- If an operator's network is deployed with a PKI system, it is recommended that the same operator-issued device certificate be used for IPsec authentication, SSL authentication, and 802.1x-based access control.
- During automatic base station deployment by plug and play (PnP), only Huawei initial certificates can be used for authentication during 802.1x-based access control. By default, the same certificate is used for 802.1x-based access control and SSL authentication in the operation phase.
- The name of the operator-issued device certificate used by a base station during base station deployment by PnP must be **OPKIDevCert.cer**.
- During base station deployment by PnP, the time of the base station, MAE, and CA server must be consistent.
- In IPsec networking scenarios, the LiteCA cannot be used for base station deployment by PnP.
- The URLs of the active and standby CA servers to which the base station connects for certificate application and update and the URL of the OCSP server to which the base station connects for certificate status acquisition cannot contain Chinese characters. If such a URL contains Chinese characters, the connection setup will fail.

4.1.3.2.2 Base Station Controller Deployment Phase (GSM/UMTS)

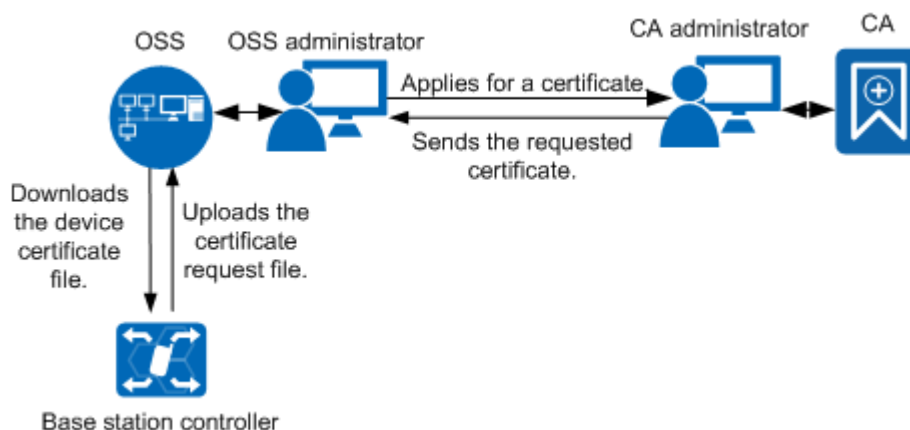
To connect to an operator's network deployed with a PKI system, the Huawei base station controller must apply for a device certificate issued by the operator's CA from the CA server. The operator-issued device certificate can be applied for using a CMPv2-based certificate application procedure or in manual mode depending on the type of preconfigured Huawei initial certificate. [Figure 4-13](#) shows an example.

Figure 4-13 Base station controller deployment procedure



A CMPv2-based certificate application procedure is triggered by the **REQ DEVCERT** command. [Figure 4-14](#) shows an example.

Figure 4-14 Manual certificate application procedure



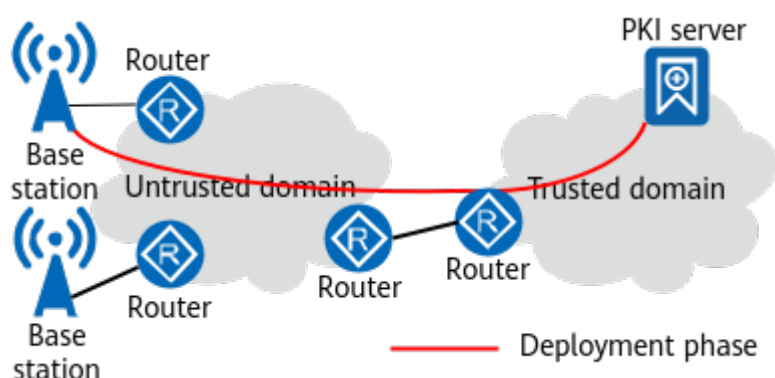
For details about CMPv2-based and manual certificate application procedures, see [5.4.4.2 Using MML Commands](#).

4.1.3.3 Operation Phase

Operations performed in the operation phase include operator-issued certificate application, certificate replacement, certificate sharing, certificate validity check, certificate update, and certificate revocation. After a board is removed from the customer network, the operator-issued certificate on the board needs to be monitored offline to prevent the certificate and private key on the board from being disclosed.

The base station/base station controller can use CMPv2 to manage certificates during certificate application and replacement. [Figure 4-15](#) shows the networking for CMPv2-based certificate management.

Figure 4-15 Networking example of CMPv2-based certificate management



As shown in [Figure 4-15](#), base stations/base station controllers communicate with the PKI server for CMPv2-based certificate management. The PKI server can be a CA, an RA, or a digital certificate&CRL database and must support CMPv2.

4.1.3.3.1 Applying for an Operator's Certificate

For details about how to apply for a certificate for the base station controller, see [4.1.3.2.2 Base Station Controller Deployment Phase \(GSM/UMTS\)](#).

In the operation phase, if a base station needs to use an operator-issued device certificate for IKE/SSL authentication but it does not have such a certificate, the base station must apply for an operator-issued device certificate from the CA server based on CMPv2.

CMPv2-based certificate application of base stations is triggered in two modes:

- Manual mode

To manually trigger the application, you need to configure information such as the certificate deployment position, CA, trust certificate, and certificate request on the base station. For details, see [4.4.3.1 Data Preparation](#). Then run the **REQ DEVCERT** command to trigger a CMPv2-based certificate application procedure. After this command is executed, the base station reports the progress of the certificate application. If an operator-issued device certificate is obtained, OM personnel are allowed to run commands such as **MOD APPCERT** to change the device certificate used by service links to the operator-issued device certificate.

To ensure that the device certificate can be used to successfully establish security channels between the base station and the following two: the OSS and the SeGW, verify that the device certificate can be used for IKE and SSL connections and then run the **CFM CB** command to enable automatic configuration data rollback before device certificate replacement. For details, see the **CFM CB** command help. The certificate test method is as follows:

- IKE connection: Run the **TST APPCERT** command to perform a certificate test.
- SSL connection: Use the MAE for a certificate test. The procedure is as follows:
 - On the MAE home page, click **Access** to enter the MAE-Access home page. Choose **Security > Certificate Authentication Management > Certificate Management** from the main menu.
 - Choose **NE Certificate Management > Device Certificate**.
 - Click the **Certificate List** tab.
 - Select an NE in the **Select NE** area.
 - In the **Certificate List** tab on the right, select the corresponding device certificate and click **Test**.

- Automatic mode

The base station obtains information about the certificate deployment position, CA, certificate request, and trust certificate from the configuration file. After a base station is restarted, a CMPv2-based certificate application procedure will be automatically triggered if IKE authentication or OMCH authentication fails. If the application fails, the base station automatically re-initiates a CMPv2-based certificate application procedure.

 NOTE

After the IKE negotiation succeeds, the base station starts checking the IKE negotiation status every 7 minutes. If IKE negotiation fails and identity authentication is performed using digital certificates, the base station checks the digital certificates used for IKE negotiation. If the digital certificates are abnormal (for example, revoked, expired, or unavailable), automatic digital certificate retrieval is triggered. If retrieval succeeds for two consecutive days (up to five successes per day) but the IKE negotiation failure persists, the base station stops automatic certificate application and waits for manual intervention.

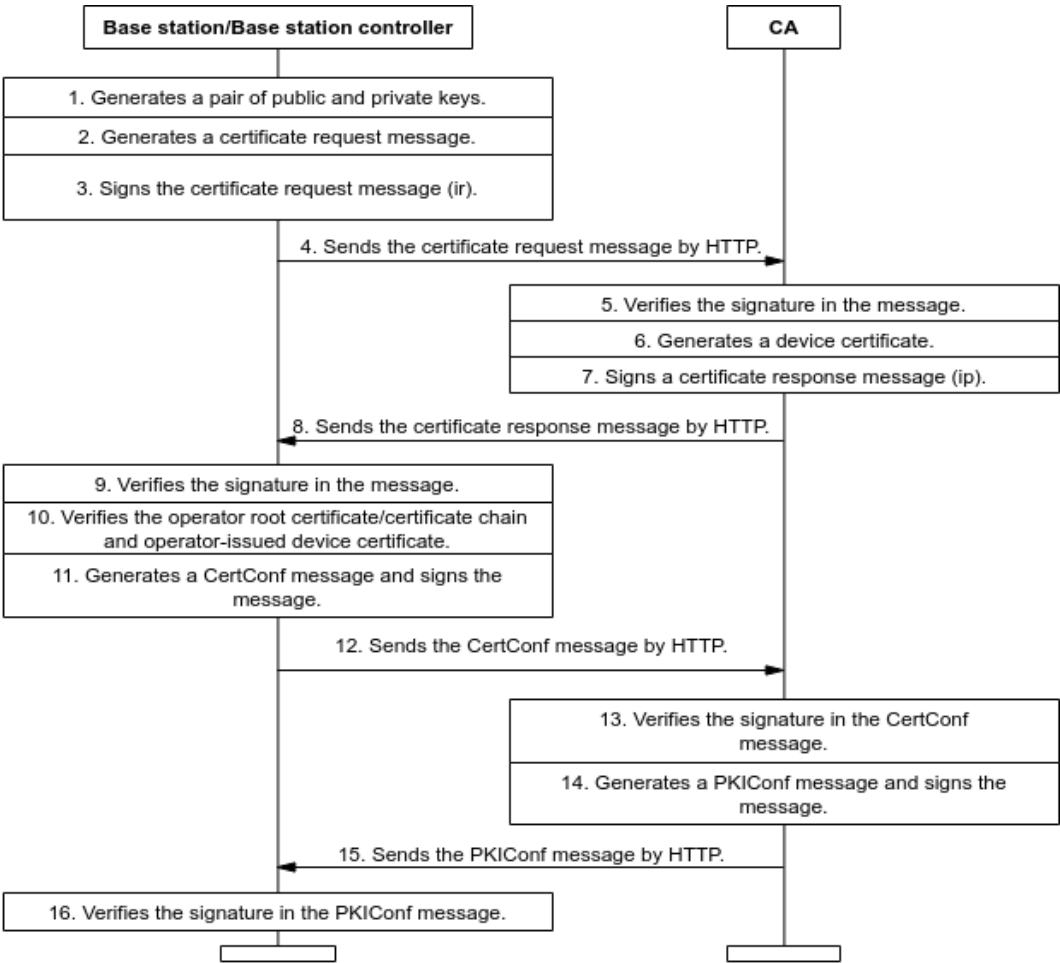
When applying for an operator-issued certificate for the first time, the base station or base station controller uses the preconfigured Huawei initial certificate as the identity credential. Therefore, the operator's CA or CMP server must be preconfigured with a Huawei-issued certificate chain to verify the Huawei initial certificate carried in a CMPv2-based request message sent from the base station or base station controller. When a Huawei initial certificate is used as an authentication credential in the application for a certificate from the CA, you can set the **CA.VENDORCERT (LTE eNodeB, 5G gNodeB)** parameter to specify which initial certificate configured is used for HTTPS authentication and signature of the certificate request message. If the board corresponding to the certificate deployment position is not preconfigured with the specified certificate or the specified certificate is damaged or lost, the system uses the other Huawei initial certificate on the board. The operator's CA also needs to include operator-issued device certificates and root certificates or certificate chains in CMPv2 response messages sent to the base stations or base station controllers.

 NOTE

Huawei certificate chain: A Huawei certificate chain involves two root CAs and three subordinate CAs of the wireless-network CA system. Therefore, when preconfiguring a Huawei certificate chain on the operator's CA or CMP server, the certificates of the two root CAs and three subordinate CAs should be preconfigured and trusted. For details about Huawei PKI hierarchy, see [4.1.3.1 Certificate Preconfiguration Phase](#).

Figure 4-16 shows how a base station or base station controller applies for a certificate based on CMPv2.

Figure 4-16 Certificate application process for a base station or base station controller



NOTE

After sending a CMPv2-based certificate request message, the base station waits for a response from the CA server. The waiting timeout interval is 60s in single-operator PKI scenarios and 20s for each PKI in multi-operator PKI scenarios. If the base station does not receive any response from the CA server before the waiting timeout interval elapses, the certificate application fails.

If the certificate issued by the CA takes effect based on the time of the CA server, the time of the base station or base station controller must be the same as that of the CA server.

- If the effective time of a certificate issued by the CA is later than the current time of the base station or base station controller, the certificate received by the base station or base station controller does not take effect until the effective time arrives.
- If the effective time of a certificate issued by the CA is earlier than the current time of the base station or base station controller, the certificate takes effect immediately.

The base station does not support certificate application through the E1/T1 port.

The format of the Common Name field in the initial certificate of the BBU in the base station is ESN.huawei.com.

The format of the Common Name field in the initial certificate of the AAU in the base station is ESN.re.huawei.com.

The format of the Organization, Organization Unit, State or Province, and Locality fields in the recipient field of the CMPv2-based certificate request message sent by the base station can be in printable or UTF8, and the former is preferred. The format of the Country field can only be in printable format.

In step 2, the message contains information such as the newly generated public key, SubjectName field of the certificate, alternative SubjectName field of the certificate, certificate signature algorithm, and local device certificate (Huawei initial certificate or operator-issued device certificate). The **SubjectName** field in a certificate request message contains the **Common Name** field. Some CAs can issue operator-issued digital certificates only when the **Common Name** field in the certificate request message is consistent with that in local device certificates.

In step 5, the CA server uses the public key in the request message to verify the signature, and uses the Huawei or operator root certificate corresponding to the base station's local device certificate to verify the local device certificate carried in the message.

NOTE

When the base station applies for a certificate from the operator's PKI system, an operator-issued device certificate can be specified for authentication and message signature between the base station and the PKI system. This solves the problem that the base station cannot apply for a certificate after the initial certificate expires.

In step 6, after the authentication succeeds, the CA server generates a device certificate for the base station or base station controller and uses the private key corresponding to the CA certificate to sign the certificate. The base station supports high-level security algorithms, and frequent certificate updates may affect network stability. Therefore, it is recommended that the CA issue device certificates with a validity period of one to three years to the base station.

In step 7, the CA server signs the certificate request response message. This message contains the device certificate issued by the CA to the base station and the operator root certificate or certificate chain (IP address).

In step 8, if the **SubjectName** field in the certificate response message is determined by the CA server's local information, the CA server must ensure that

the string conforms to protocol specifications and is recognizable by the base station. Failure to do so may result in certificate verification or application failure at the base station.

The algorithms related to certificate application are as follows:

- CMPv2 messages support the following signature algorithms: SHA1, SHA256, and SHA384.
- The certificate request template supports the following signature algorithms: SHA1, SHA256, and SHA384.
- The key types supported by the certificate request template include RSA and ECDSA.
- The RSA key sizes supported by the certificate request template include 1024, 2048, 3072, and 4096 bits.
- The ECC curve types supported by the certificate request template include P256, P384, and ED25519.

NOTE

An ECDSA digital certificate features high security, fast computing, small key length, and small storage space. Therefore, if all the operator's devices support ECDSA digital certificates, the ECDSA key type is recommended.

SHA1 and RSA1024 are considered insecure in the industry. If these algorithms are used, transmitted data and issued certificates may be forged by attackers. You are advised to use high-security signature algorithms such as SHA256 and SHA384 and high-security key sizes such as RSA3072, RSA4096, and P256.

Unlike device certificates, trust certificates and CRLs also support the NIST P521 curve and SHA512withECDSA signature algorithm.

In compliance with 3GPP specifications, the TLS application on the base station checks whether a certificate contains an extended key during certificate verification. If not, verification proceeds. If yes, the TLS application checks whether the TLS server certificate contains the extended key usage **id-kp-serverAuth** and whether the TLS client certificate contains the extended key usage **id-kp-clientAuth**. If yes, verification proceeds; if not, certificate verification fails. To ensure normal base station services, it is recommended that a certificate issued to the base station do not contain extended key usages. If extended key usages are required, the certificate should contain **id-kp-serverAuth** and **id-kp-clientAuth**. Avoid using extended key usage **any**.

Support for Automatic Re-Application When the Certificate Issuer Changes

The issuer of a certificate file needs to be the same as the value of **CA.CANAME (LTE eNodeB, 5G gNodeB)** corresponding to the certificate file.

Assume that a base station is operating properly, the base station uses certificates for identity authentication, and the **CERTCHKTSK.AUTOREAPPLYSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**. If SSL link setup or IKE negotiation fails, the base station checks the certificate status at an interval of 1 minute for SSL certificates and 7 minutes for IKE certificates. (If the certificate application is successful for two consecutive days but the SSL link setup or IKE negotiation failure persists, the base station stops automatic certificate application and waits for manual intervention.) If the issuer of a certificate file is found to be inconsistent with the corresponding **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter value, the base station triggers automatic certificate re-application (up to five successes per day) to recover the link. An inconsistency will occur in the following scenarios:

- The base station is running properly, and the issuer of the certificate file is consistent with the corresponding **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter value. However, the **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter value is then modified.
- The base station is running properly, and the issuer of the certificate file is inconsistent with the corresponding **CA.CANAME (LTE eNodeB, 5G gNodeB)** parameter value.

4.1.3.3.2 Replacing Certificates

Certificate replacement is a process of replacing the certificate currently used by applications such as IKE/SSL on an NE with a new certificate. Main scenarios include:

- The Huawei initial certificate in use needs to be replaced with an operator-issued certificate.
- The operator's CA system is changed. The certificates issued by the old CA on the NE need to be replaced with the certificates issued by the new CA.
- NEs are migrated from one CA domain to another CA domain.

Replacing the Initial Certificate

A preconfigured Huawei initial certificate is used for secure access to an operator's network during site deployment. The preconfigured Huawei initial certificate has a validity period. After the validity period expires, certificate update, application, or revocation is not supported. Therefore, such a certificate on the live network needs to be replaced with a certificate issued by the operator's CA. Alternatively, a new device certificate issued by the LiteCA can be used to replace the preconfigured Huawei initial certificate.

The initial certificate can be replaced with the certificate issued by the operator's CA during base station deployment by PnP or during the course of operation. If an operator has deployed an operator-issued device certificate and the initial certificate has been replaced with the operator-issued device certificate, no further replacement is required.

NOTE

In IPsec networking scenarios, the LiteCA cannot be used. In this scenario, a Huawei initial certificate can be replaced only by using the certificate replacement function after site deployment.

For details about how to replace initial certificates, see [4.4.8 Replacing the Initial Certificate](#).

The base station and base station controller use the following failure check methods:

- On the base station, the **CERTCHKTSK.VENDORCERTCHKSW (LTE eNodeB, 5G gNodeB)** parameter value determines whether to perform validity checks on initial certificates and their corresponding trust certificates. The check items include whether the certificates are about to expire, have expired, are damaged, or are lost. The **VENDORCERT.VENDORCERTCHKSW (5G gNodeB, LTE eNodeB)** parameter can be used to set different validity check policies for the two initial certificates (**appcert.pem** and **vendorcert.pem**) preconfigured

on a board. For an initial certificate, if **VENDORCERT.CERTCHKTSKPOLICY (LTE eNodeB, 5G gNodeB)** is set to **GLOBAL**, the value of **CERTCHKTSK.VENDORCERTCHKSW (LTE eNodeB, 5G gNodeB)** determines whether to perform validity checks; if **VENDORCERT.CERTCHKTSKPOLICY (LTE eNodeB, 5G gNodeB)** is set to **CUSTOM**, the value of **VENDORCERT.VENDORCERTCHKSW (5G gNodeB, LTE eNodeB)** determines whether to perform validity checks. When the validity check function is enabled, the following principles are used:

- A validity period check, which determines whether an initial certificate is about to expire or has expired, is performed only when the certificate is being used. An initial certificate is being used in any of the following scenarios:
 - The value of **Application Type** is **SSL** and the value of **Certificate File Name** is **appcert.pem** in the **LST APPCERT** command output for the certificate.
 - DTLS is enabled. In addition, the value of **Application Type** is **DTLS** and the value of **Certificate File Name** is **appcert.pem** in the **LST APPCERT** command output for the certificate. For details about DTLS, see *DTLS*.
 - IPsec is enabled. In addition, the initial certificate is used in an **IKEPEER** or **SECURITYTEMPLATE** MO. For details about IPsec, see *IPsec*.
 - An initial certificate is used in an **IKEPEER** MO when **IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)** is set to **CERTMK** and **IKEPEER.CERTNAME (5G gNodeB, LTE eNodeB)** is set to **appcert.pem**, or when **IKEPEER.CERTSOURCE (5G gNodeB, LTE eNodeB)** is set to **APPCERT** and the value of **Application Type** is **IKE** and the value of **Certificate File Name** is **appcert.pem** in the **LST APPCERT** command output for the certificate.
 - An initial certificate is used in a **SECURITYTEMPLATE** MO when **SECURITYTEMPLATE.CERTSOURCE (LTE eNodeB, 5G gNodeB)** is set to **CERTMK** and **SECURITYTEMPLATE.CERTNAME (LTE eNodeB, 5G gNodeB)** is set to **appcert.pem**, or when **SECURITYTEMPLATE.CERTSOURCE (LTE eNodeB, 5G gNodeB)** is set to **APPCERT** and the value of **Application Type** is **IKE** and the value of **Certificate File Name** is **appcert.pem** in the **LST APPCERT** command output for the certificate.
 - The **ADD CRLTSK** command is executed with **CRLTSK.CERTNAME (LTE eNodeB, 5G gNodeB)** not specified or set to **appcert.pem**.
 - A BBP supporting eCPRI ports is not using an operator-issued device certificate (the certificate with **Application Type** set to **SSL** in the **APPCERT** MO is an initial certificate or **APPCERT.RRUCERTREQSW (LTE eNodeB, 5G gNodeB)** is set to **OFF**). In addition, the value of **RRU TLS Connection Type** is not **Only PlainText** in the **LST SSLCONF** command output, or the value of **MACsec Connection Type** is not **PLAINTEXT** in the **LST CPRIPORT** command output.

- An AAU is not using an operator-issued device certificate (the certificate with **Application Type** set to **SSL** in the **APPCERT MO** is an initial certificate or **APPCERT.RRUCERTREQSW (LTE eNodeB, 5G gNodeB)** is set to **OFF**). In addition, the value of **RRU TLS Connection Type** is not **Only PlainText** in the **LST SSLCONF** command output, or the value of **MACsec Connection Type** is not **PLAINTEXT** in the **LST CPRIPORT** command output.
- If the interval between the expiration time of an initial certificate and the current time of the base station is less than the value of the **CERTCHKTSK.VENDORALMRNG (LTE eNodeB, 5G gNodeB)** parameter, the base station reports ALM-26840 Imminent Certificate Expiry, and the RF unit using eCPRI ports reports ALM-26567 RF Unit Imminent Certificate Expiry.
- If the initial certificate is damaged, lost, or expired, the base station reports ALM-26841 Certificate Invalid, and the RF unit using eCPRI ports reports ALM-26565 RF Unit Certificate Fault.
- In GSM or UMTS, the **CERTCHKTSK.VENDORCERTCHKSW** parameter specifies whether the BSC6910 checks the validity of initial certificates. When the validity check function is enabled, the following principles are used:
 - If the interval between the expiration time of an initial certificate in use and the current time of the BSC6910 is less than the value of the **CERTCHKTSK.VENDORALMRNG** parameter, the BSC6910 reports ALM-20850 Digital Certificate Will Be out of Valid Time.
 - If the initial certificate is lost, damaged, or expired and the initial certificate has not been replaced, the BSC6910 reports ALM-20851 Digital Certificate Loss, Expiry, or Damage.
- In GSM or UMTS, the **CERTCHKTSK.VENDORCERTCHKSW** parameter specifies whether the BSC6960 checks the validity of initial certificates and their associated trust certificates. When the validity check function is enabled, the following principles are used:
 - If the interval between the expiration time of an initial certificate in use and the current time of the BSC6960 is less than the value of the **CERTCHKTSK.VENDORALMRNG** parameter, the BSC6960 reports ALM-21030 Imminent Certificate Expiry.
 - If the initial certificate is damaged, lost, or expired, the BSC6960 reports ALM-21031 Certificate Invalid.

For details about how to enable the LiteCA, see *iMaster MAE Product Documentation*.

Replacing Certificates in Batches

If you need to replace the certificates on devices or change the operator's CA, you can use the batch certificate replacement function.

The **Certificate Management** function of the OSS can be used to query the certificate information on a board, such as the current status and expiration time. To navigate to the **Certificate Management** page, in the **Apps** area of the MAE home page, click **Access** to enter the MAE-Access home page. Choose **Security > Certificate Authentication Management > Certificate Management > Task Management** from the main menu.

Huawei initial certificates on boards of base stations and base station controllers can be replaced with operator-issued device certificates issued by the LiteCA or operator's third-party CA. The device certificates issued by the LiteCA and operator's third-party CA can be replaced with each other. For details, see "Replacing NE Device Certificates in Batches" in *iMaster MAE Product Documentation*.

NOTE

A device certificate can be replaced only before it expires.

During certificate replacement, it is recommended that this operation be performed for all NEs and OSSs on the entire network to avoid management cost increase when there are certificates issued by multiple CAs on the same network. After an automatic initial certificate replacement task is created for a disconnected NE on the MAE, the MAE automatically replaces the NE's initial certificate with an operator-issued certificate once the link between the NE and MAE is restored.

Only SSL and RRU-SSL certificates can be replaced in batches.

4.1.3.3 Certificate Sharing

BSC6900/BSC6910 (GSM/UMTS)

The BSC6900/BSC6910 uses the active OMU board to apply for device certificates during base station controller deployment. If needed, the standby OMU or SAU board can obtain digital certificates from the active OMU board.

The active/standby status of OMU boards is not fixed. During base station controller deployment, the ESN of the currently active OMU board is used to apply for a digital certificate. If this board is subsequently replaced out of the subrack due to a fault, you are advised to re-apply for a digital certificate using the ESN of a functional OMU board.

Certificate sharing of BSC6900/BSC6910 applies to the following scenarios:

- Active and standby OMU boards are switched over. The currently active OMU board can use the digital certificate on the previously active OMU board to set up an SSL connection with the MAE.
- The SAU board obtains the digital certificate from the active OMU board and use the certificate to set up an SSL connection with the Nastar.

4.1.3.3.4 Checking Certificate Validity

A certificate has a validity period. After a certificate expires, a base station/base station controller that uses the certificate cannot be authenticated by the peer end and therefore cannot access the operator's network. To prevent this problem, base stations/base station controllers periodically check the validity of certificates. The **CERTCHKTSK.PERIOD (5G gNodeB, LTE eNodeB)** parameter specifies the interval between two consecutive certificate validity checks. The base station/base station controller periodically checks certificate validity as follows:

- Upon detecting that the period remaining until a certificate expires is less than the value of the **CERTCHKTSK.ALMRNG (5G gNodeB, LTE eNodeB)** parameter, the base station/base station controller determines that the certificate is about to expire.

- Upon detecting that the expiration time of a certificate is earlier than the current time, the base station/base station controller determines that the certificate has expired.
- If an initial certificate of the base station is not being used, the base station does not check its validity period. For details about the scenarios where an initial certificate is being used, see [4.1.3.3.2 Replacing Certificates](#).

Table 4-1 describes how the base station or base station controller processes an abnormal device certificate.

 NOTE

When a certificate is imported to a base station for the first time, the base station checks whether the certificate is about to expire. If the certificate is about to expire, the base station reports ALM-26840 Imminent Certificate Expiry.

Table 4-1 Processing performed when a device certificate is detected abnormal

Certificate Validity Status	Certificate Is in Use or Not	Value of CERTCHKTS K.UPDATE METHOD (5G gNodeB, LTE eNodeB)	Processing
To be expired	In use	CMP	The base station/base station controller automatically triggers a CMPv2-based certificate update procedure. If the certificate update fails, the base station reports ALM-26842 Automatic Certificate Update Failed, the BSC6900/BSC6910 reports ALM-20803 Certificate Auto-update Failed, the BSC6960 reports ALM-21032 Automatic Certificate Update Failed, and the AAU reports ALM-26565 RF Unit Certificate Fault. If the certificate is successfully updated subsequently or is no longer used, the alarm is cleared.

Certificate Validity Status	Certificate Is in Use or Not	Value of CERTCHKTS K.UPDATE METHOD (5G gNodeB, LTE eNodeB)	Processing
	In use	MANUAL	The base station reports ALM-26840 Imminent Certificate Expiry, the BSC6900/BSC6910 reports ALM-20850 Digital Certificate Will Be out of Valid Time, and the BSC6960 reports ALM-21030 Imminent Certificate Expiry. If the certificate is updated subsequently or is no longer used, the alarm is cleared.
	Not in use	N/A	
Expired	N/A	N/A	The base station reports ALM-26841 Certificate Invalid, the BSC6900/BSC6910 reports ALM-20851 Digital Certificate Loss, Expiry, or Damage, the BSC6960 reports ALM-21031 Certificate Invalid, and the AAU reports ALM-26565 RF Unit Certificate Fault to the OSS. These alarms remind the OM personnel of determining the cause and updating the digital certificate in a timely manner. If the certificate is updated subsequently or is no longer used, the alarm is cleared.

 NOTE

- Device certificates in use are those referenced by other MOs, including the device certificates specified by **APPCERT.APPCERT (LTE eNodeB, 5G gNodeB)**, **IKEPEER.CERTNAME (LTE eNodeB, 5G gNodeB)**, **CRLTSK.CERTNAME (LTE eNodeB, 5G gNodeB)**, and **CA.CERTNAME (LTE eNodeB, 5G gNodeB)**.
- Certificate validity checks require that the system time of the base station/base station controller be valid. Alarms may fail to be reported if the system time is invalid. If the system time is invalid (for example, ALM-26266 Time Synchronization Failure is reported), the certificate status on the base station/base station controller may be displayed as normal by default.
- For ALM-26840 Imminent Certificate Expiry of the base station and ALM-26567 RF Unit Imminent Certificate Expiry of the AAU, the alarm severity is minor if the remaining validity time of the certificate is less than the alarm threshold, greater than 15 days, and greater than 30% of the alarm threshold; if the remaining validity time of the certificate is less than or equal to 30% of the alarm threshold or the smaller value between the alarm threshold and 15 days, the alarm severity automatically changes to major.
- For ALM-21030 Imminent Certificate Expiry of the BSC6960, the alarm severity is minor if the remaining validity time of the certificate is less than the alarm threshold, greater than 15 days, and greater than 30% of the alarm threshold; if the remaining validity time of the certificate is less than or equal to 30% of the alarm threshold or the smaller value between the alarm threshold and 15 days, the alarm severity automatically changes to major.
- Each time a base station is restarted/upgraded, a certificate validity check task is added to immediately check certificate validity.
- If the base station detects that an operator-issued device certificate in use has become invalid, it applies for a new operator-issued device certificate to restore the certificate.

4.1.3.3.5 Updating Certificates

Certificate update includes operator-issued device certificate update and operator's CA certificate update. Certificate update described in other sections refers to the update of operator-issued device certificates.

Certificates used by base stations/base station controllers are Huawei-issued certificates and operator-issued certificates. Huawei-issued certificates cannot be updated because:

- Huawei initial certificates and CA certificates are used to ensure security during certificate application.
- Generally, Huawei initial certificates and CA certificates are used only during site deployment.
- The lifecycle of Huawei initial certificates is usually longer than that of the devices.
- The Huawei factory CA system is used only to preconfigure certificates for devices during production and cannot be accessed by devices on live networks.
- After a base station accesses a network, it cannot connect to the Huawei factory CA system. Therefore, Huawei initial certificates and CA certificates of the base station cannot be automatically updated.

Preconfigured Huawei initial certificates should be replaced with operator-issued device certificates in a timely manner to ensure that they are updated before expiration.

NOTICE

If the automatic certificate update of a base station fails, the base station reports ALM-26842 Automatic Certificate Update Failed.

If the automatic certificate update of a base station controller fails, the BSC6900/BSC6910 reports ALM-20803 Digital Certificate Auto-update Failed while the BSC6960 reports ALM-21032 Automatic Certificate Update Failed.

Scenarios of Updating the Operator-issued Device Certificates

A certificate used by a base station/base station controller must be updated in the following scenarios:

- The certificate is nearing expiration.
- Base station/base station controller information, such as the time and geographical location, has changed.

Updating the Operator-issued Device Certificates of the Base Station/Base Station Controller

A certificate update on the base station or base station controller can be triggered automatically or manually.

- Automatic triggering

A task of periodically checking the certificate validity is configured on the base station/base station controller with the parameter

CERTCHKTSK.UPDATEMETHOD (LTE eNodeB, 5G gNodeB) (for the eGBTS/NodeB/eNodeB/gNodeB)/**BTSCERTCHKTSK.UPDATEMETHOD** (for the GBTS) set to **CMP**. Upon detecting that a certificate is nearing expiration, the base station or base station controller automatically triggers a CMPv2-based certificate update. In this mode, the private-public key pair of the certificate is automatically updated during the certificate update.

NOTE

If the configured key type is different from the actual key type, automatic certificate update will fail. In this case, you need to run the **UPD DEVCERT** command with **Renew Key** set to **YES** to manually update the key.

During an automatic certificate update procedure, if the certificate update fails due to intermittent transmission interruption or network congestion, the base station automatically retries certificate update for at most twice with an interval of 10 minutes.

- Manual triggering

Users can run the **UPD DEVCERT** command to trigger a CMPv2-based certificate update. In this command, the **APPCERT (LTE eNodeB, 5G gNodeB)** parameter specifies the certificate to be updated, and the **REKEY** parameter specifies whether to update the private-public key pair. After this command is executed, the base station or base station controller reports the certificate update progress.

During the certificate update, the base station or base station controller automatically configures a new certificate and tests it. If the configuration or test of the new certificate fails, the base station or base station controller will use the

original certificate. If the automatic certificate update procedure is triggered, the base station will report ALM-26842 Automatic Certificate Update Failed, the BSC6900/BSC6910 will report ALM-20803 Certificate Auto-update Failed, and the BSC6960 will report ALM-21032 Automatic Certificate Update Failed. These alarms will be cleared until the automatic certificate update succeeds. If the manual certificate update is triggered, the base station or base station controller will report an update failure.

NOTE

When the base station updates a device certificate, if the intermediate subordinate CA certificate is also updated and the message returned by the CA to the base station carries the updated intermediate subordinate CA certificate, the base station will update the intermediate subordinate CA certificate stored locally.

Certificate tests are performed in IPsec and SSL scenarios. The base station controller supports only SSL scenarios.

- In IPsec scenarios, certificate testing refers to using a new certificate for IKE renegotiation authentication. If the IKE reconnection fails, the base station uses the original certificate.
- In SSL scenarios, certificate testing refers to using a new certificate for SSL reconnection. If the SSL reconnection fails, the base station or base station controller uses the original certificate.

Bidirectional authentication is forcibly used for SSL certificate testing. That is, the NE and MAE authenticate each other's device certificates to ensure that the test result directly reflects certificate availability.

Scenarios of Updating the Operator CA Certificates

The operator subordinate CA and root CA certificates on the base station/base station controller need to be updated in the following scenarios:

- The validity period of an operator subordinate CA certificate or root CA certificate is about to expire.
- The time or geographical location of an operator subordinate CA or root CA is changed.

Updating the Operator CA Certificates

Updating operator subordinate CA or root CA certificates affects all NEs that use these CA certificates. During the transition between the old and new operator subordinate CA or root CA certificates, authentication between NEs using their device certificates may fail. Therefore, it is recommended that a dedicated team be assigned to manage updates of operator subordinate CA and root CA certificates and develop a plan to complete the updates across the entire network as soon as possible.

During updates of operator subordinate CA or operator root CA certificates, you are advised not to modify their public or private keys. This helps reduce inter-NE authentication failures during the transition between the old and new certificates.

There are two scenarios for updating the operator's CA certificates: updating only the operator subordinate CA certificates, or updating both the operator

subordinate CA certificates and the operator root CA certificate. [Table 4-2](#) describes the update procedure for each scenario.

Table 4-2 Procedure for updating the operator's CA certificates

NE	Updating Only the Operator Subordinate CA Certificates (CMP Messages Containing Subordinate CA Certificates)	Updating Only the Operator Subordinate CA Certificates (CMP Messages Not Containing Subordinate CA Certificates)	Updating Certificates of Both Operator Root CA and Operator Subordinate CAs
Base station	Run the UPD DEVCERT command.	1. Add all new operator subordinate CA certificates. 2. Update device certificates. 3. Delete the old operator subordinate CA certificates.	1. Add all new operator root CA certificates and new operator subordinate CA certificates to all network devices. 2. Update the device certificates of all NEs on the network.
Base station controller	1. Add all new operator subordinate CA certificates. 2. Update device certificates. 3. Delete the old operator subordinate CA certificates.	1. Add all new operator subordinate CA certificates. 2. Update device certificates. 3. Delete the old operator subordinate CA certificates.	3. Delete the old operator subordinate CA certificates and the old operator root CA certificate.

NOTE

In scenarios of UMPT+UMPT cold backup or transmission link backup of main control boards (LTE/NR), the operator-issued certificates on both the active and standby boards must be updated after the subordinate CA certificate is updated. If only the active board is updated while the standby board is not, link setup for protocols such as IPsec, SSL, and DTLS may be affected after an active/standby switchover. This is because the active board uses the new subordinate CA certificate and device certificate and no longer retains the old subordinate CA certificate, whereas the standby board contains only the device certificate issued by the old subordinate CA and does not contain the old subordinate CA certificate.

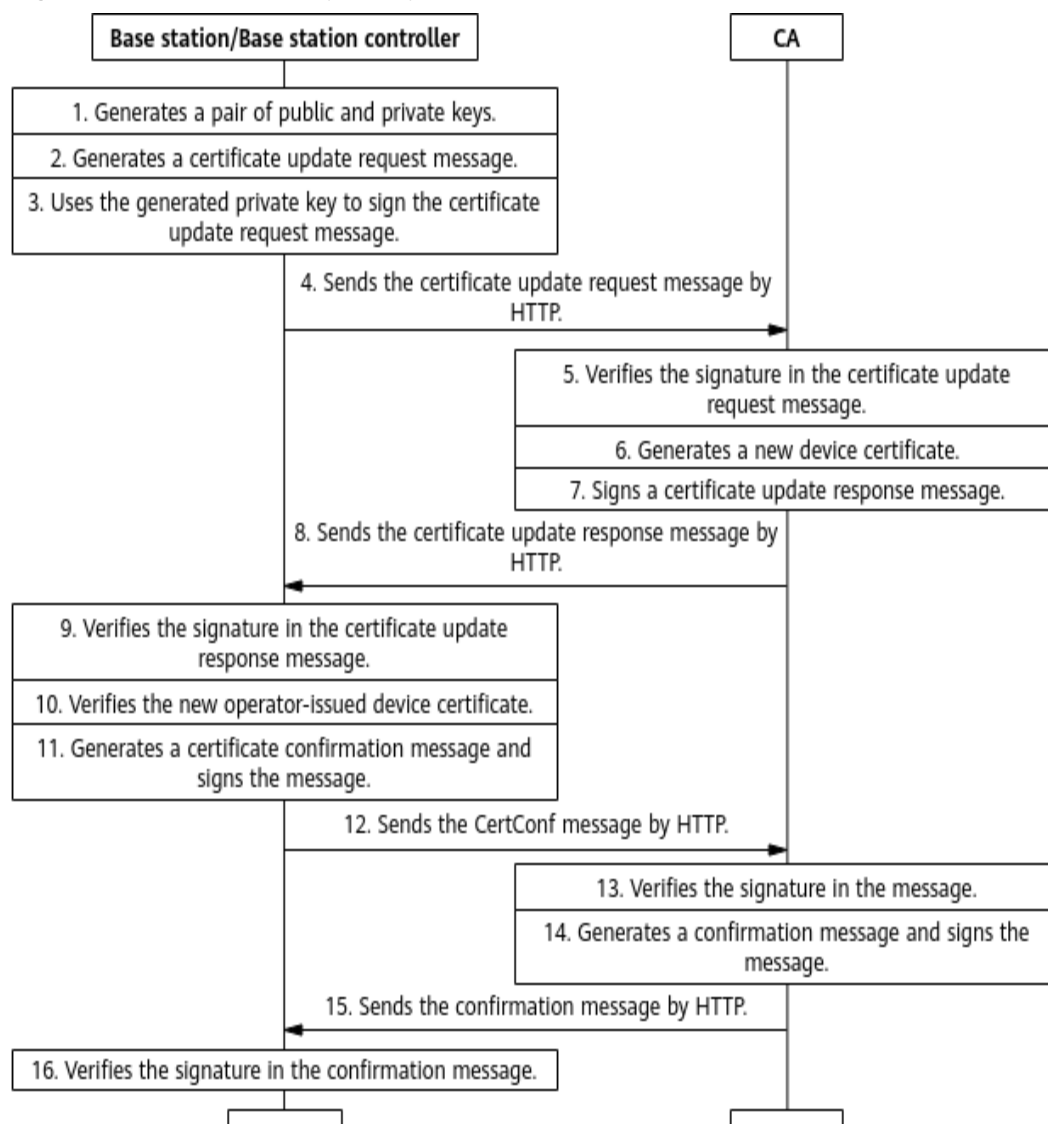
Updating Certificates Using CMPv2

Different from the certificate application process of the base station/base station controller, the operator's CA and base station/base station controller authenticate each other using the operator-issued certificate and operator certificate chain or

root certificate during certificate update. Huawei initial certificate and Huawei root certificate are no longer used.

Figure 4-17 shows how a base station or base station controller applies for a certificate based on CMPv2.

Figure 4-17 Certificate update process of a base station/base station controller



In step 2, the key update request message is also the certificate update request. This message includes the new public key and the operator-issued device certificate to be updated.

In step 3, the base station or base station controller uses the private key corresponding to the operator-issued device certificate to be updated to sign the key update request message.

In step 5, the CA uses the public key of the operator-issued device certificate carried in the key update request message to verify the signature in the message. In addition, the CA uses the operator root certificate or certificate chain to verify the operator-issued device certificate carried in the message.

In step 6, after the authentication succeeds, the CA server generates a new device certificate for the base station or base station controller and uses the private key corresponding to the CA certificate to sign the certificate.

In step 7, the CA signs the key update response message. This message contains the new certificate issued by the CA to the base station or base station controller.

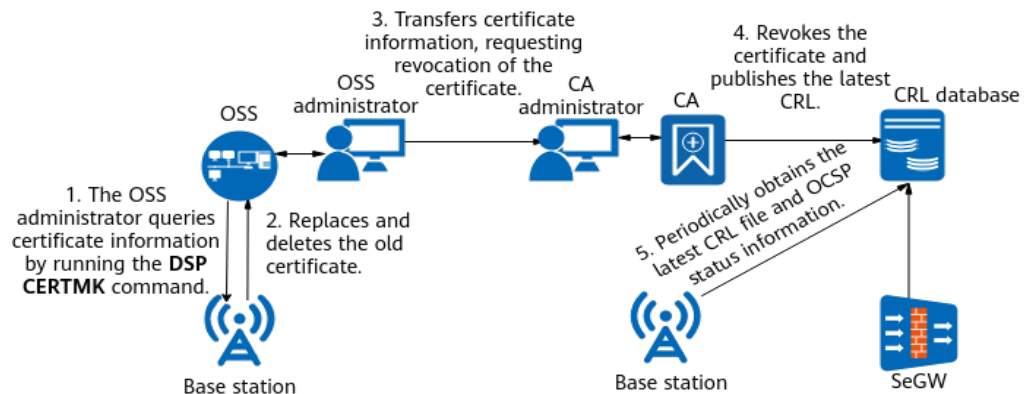
For details about the CMPv2 message structure and the process of exchanging CMPv2 messages, see RFC 4210 and RFC 4211.

4.1.3.3.6 Revoking Certificates

Certificate Revocation Process

If a base station/base station controller is no longer used or the private key of its device certificate is disclosed or cracked before the certificate expires, the administrator must apply to the CA for revoking the certificate in a timely manner to prevent unauthorized use of the certificate. Currently, the base station/base station controller does not support online certificate revocation. Certificates must be manually revoked. [Figure 4-18](#) shows the certificate revocation process of the base station. The certificate revocation process of the base station controller is the same as that of the base station.

Figure 4-18 Certificate revocation process of the base station



NOTICE

Certificate revocation may affect the normal communication of the application services that use the certificate. You are advised to replace a certificate before revoking it to ensure that services are running properly using the new certificate. Then, delete and revoke the old certificate. The following faults may occur when a base station downloads a CRL or obtains the OCSP status:

- The base station's downloading fails and it reports ALM-26842 Automatic Certificate Update Failed when both the base station and CRL server support FTPS, LDAPS, and HTTPS, the base station's local device certificate is revoked, and the CRL server authenticates the base station.
- The base station fails to obtain the OCSP status and reports ALM-26842 Automatic Certificate Update Failed when the base station and OCSP server support HTTP and HTTPS, the base station uses the local device certificate for which the OCSP status needs to be obtained for authentication by the peer OCSP server when HTTPS is used, the local device certificate of the base station is revoked, and the OCSP server authenticates the base station.

In the preceding scenarios, resolve the certificate issue on the base station side first. For example, apply for a new device certificate for the base station. If the base station has downloaded the CRL and finds that the operator-issued device certificate has been revoked, the base station will initiate a certificate application procedure again to ensure normal services.

CRL Acquisition

A base station/base station controller periodically obtains CRLs from the certificate & CRL database. The CRLs are used to verify the validity of the certificate of the peer device to ensure network security.

Table 4-3 lists the methods for a base station/base station controller to obtain CRLs.

Table 4-3 Methods to obtain CRLs

Category	Method to Obtain CRLs	Type of CRL Server
Manual	Users run MML commands to enable the base station or base station controller to obtain the CRLs from the FTP server.	FTP server
Automatic	Scheduled tasks are configured so that CRLs can be automatically obtained.	LDAP server
		FTP server
		HTTP server (supported only by base stations)

To enable the base station/base station controller to automatically obtain CRLs during normal operation, set the CRL server IP address (**CRLTSK.IP (LTE eNodeB,**

5G gNodeB)), retrieval mode (**CRLTSK.CRLGETMETHOD (LTE eNodeB, 5G gNodeB)**), whether to update the CRL at the next update time (**CRLTSK.ISCRLTIME (LTE eNodeB, 5G gNodeB)**), and CRL update period (**CRLTSK.PERIOD (LTE eNodeB, 5G gNodeB)**). It is recommended that the CRL update period be set to 1/3 of the CRL validity period (interval between the effective date of the CRL and the next update time). For example, if the CRL validity period is 24 hours, you are advised to set the CRL update period to 8 hours.

NOTE

CRL specifications for the base station are as follows: The aggregate size of all CRLs must not exceed 2 MB. A CRL in PEM format must be no larger than 256 KB, while that in DER format must be no larger than 2 MB.

When a base station connects to multiple CRL servers, it stores CRLs directly from these servers. To avoid conflicts, ensure that CRL file names are unique across all servers.

In addition to the preceding parameters, set other parameters as follows for the base station/base station controller to automatically obtain CRLs based on the CRL storage server type.

- If LDAP is used to obtain CRLs, also set **CRLTSK.SEARCHDN (LTE eNodeB, 5G gNodeB)** and **CRLTSK.PORT (LTE eNodeB, 5G gNodeB)**.

If SSL-protected transmission mode is used, also set **CRLTSK.CONNMODE (LTE eNodeB, 5G gNodeB)** and **CRLTSK.AUTHPEER (LTE eNodeB, 5G gNodeB)**. If **CRLTSK.AUTHPEER (LTE eNodeB, 5G gNodeB)** is set to **ENABLE**, both the base station/base station controller and the CRL server must be configured with the device certificates used for SSL authentication and peer CA trust certificates.

NOTE

Base stations and base station controllers support only LDAP V3. Therefore, if LDAP is used to obtain CRLs, the CRL server must support LDAP V3. For details, see RFC 4511 and RFC 4523.

When the base station obtains CRLs over LDAP, the values of **CRLTSK.USR (5G gNodeB, LTE eNodeB)** and **CRLTSK.SLVUSR (5G gNodeB, LTE eNodeB)** must be a string of English characters and should not contain Chinese characters.

- If FTP/FTPS is used to obtain CRLs, the FTP/FTPS function must be enabled on the CRL server. When **FTPCLT.ENCRYMODE (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **FTPSCLT.ENCRYMODE** (for the GBTS/GBSC/RNC) is set to **Plaintext**, the plaintext FTP mode is used. When the parameter is set to **Encrypted**, the SSL-encrypted FTPS mode is used. In addition, ensure that all FTP servers interacting with the base station/base station controller support FTP/FTPS.

If FTPS is used to obtain CRLs and the CRL server needs to be authenticated, set **FTPCLT.SSLCERTAUTH (LTE eNodeB, 5G gNodeB)** (for the eGBTS/NodeB/eNodeB/gNodeB) or **FTPSCLT.SSLCERTAUTH** (for the GBTS/GBSC/RNC) to **YES**. In addition, ensure that the base station/base station controller has been configured with the peer CA trust certificate and the CRL server has been configured with a device certificate for SSL authentication.

- If HTTP is used to obtain CRLs, **CRLTSK.PORT (LTE eNodeB, 5G gNodeB)** must be configured to specify the port number used to find the desired HTTP server. If HTTPS is used, **CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB)**

must also be configured and **CRLTSK.AUTHPEER** (*5G gNodeB, LTE eNodeB*) must be set to **ENABLE** in addition to the port number.

 NOTE

When the base station uses HTTP to obtain CRLs, the HTTP URL for obtaining the CRL consists of the HTTP packet header, connection mode (**CRLTSK.CONNMODE** (*5G gNodeB, LTE eNodeB*)), colon (:) and slash (/), IP address (**CRLTSK.IP** (*5G gNodeB, LTE eNodeB*)) or IPv6 address (**CRLTSK.IP6** (*LTE eNodeB, 5G gNodeB*)), port number (**CRLTSK.PORT** (*LTE eNodeB, 5G gNodeB*)), and CRL file name (**CRLTSK.FILENAME** (*LTE eNodeB, 5G gNodeB*)). For example, if the connection mode is SSL, the IP address is 192.168.23.112, the port number is 443, and the CRL file name is /test.crl, then the HTTP URL is HTTPS://192.168.23.112:443/test.crl.

When the base station uses HTTP to obtain CRLs, the default certificate deployment position should be used, and neither username nor password is required. The slave IP address (**CRLTSK.SLVIP** (*5G gNodeB, LTE eNodeB*)) or slave IPv6 address (**CRLTSK.SLVIP6** (*LTE eNodeB, 5G gNodeB*)) cannot be set.

During an automatic CRL update, if the update fails due to intermittent transmission interruption or network congestion, the base station retries the update up to two times at 20-second intervals.

CRL Validity Period Check

The validity period of a CRL refers to the interval between the effective date and the next update time of the CRL. After the CRL expires, certificate revocation verification cannot be accurately performed. As a result, devices with invalid certificates may access the operator's network. To address this issue, the base station periodically checks the validity periods of CRLs. The base station performs periodic checks as follows:

- The base station checks the CRL obtained manually every 12 hours and the CRL obtained automatically every hour.
- Alarm generation: The system checks the expiration time of the loaded CRL, that is, the interval between the current time and the next CRL update time.
 - If the number of days before the CRL expires is less than or equal to the smaller value between 30% of the validity period and 7 days, ALM-26840 Imminent Certificate Expiry with the minor severity is reported.
 - If the number of days before the CRL expires is less than or equal to the smaller value between 15% of the validity period and 4 days, ALM-26840 Imminent Certificate Expiry with the major severity is reported.

If an automatically obtained CRL will be automatically updated before it expires, no alarm is generated.

- Alarm clearing: This alarm is cleared when the CRL is deleted or the remaining days before expiration of the CRL is not within the preceding ranges.

 NOTE

When the **SET CERTCHKTSK** command is executed to change any parameter value, the system immediately checks the validity period of the CRL obtained manually and repeats the check every 12 hours.

OCSP Status Acquisition

The base station can only query the OCSP status of device certificates. The OCSP server address can be configured using the **CA.OCSPURL** (*LTE eNodeB, 5G*

gNodeB) parameter, which can be either an IPv4 or IPv6 address. If the base station fails to obtain the OCSP status because the OCSP server's URL is unreachable or the OCSP status verification fails, it will report ALM-26842 Automatic Certificate Update Failed.

The base station caches the information about the device certificate's OCSP status upon successful acquisition of this information. The **DSP CERTMK** and **DSP BRDCERTMK** commands can be executed to query the cached OCSP status information corresponding to the current device certificate. [Table 4-4](#) lists the output items of the cache information.

Table 4-4 OCSP status output items

Output Item	Description
Certificate OCSP Status	This parameter indicates the OCSP status of the obtained certificate.
Time of Current Certificate OCSP Status Update	This parameter indicates the time when the current OCSP status of the certificate is obtained. The OCSP status of the certificate takes effect as of this time.
Time of Next Certificate OCSP Status Update	This parameter indicates the time when the OCSP status of the certificate is updated for the next time. When the time for the next update arrives, the current OCSP status of the certificate becomes invalid and needs to be updated from the OCSP server again.

The base station determines the status of the local device certificate based on the CRL and OCSP information. If the OCSP status is revoked or unknown, or there is a revocation record of this certificate in the CRL, it considers that the certificate status is invalid and reports ALM-26841 Certificate Invalid.

NOTE

The OCSP status check items of the base station are as follows:

- Checking whether the serial number of the certificate in the received OCSP response is the same as that of the requested certificate
- Checking whether the signature on the OCSP response is valid
- Checking whether the signer is the CA that issues the device certificate
- Checking whether the current base station time is between **Time of Current Certificate OCSP Status Update** and **Time of Next Certificate OCSP Status Update**

The base station supports the following signature algorithms for verifying OCSP responses: RSA_PKCS1_SHA256, RSA_PKCS1_SHA384, ECDSA_SECP256R1_SHA256, ECDSA_SECP384R1_SHA384, and ED25519.

The OCSP status information cached on the base station is checked every 5 minutes.

The timeout duration for the connection between the base station and the OCSP server is 15 seconds. If the connection times out, the base station reports ALM-26842 Automatic Certificate Update Failed.

For details, see RFC 6960.

4.1.3.3.7 Offline Certificate Monitoring

If a board leaves the customer's network due to board repair or retirement, the certificate and private key on the board may be disclosed. Therefore, the offline certificate monitoring function is required. This function allows users to use the MAE to query and export basic information about abnormal certificates, including the base station name, certificate issuer name, certificate serial number, status, and time when the certificate is detected abnormal. Currently, this function can take effect only for offline certificates on base stations.

NOTE

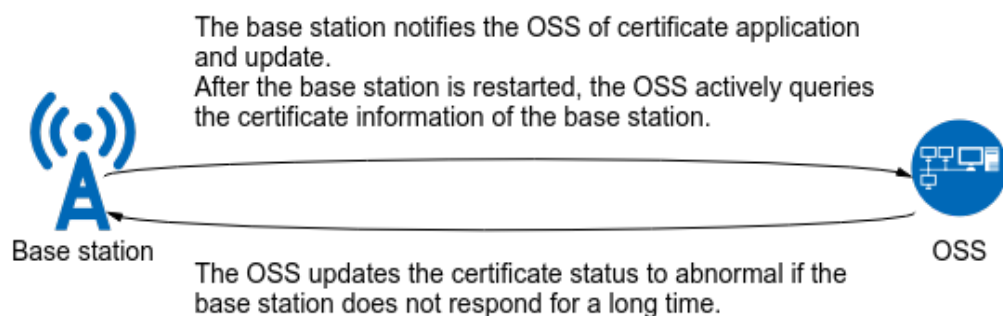
Basic information about abnormal certificates will be saved on the MAE for 30 days and then be automatically removed. The purpose is to avoid repeated exporting of certificate information.

The status of a certificate may be abnormal in the following conditions:

- The certificate does not exist, for example, when the board is returned for repair.
- The MAE deletes the NE where the certificate is deployed.
- The NE where the certificate is deployed cannot be accessed for a long time (for example, when a base station exits the network).
- The certificate is deleted manually. The operator needs to ensure that a manually deleted certificate has been revoked by the CA. The information about a deleted certificate will be saved on the MAE for a period. After the period expires, the information is automatically deleted.

Figure 4-19 shows the principle of offline certificate monitoring.

Figure 4-19 Principle of offline certificate monitoring



In the following scenarios, the device certificate on the base station does not need to be revoked although the device certificate status is abnormal on the MAE:

- The base station operates normally but cannot communicate with the MAE for a long time.
- The base station or its board is transferred to another MAE for management. In this case, the device certificate on the base station or board is recorded as abnormal on the original MAE.

If the certificate must be revoked, you need to manually run the certificate revocation command on the CA.

The offline certificate monitoring function cannot be used in the following conditions:

- This function does not take effect for the preconfigured Huawei initial certificates. The name of the issuer of the preconfigured Huawei initial certificates starts with "Huawei". Therefore, it is not recommended that the name of the issuer of operator-issued device certificates start with "Huawei".
- If the base station cannot communicate with the MAE after obtaining a device certificate, the MAE cannot record the information about the device certificate. In this case, the MAE cannot monitor the device certificate.
- During the period when base station software is rolled back to a version not supporting offline certificate monitoring, the MAE cannot update certificate status. After base station software is upgraded to a version supporting offline certificate monitoring, the MAE can update certificate status.

The offline certificate monitoring function does not need to be activated. You can query and export the basic information about abnormal certificates on the MAE. For details, see [4.4.3.4 Activation Verification](#).

4.1.3.4 Application of Digital Certificates

4.1.3.4.1 Certificate Usage in UMPT+UMPT Cold Backup Mode

In UMPT+UMPT cold backup mode, only one UMPT is working at a time. The two boards are deployed in the same logical slot and configured with the same logical slot number. The data configuration for the certificate deployment position specifies the bound logical slot number. For details of UMPT cold backup and logical slot numbers, see *Base Station Equipment Reliability*.

The two UMPT boards use and manage their own device certificates. The certificate validity period check task also applies to the standby UMPT. During site deployment, the active UMPT applies for an operator-issued device certificate. After deployment is complete and the configuration is activated, the standby UMPT periodically checks the status of its standby certificate corresponding to the certificate used by the active UMPT. If the standby certificate is abnormal (expired, damaged, lost, revoked, or invalid), the standby UMPT automatically applies for a new certificate. If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD (5G gNodeB, LTE eNodeB)** is set to **CMP**, the certificate is automatically updated.

An operator-issued device certificate for the standby UMPT can be manually requested using the **REQ DEVCERT** command on the base station, with the cabinet, subrack, and slot numbers of the standby UMPT specified. If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD (5G gNodeB, LTE eNodeB)** is set to **MANUAL**, users can update the operator-issued device certificate on the standby UMPT by running the **UPD DEVCERT** command on the base station with the cabinet, subrack, and slot numbers of the standby UMPT specified.

In the operation phase, if the active UMPT malfunctions, a switchover will take place and the standby UMPT will take over the ongoing services. After the switchover, the standby UMPT automatically requests an operator-issued device certificate that was unavailable before the switchover according to the configuration file, triggering a CMPv2-based certificate application. During

certificate application, if the CA needs to check the ID of the certificate that the base station applies for, for example, the *Common Name* field or the *dNS* name in the *Subject Alternative Name* extension field of the certificate, the *Subject* identity information of the certificate of the standby board needs to be imported to the CA system. Therefore, if UMPT boards work in cold backup mode on a live network and the CA checks the applicant ID when issuing a certificate, you are advised to import the certificate IDs of all UMPT boards to the CA in advance. Otherwise, the certificate application fails after an active/standby switchover because the certificate ID of the previous standby UMPT board is not imported to the CA. As a result, the base station fails to access the network after the board switchover.

In UMPT+UMPT cold backup mode, if both IPsec and PKI are deployed, the **IKEPEER.IDTYPE (LTE eNodeB, 5G gNodeB)** parameter can be set to **IPv4**, **IPv6**, **FQDN**, or **DN** on the base station side. If this parameter is set to **FQDN** or **DN**, the SeGW does not verify the ID of the base station or changes the ID of the corresponding base station on the SeGW.

NOTE

When running the **REQ DEVCERT** command to request an operator-issued device certificate for the standby UMPT, ensure that the certificate file name (**CERTMK.APPCERT (LTE eNodeB, 5G gNodeB)**) matches the file name configured on the active UMPT.

The operator's trust certificate, cross-certificate, and CRL are managed at the base station level rather than at the board level.

4.1.3.4.2 Estimation of Certificate Reconfiguration Impact

This function estimates the impact of certificate reconfiguration on services based on the existing IPsec and certificate configuration data. If a certificate reconfiguration may lead to a site disconnection, the system displays a message indicating the high-risk issues or terminates the operation to avoid configuration impact on services.

Estimation of certificate reconfiguration impact involves the following operations:

- Check whether the value of **Local IP** in the certificate request is the same as the value of **Local IP** in the effective **IKEPEER** MO before running any of the following commands:
 - **MOD CERTREQ**
 - **ADD CA**
 - **MOD CA**
 - **MOD APPCERT**
 - **MOD CERTMK**
 - **MOD IKEPEER**
 - **ADD IPSECBIND (old model)/ADD IPSECBINDITF (new model)**
 - **ADD IPSECPOLICY**
 - **MOD IPSECPOLICY**

If the two **Local IP** values are different, certificate reconfiguration impact cannot be estimated.

- Check whether the effective IKE/SSL certificate is using the local trust certificate to be deleted (by running the **RMV TRUSTCERT** command).

 NOTE

This function cannot determine whether the peer trust certificate is deleted.

Estimation of certificate reconfiguration impact applies to the following configuration scenarios: MML command configuration and MAE-Deployment XML batch configuration. Estimation of trust certificate deletion applies only to MML command configuration scenarios.

- **MML command configuration**

Whether to estimate certificate reconfiguration impact is controlled by the **Support Forcible Execution** parameter in the MML commands listed above.

- If this parameter is set to **NO** in a command, certificate reconfiguration impact is estimated after this command is delivered:
 - If services will be affected, a failure message is returned and the failure cause is described.
 - If services will not be affected, certificate reconfiguration is performed.
- If this parameter is set to **YES** in a command, certificate reconfiguration is performed without estimating the impact.

This function increases the execution time of a single MML command by up to 2s.

- **MAE-Deployment XML batch configuration**

The **Precise Check** option in the MAE-Deployment script export function determines whether to estimate the impact of certificate reconfiguration on services. This option is selected by default. When an operator exports a script, the MAE-Deployment estimates the impact on services. If services will be affected, the estimation results are displayed. If services will not be affected, certificate reconfiguration is performed.

 NOTE

The **Support Forcible Execution** parameter is not supported on the certificate management GUI of the MAE. If operations on the GUI fail due to estimation, users can run the following MML commands to perform risky reconfigurations after determining that the reconfiguration risks are acceptable.

- **MOD CERTREQ**
- **ADD CA**
- **MOD CA**
- **MOD APPCERT**
- **MOD CERTMK**

This function does not apply to base stations using a GTMUc as the main control board.

4.1.3.4.3 Certificate Application of eCPRI Connections

In the boards with eCPRI ports, the BBP and AAU are preconfigured with Huawei initial certificates. If a PKI system is deployed on the operator's network, an AAU can apply for an operator-issued device certificate and use the operator-issued device certificate to establish a secure eCPRI connection with the BBU configured with a CA.

As a remote unit of a base station, the AAU cannot directly access the CA. The BBU must function as a proxy to apply for a certificate for the AAU. If the AAU certificate needs to be replaced by an operator-issued certificate, check whether the operator's CA supports this proxy-based certificate application method and ensure that the IOT test is passed. If the operator's CA needs to verify the Subject information of the requested certificate, the Subject information of the AAU certificate must be imported to the CA system to ensure that the AAU can successfully apply for an operator-issued device certificate. During the application and update, the BBU must have obtained an operator-issued device certificate, and the CA of the BBU must be configured as the operator's CA corresponding to the operator-issued device certificate.

The **APPCERT.RRUCERTREQSW (LTE eNodeB, 5G gNodeB)** parameter specifies whether an AAU applies for an operator-issued device certificate. When the **SSL.RRUTLSCONNTYPE (LTE eNodeB, 5G gNodeB)** parameter is set to **PLAINTEXT**, a BBU or AAU using eCPRI ports does not use an operator-issued certificate. In this case, you are advised not to change the **APPCERT.RRUCERTREQSW (LTE eNodeB, 5G gNodeB)** parameter setting.

Currently, an AAU does not support manual application for an operator-issued device certificate. It can only use CMPv2 to apply for an operator-issued device certificate from the operator's CA through the BBU (acting as a proxy).

- Step 1** An AAU generates the certificate private key and certificate request file based on the parameters in the **CERTREQ MO**.

For an AAU, the certificate common name is always the ESN, and the local name and local IP address do not take effect. When the signature algorithm is set to SHA1, the AAU uses SHA256 as the signature algorithm for certificate request configuration information. When the key size is set to 1024 bits, the AAU uses the 2048-bit key algorithm.

- Step 2** The AAU sends the certificate request file, AAU initial certificate, and certificate chain to the BBU.

- Step 3** The BBU uses the AAU's certificate request file, AAU initial certificate, and certificate chain to apply for an operator-issued device certificate for the AAU from the CA, and then delivers the certificate to the AAU.

----End

When a Huawei initial certificate is used as an authentication credential in the application for a certificate from the CA, you can set the **CA.VENDORCERT (LTE eNodeB, 5G gNodeB)** parameter to specify which initial certificate configured is used for HTTPS authentication and signature of the certificate request message. If the board corresponding to the certificate deployment position is not preconfigured with the specified certificate or the specified certificate is damaged or lost, the system uses the other Huawei initial certificate on the board. Since the BBU functions as a proxy to apply for a certificate for the AAU, the BBU's certificate is used for HTTPS authentication. During proxy certificate application, if the AAU or BBU does not have the initial certificate specified by **CA.VENDORCERT (LTE eNodeB, 5G gNodeB)** or the initial certificate is damaged or lost, the initial certificate used for HTTPS authentication may be inconsistent with that used for signing the certificate request message. In this case, two Huawei initial certificates must be added and trusted on the CA side (under HTTPS and CMP authentications). Otherwise, the AAU may fail to apply for an operator-issued device certificate.

The BBU controls whether to allow AAUs to use Huawei-issued certificates for connection. After the certificates of all AAUs connected to the BBU are replaced with the operator-issued device certificates, you need to run the **SET RRUSECPOLICY** command to disable authentication based on Huawei initial certificates. In this case, the BBU accepts only the connection requests of AAUs using operator-issued device certificates. If the BBU requires an AAU to apply for an operator-issued device certificate, the AAU uses the operator-issued certificate to establish a secure connection with the BBU. In other scenarios, an AAU uses a Huawei initial certificate to establish a secure connection with the BBU. In secure networking using certificate-based authentication, if a new AAU is connected to the network, authentication based on Huawei initial certificates must be enabled on the BBU. Otherwise, the AAU may fail to apply for an operator-issued device certificate.

ALM-26565 RF Unit Certificate Fault will be reported one hour after an AAU certificate application or update fails or the AAU authentication mode is changed to anonymous authentication.

The AAU will report ALM-26567 RF Unit Imminent Certificate Expiry one hour after detecting that the AAU certificate is about to expire.

NOTE

When the CA needs to issue certificates for both the BBU and AAU, the BBU can successfully apply for certificates if the CA does not support the SHA256 or 2048-bit key algorithm and the certificate request configuration information of the base station indicates SHA1 or 1024-bit key algorithm; however, the AAU's application fails. After the AAU certificate application is started, you can observe that the AAU completes the certificate application in about 30 minutes. The AAU checks the certificate validity once a day.

Changing the CA That Issues the AAU Certificate

The CA that issues the AAU certificate is the CA that issues the TLS certificate of the BBU. To change the CA that issues the AAU certificate, you must change the CA that issues the TLS certificate of the BBU. Changing the CA affects the communication between the AAU and the BBU and between the BBU and other external devices. Therefore, ensure that the corresponding CA and root CA certificate have been configured on the peer devices.

To change the CA, do as follows: Add a CA, apply for the device certificate from the CA, and run the **MOD APPCERT** command to replace the original BBU TLS certificate with the new one. This is because the issuer of a BBU certificate file must be the same as the value of **CA.CANAME (LTE eNodeB, 5G gNodeB)** corresponding to this certificate file.

NOTICE

If an AAU with an ECDSA certificate is rolled back to a version earlier than SRAN16.1 and the CA remains unchanged, the certificate needs to be manually deleted before an RSA certificate request from this CA.

4.1.3.4.4 Certificate Application When Transmission Link Backup of Main Control Boards (LTE/NR) Is Deployed

Transmission link backup of main control boards (LTE/NR) implements transmission link backup by deploying transmission links on two main control boards of the same model. In this scenario, the certificate deployment position and the source IP addresses for certificate management on the two boards must be specified. For details about transmission link backup of main control boards (LTE/NR), see *Base Station Equipment Reliability*.

The two UMPT boards use and manage their own device certificates. The certificate validity period check task also applies to the standby UMPT. During site deployment, the active UMPT applies for an operator-issued device certificate. After deployment is complete and the configuration is activated, the standby UMPT periodically checks the status of its standby certificate corresponding to the certificate used by the active UMPT. If the standby certificate is abnormal (expired, damaged, lost, revoked, or invalid), the standby UMPT automatically applies for a new certificate. If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD** is set to **CMP**, the certificate is automatically updated.

An operator-issued device certificate for the standby UMPT can be manually requested using the **REQ DEVCERT** command on the base station, with the cabinet, subrack, and slot numbers of the standby UMPT specified. If the standby operator-issued device certificate is nearing expiration and **CERTCHKTSK.UPDATEMETHOD** is set to **MANUAL**, users can update the operator-issued device certificate on the standby UMPT by running the **UPD DEVCERT** command on the base station with the cabinet, subrack, and slot numbers of the standby UMPT specified.

NOTE

When running the **REQ DEVCERT** command to request an operator-issued device certificate for the standby UMPT, ensure that the certificate file name (**CERTMK.APPCERT**) matches the file name configured on the active UMPT.

The operator's trust certificate, cross-certificate, and CRL are managed at the base station level rather than at the board level.

Certificate Deployment Position Requirements

CERTDEPLOY.DEPLOYTYPE (LTE eNodeB, 5G gNodeB) must be set to **DEFAULT**. This ensures that the transmission link backup of main control boards (LTE/NR) function is not coupled with the active/standby states of the main control boards. That is, if one main control board is faulty, the board specified at the certificate deployment position is still working properly.

Configuration Requirements for the Certificate Management Source IP Address of a Board

Either a specified IP address or the O&M IP address can be configured as the source IP address for certificate management.

- Specify an IP address as the source IP address for certificate management (recommended)

- Two source IP addresses (active and standby) need to be configured in the **CRLTSK** MO and be separately deployed on two different main control boards.
- The **CA.MODE (LTE eNodeB, 5G gNodeB)** parameter is set to **CFG_UPD_SIP** or **CFG_INIT_UPD_ADDR**. In this case, the active and standby source IP addresses need to be configured separately on different main control boards.

 NOTE

- If an IP address is specified as the source IP address for certificate management, certificate management during routine management supports two certificate management links: one between the active source IP address and the CA server address, and the other between the standby source IP address and the CA server address. If the active certificate management link is faulty, the base station attempts to use the standby certificate management link.
- If an IP address is specified as the source IP address for certificate management, the new transmission model must be used (that is, **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** must be set to **NEW**), and the address cannot be set to the standby IP address of the CA server.
- Using the O&M IP address as the source IP address for certificate management
 - **CRLTSK.SIP (LTE eNodeB, 5G gNodeB)** or **CRLTSK.SIP6 (LTE eNodeB, 5G gNodeB)** is set to an all-0 address.
 - **CA.MODE (LTE eNodeB, 5G gNodeB)** is set to **DEFAULT_MODE**.

 NOTE

If the O&M IP address is used as the source IP address for certificate management, only one link is available for routine certificate management, that is, the communication link between the effective O&M IP address of the base station and the CA server address.

4.2 Network Analysis

4.2.1 Benefits

A PKI system manages digital certificates for network equipment. This helps operators establish a trusted security domain and gain trust relationships with different vendors to jointly establish a secure network environment.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.1.3 Function Impacts](#), [4.3.2.2.3 Function Impacts](#), [4.3.2.3.3 Function Impacts](#), [4.3.2.4.3 Function Impacts](#), [4.3.2.5.3 Function Impacts](#), [4.3.2.6.3 Function Impacts](#), [4.3.2.7.3 Function Impacts](#), and [4.3.2.8.3 Function Impacts](#).

A certificate application process prolongs base station deployment by approximately 10s. If the requested certificate key type is RSA3072 or RSA4096, the entire deployment process will be further prolonged by approximately 60s, resulting in 70s prolonged in total.

A certificate application process prolongs base station controller deployment by approximately 10s. If the requested certificate key type is RSA3072 or RSA4096,

the entire deployment process will be further prolonged by approximately 60s, resulting in 70s prolonged in total.

During the execution of a certificate validity check task, the CPU usage may temporarily increase.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 GBFD-113526 BTS Supporting PKI

4.3.2.1.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
GS M	Abis over IP	None	<i>IPv4 Transmission</i>

4.3.2.1.2 Mutually Exclusive Functions

None

4.3.2.1.3 Function Impacts

None

4.3.2.2 WRFD-140210 NodeB PKI Support

4.3.2.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
UM TS	IP Transmission Introduction on lub Interface	None	<i>IPv4 Transmission</i>

4.3.2.2.2 Mutually Exclusive Functions

None

4.3.2.2.3 Function Impacts

None

4.3.2.3 GBFD-160211 BSC Supporting PKI

4.3.2.3.1 Prerequisite Functions

None

4.3.2.3.2 Mutually Exclusive Functions

None

4.3.2.3.3 Function Impacts

RA T	Function Name	Function Switch	Reference
GS M	Encrypted Network Managemen t	None	<i>SSL</i>

4.3.2.4 WRFD-160276 RNC Supporting PKI

4.3.2.4.1 Prerequisite Functions

None

4.3.2.4.2 Mutually Exclusive Functions

None

4.3.2.4.3 Function Impacts

RA T	Function Name	Function Switch	Reference
UM TS	Security Managemen t	None	<i>Equipment Security</i>

4.3.2.5 LBFD-003010 Public Key Infrastructure (PKI)

4.3.2.5.1 Prerequisite Functions

None

4.3.2.5.2 Mutually Exclusive Functions

None

4.3.2.5.3 Function Impacts

None

4.3.2.6 MLBFD-12000312 Public Key Infrastructure (PKI)

4.3.2.6.1 Prerequisite Functions

None

4.3.2.6.2 Mutually Exclusive Functions

None

4.3.2.6.3 Function Impacts

None

4.3.2.7 TDLBFD-003010 Public Key Infrastructure (PKI)

4.3.2.7.1 Prerequisite Functions

None

4.3.2.7.2 Mutually Exclusive Functions

None

4.3.2.7.3 Function Impacts

None

4.3.2.8 FBFD-010023 Security Mechanism (PKI)

4.3.2.8.1 Prerequisite Functions

None

4.3.2.8.2 Mutually Exclusive Functions

None

4.3.2.8.3 Function Impacts

None

4.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

The following table lists the hardware to be configured in base stations to support PKI.

Table 4-5 PKI hardware requirements

NE	Hardware
GBTS	GTMUc+UMPT_L
eGBTS	UMPT_G/UMDU_G/MDUC_G
NodeB	UMPT_U/UTRPc/UMDU_U/MDUC_U
eNodeB	UMPT_L/UMPT_T/UTRPc/UMDU_L/UMDU_T
gNodeB	UMPT_N

NE	Hardware
Multimode base station	UMPT_G/UMDU_G/MDUC_G/UMPT_U/UMDU_U/MDUC_U/ UMPT_L/UMPT_T/UMDU_L/UMDU_T/UTRPc
Base station controller	BSC6900/BSC6910 • OMU • SAU (the SAU obtains digital certificates from the OMU) BSC6960: CMPU

For the boards providing eCPRI ports, see "Hardware" in *eCPRI*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

- A PKI server is deployed on the operator's network.
- The device certificates and CRL issued by the operator's CA server comply with RFC 5280.
- The operator's CA server supports CMPv2 defined in RFC 4210. The format of certificate request messages complies with RFC 4211.
- The operator's CA server meets the following requirements in 3GPP TS 33.310: The certificate request response message carries the operator certificate chain.
- The operator's CA server is preconfigured with a Huawei trust certificate chain.
- Deploying PKI over eCPRI ports is under the same networking constraints as the eCPRI interface. For details, see *RF Unit and Topology Management*.
- When an NE is interconnected with the LiteCA, the certificate request template cannot use the SHA1 signature and RSA1024. To support PnP-based deployment for NEs earlier than SRAN16.1, the SHA1 signature algorithm for CMP messages needs to be enabled on the LiteCA.
- An operator can enable the OCSP function only after deploying an OCSP server on the live network. If the public key of the OCSP server is trusted by the base station, ensure that the response from the OCSP server contains a certificate chain. Otherwise, the base station's check on the OCSP status fails and it reports ALM-26842 Automatic Certificate Update Failed.
- When HTTP is used to obtain CRLs, the operator's CA server must support the mechanism defined in RFC 2585 to obtain CRLs over HTTP.

4.4 Operation and Maintenance

4.4.1 When to Use

A Huawei initial certificate can meet the basic transmission security requirements, but it does not support online update. Therefore, directly using a Huawei initial

certificate on the network has security risks. It is recommended that the operator deploy a PKI system on the live network and use the operator-issued device certificate to replace the Huawei initial certificate, so that the operator-issued device certificate can be updated online, which minimizes security risks.

If the base stations and base station controllers on the live network need to interconnect with the PKI system, the PKI feature needs to be enabled for the base stations and base station controllers.

4.4.2 Precautions

Before deploying the PKI feature for a base station or base station controller, engineering personnel must obtain CA information from CA maintenance personnel. The following table lists the CA information that needs to be collected.

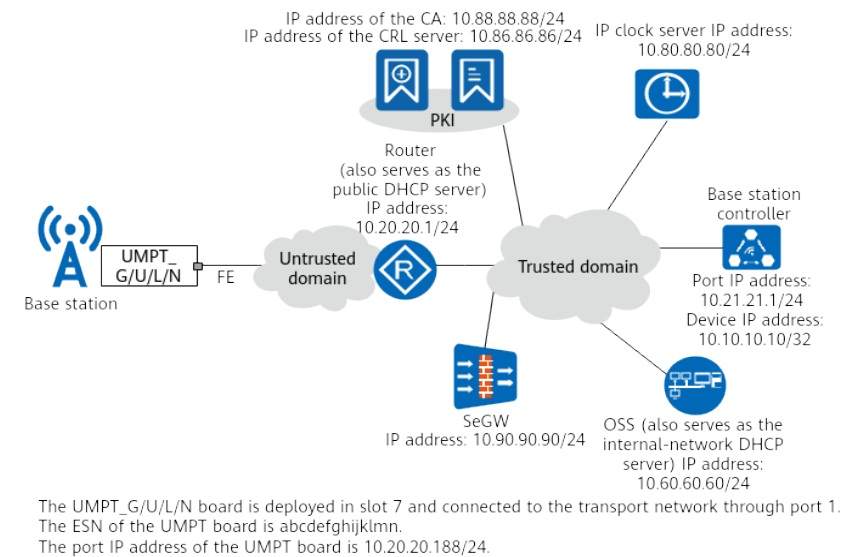
Item to Be Collected	Required Parameter on the Base Station or Base Station Controller Side
CA name	<i>CA.CANAME (5G gNodeB, LTE eNodeB)</i>
RA name	<i>CA.RANAME (5G gNodeB, LTE eNodeB)</i>
Uniform resource locator (URL) of the CA	<i>CA.URL (5G gNodeB, LTE eNodeB)</i> (Optional) <i>CA.INITREQURL (5G gNodeB, LTE eNodeB)</i>
Signature algorithm for CMP messages	<i>CA.SIGNALG (5G gNodeB, LTE eNodeB)</i>
OCSP URL	<i>CA.OCSPURL (LTE eNodeB, 5G gNodeB)</i>
Signature algorithm for the CA to issue certificates	<i>CERTREQ.SIGNALG (5G gNodeB, LTE eNodeB)</i>
Size of the certificate key	<i>CERTREQ.KEYSIZE (5G gNodeB, LTE eNodeB)</i>
Use of the certificate key	<i>CERTREQ.KEYUSAGE (5G gNodeB, LTE eNodeB)</i>
File name of the root certificate or certificate chain	<i>TRUSTCERT.CERTNAME (5G gNodeB, LTE eNodeB)</i>
(Optional) IP address of the CRL server	<i>CRLTSK.IP (5G gNodeB, LTE eNodeB)</i>
(Optional) User name for logging in to the CRL server	<i>CRLTSK.USR (5G gNodeB, LTE eNodeB)</i>
(Optional) Password for logging in to the CRL server	<i>CRLTSK.PWD (5G gNodeB, LTE eNodeB)</i>
(Optional) CRL file name	<i>CRLTSK.FILENAME (5G gNodeB, LTE eNodeB)</i>
(Optional) Method of obtaining the CRL file	<i>CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB)</i>

Item to Be Collected	Required Parameter on the Base Station or Base Station Controller Side
(Optional) Distinguish name of the CRL server	CRLTSK.SEARCHDN (5G gNodeB, LTE eNodeB)
(Optional) Port number of the CRL server	CRLTSK.PORT (5G gNodeB, LTE eNodeB)

4.4.3 Deployment of PKI on the GBTS/eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station

This section uses the networking illustrated in [Figure 4-20](#) as an example to describe how to deploy the PKI feature on the GBTS/eGBTS/NodeB/eNodeB/gNodeB/multimode base station. A UMDU/MDUC can also be used in a co-MPT multimode base station in the secure networking shown in this figure. However, a UMDU/MDUC cannot be used in a separate-MPT multimode base station. The descriptions in this section are applicable to an eGBTS using a UMPT, UMDU, or MDUC.

Figure 4-20 Example of the secure networking for the GBTS/eGBTS/NodeB/eNodeB/gNodeB/multimode base station



4.4.3.1 Data Preparation

"-" in the following tables in this section indicates that there is no special requirement for parameter setting. Set the parameters according to network plans.

[Table 4-6](#) lists the data to be prepared for the certificate deployment position on the base station (corresponding to the **CERTDEPLOY** MO).

Table 4-6 Data to be prepared for the deployment position of a certificate

Parameter Name	Parameter ID	Setting Notes
Certification Deploy Position Type	CERTDEPLOY.DEPLOY TYPE (LTE eNodeB, 5G gNodeB)	The value DEFAULT is recommended.
Cabinet No.	CERTDEPLOY.CN (5G gNodeB, LTE eNodeB)	-
Subrack No.	CERTDEPLOY.SRN (5G gNodeB, LTE eNodeB)	
Slot No.	CERTDEPLOY.SN (5G gNodeB, LTE eNodeB)	

Table 4-7 lists the data to be prepared for a certificate request template of the base station (corresponding to the **CERTREQ MO**).

Table 4-7 Data to be prepared for a certificate request template

Parameter Name	Parameter ID	Setting Notes
Common Name Additional Info.	CERTREQ.USERADDI NFO (LTE eNodeB, 5G gNodeB)	-
Country	CERTREQ.COUNTRY (LTE eNodeB, 5G gNodeB)	-
Organization	CERTREQ.ORG (LTE eNodeB, 5G gNodeB)	-
Organizational Unit	CERTREQ.ORGUNIT (LTE eNodeB, 5G gNodeB)	-
State or Province	CERTREQ.STATEPRO VINCENAME (LTE eNodeB, 5G gNodeB)	-
Locality	CERTREQ.LOCALITY (LTE eNodeB, 5G gNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Key Usage	CERTREQ.KEYUSAGE (LTE eNodeB, 5G gNodeB)	It is recommended that at least the DIGITAL_SIGNATURE option be selected. A digital signature is used to verify the peer's digital signature during CMPv2-based certificate application and update, IKE negotiation, and SSL authentication.
Signature Algorithm	CERTREQ.SIGNALG (LTE eNodeB, 5G gNodeB)	- The value SHA256 is recommended. - The SHA1 algorithm is insufficiently secure. If this parameter is set to SHA1, certificates may be forged. - You can run SET SECPY to adjust the configuration security level to prohibit the parameter from being set to SHA1.
Key Size	CERTREQ.KEYSIZE (5G gNodeB, LTE eNodeB)	This parameter is valid only when the CERTREQ.KEYTYPE (5G gNodeB, LTE eNodeB) parameter is set to RSA. If this parameter is set to KEYSIZE1024, the key is considered insufficiently secure and the certificate may be forged. Therefore, it is recommended that this parameter not be set to KEYSIZE1024. You can run SET SECPY to adjust the configuration security level to prohibit the parameter from being set to KEYSIZE1024.
Local Name	CERTREQ.LOCALNAME (LTE eNodeB, 5G gNodeB)	If this parameter is not set, the value of the Common Name field in a certificate is used by default. If this parameter is set, the value of the Local Name field in a certificate must be the same as the value of this parameter.
Local IP	CERTREQ.LOCALIP (LTE eNodeB, 5G gNodeB)	If the IDTYPE (5G gNodeB, LTE eNodeB) parameter is set to IP , the value of this parameter must be the same as the value of LOCALIP in the IKEPEER MO.
Local IPv6	CERTREQ.LOCALIP6 (5G gNodeB, LTE eNodeB)	In IPv6 networking, if the IDTYPE (5G gNodeB, LTE eNodeB) parameter is set to IP , the value of this parameter must be the same as the value of LOCALIP6 in the TUNNELITF MO.

Parameter Name	Parameter ID	Setting Notes
Key Type	CERTREQ.KEYTYPE (5G gNodeB, LTE eNodeB)	An ECDSA digital certificate features fast computing, a short key length, and a small space for storage. If all the operator's devices support ECDSA digital certificates, it is recommended that this parameter be set to ECDSA .
ECC Curve	CERTREQ.ECCCURVE (5G gNodeB, LTE eNodeB)	If this parameter is set to P256 , the security strength meets the security requirements. The recommended value is therefore P256 . If the operator's devices support the P384 and ED25519 algorithms, P384 or ED25519 can also be used.

The base station must be configured with CA server information to apply for a certificate from the CA server. [Table 4-8](#) lists the data to be prepared for the CA server (corresponding to the **CA MO**).

Table 4-8 Data to be prepared for the CA

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME (LTE eNodeB, 5G gNodeB)	This parameter indicates the name of the CA. The Common Name field of the CA name cannot contain any of the following characters: backslash (\), slash (/), colon (:), asterisk (*), question mark (?), double quotation marks ("), left angle bracket (<), right angle bracket (>), and bar (). Otherwise, an error occurs when you run the REQ DEVCERT command to apply for a device certificate. For example: The Subject name of the certificate used for signing CMP messages on the CA/RA is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com, and the corresponding CANAME is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com.

Parameter Name	Parameter ID	Setting Notes
Certificate Authority URL	CA.URL (LTE eNodeB, 5G gNodeB)	Currently, base stations cannot translate domain names. Therefore, use an IP address instead of a domain name in the URL. The TCP port number is determined by the CA server. For example, the URL domain name of the CA server can be set to http://10.88.88.88:80/pkix/. For an IPv6 URL address, the format "[IP address]:Port Number" must be used to separate the IP address from the port number, for example, https://[2050::20:23:123]:34481/pkix/.
Signature Algorithm	CA.SIGNALG (LTE eNodeB, 5G gNodeB)	• The value SHA256 is recommended. • The SHA1 algorithm is insufficiently secure. If this parameter is set to SHA1, CMPv2 messages may be tampered with. • You can run SET SECPLY to adjust the configuration security level to prohibit the parameter from being set to SHA1.
Certificate Registration Name	CA.RANAME (5G gNodeB, LTE eNodeB)	Set this parameter if the customer's certificate registration server requires that the value of the recipient field in the received CMPv2-based certificate message be the name of the certificate registration server.

Parameter Name	Parameter ID	Setting Notes
Certificate Fetch Mode	CA.MODE (LTE eNodeB, 5G gNodeB)	- If this parameter is set to DEFAULT_MODE, the UPDSIP (5G gNodeB, LTE eNodeB), INITREQURL (5G gNodeB, LTE eNodeB), and INITREQSIP (5G gNodeB, LTE eNodeB) parameters do not need to be set. The base station uses the OM IP address and URL as the source and destination IP addresses, respectively, for routine certificate management. Routine certificate management involves certificate application and certificate update, both of which can be performed automatically or manually. When applying for a certificate for the first time during base station deployment, the base station uses the interface IP address (deployment by PnP) or OM IP address (deployment in other modes) as the source IP address, and uses the URL as the destination IP address. - If this parameter is set to CFG_UPD_SIP, the INITREQURL and INITREQSIP (5G gNodeB, LTE eNodeB) parameters do not need to be set. The base station uses the values of the UPDSIP (5G gNodeB, LTE eNodeB) and URL (5G gNodeB, LTE eNodeB) parameters as the source and destination IP addresses, respectively, for routine certificate management. When applying for a certificate for the first time during base station deployment, the base station uses the interface IP address (for deployment by PnP) or the OM IP address (for other deployment modes) as the source IP address, and the URL as the destination IP address. - If this parameter is set to CFG_INIT_UPD_ADDR on a base station:

Parameter Name	Parameter ID	Setting Notes
		- The base station uses the values of the UPDSIP (5G gNodeB, LTE eNodeB) and URL (5G gNodeB, LTE eNodeB) parameters as the source and destination IP addresses, respectively, for routine certificate management. - When applying for a certificate for the first time during base station deployment, the base station uses the interface IP address (deployment by PnP) or INITREQSIP (5G gNodeB, LTE eNodeB) value (deployment in other modes) as the source IP address, and uses the INITREQURL value as the destination IP address.
Certificate Update Source IP	CA.UPDSIP (LTE eNodeB, 5G gNodeB)	-
CA URL During Site Deployment	CA.INITREQURL (LTE eNodeB, 5G gNodeB)	-
Source IP for Applying for a Certificate During Site Deployment	CA.INITREQSIP (LTE eNodeB, 5G gNodeB)	-
Certificate Update Source IPv6 Address	CA.UPDSIP6 (5G gNodeB, LTE eNodeB)	-
Source IPv6 Address for Applying for a Certificate During Site Deployment	CA.INITREQSIP6 (5G gNodeB, LTE eNodeB)	-
Slave Certificate Authority URL	CA.SLVURL (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Slave CA URL During Site Deployment	CA.SLVINITREQURL (5G gNodeB, LTE eNodeB)	-
Certificate Request Switch	CA.CERTREQSW (5G gNodeB, LTE eNodeB)	- In single-operator scenarios, it is recommended that this parameter be set to DEFAULT. - In multi-operator scenarios, it is recommended that this parameter be set to USERDEFINE to allow different operators to use different certificate templates.
Local Name	CA.LOCALNAME (5G gNodeB, LTE eNodeB)	-
Local IP	CA.LOCALIP (5G gNodeB, LTE eNodeB)	-
Local IPv6 Address	CA.LOCALIP6 (5G gNodeB, LTE eNodeB)	-
Key Type	CA.KEYTYPE (5G gNodeB, LTE eNodeB)	An ECDSA digital certificate features fast computing, a short key length, and a small space for storage. If all the operator's devices support ECDSA digital certificates, it is recommended that this parameter be set to ECDSA .
ECC Curve	CA.ECCCURVE (5G gNodeB, LTE eNodeB)	If this parameter is set to P256 , the security strength meets the security requirements. The recommended value is therefore P256 . If the operator's devices support the P384 and ED25519 algorithms, P384 or ED25519 can also be used.
Certificate Request Signature Algorithm	CA.CERTREQSIGNALG (LTE eNodeB, 5G gNodeB)	- The value SHA256 is recommended. - The SHA1 algorithm is insufficiently secure. If this parameter is set to SHA1, CMPv2 messages may be tampered with. - You can run SET SECPLY to adjust the configuration security level to prohibit the parameter from being set to SHA1.

Parameter Name	Parameter ID	Setting Notes
Key Size	CA.KEYSIZE (<i>LTE eNodeB, 5G gNodeB</i>)	- This parameter is valid only when the CA.KEYTYPE (<i>5G gNodeB, LTE eNodeB</i>) parameter is set to RSA. - If this parameter is set to KEYSIZE1024, the key is considered insufficiently secure and the certificate may be forged. Therefore, it is recommended that this parameter not be set to KEYSIZE1024. - You can run SET SECPLY to adjust the configuration security level to prohibit the parameter from being set to KEYSIZE1024.
Certificate File Name	CA.CERTNAME (<i>LTE eNodeB, 5G gNodeB</i>)	After an alarm indicating that a Huawei initial certificate is about to expire is reported, an operator-issued device certificate must be configured before the certificate expires.
Initial Certificate File Name	CA.VENDORCERT (<i>5G gNodeB, LTE eNodeB</i>)	This parameter indicates which device certificate configured in CA.VENDORCERT (<i>5G gNodeB, LTE eNodeB</i>) is used for HTTPS authentication and signature of the certificate request message when the base station uses a Huawei initial certificate to apply for a certificate from the CA. If the board corresponding to the certificate deployment position is not preconfigured with the specified certificate or the specified certificate is damaged or lost, the system uses the other Huawei initial certificate on the board. Corresponding Huawei trust certificate needs to be installed on the CA server.

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	CA.TRUSTGROUPID <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the ID of a trust group. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust certificate group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.
OCSP URL	CA.OCSPURL <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies the URL of the OCSP server of the CA. The URL can be either an http or https URL. The IP address in the URL must be a valid IP address. The default port number is 80 for http or 443 for https. If the port number is not specified, the system automatically adds the default port number to the URL.
Support Forcible Execution	CA.FORCEEXECUTE <i>(5G gNodeB, LTE eNodeB)</i>	For setting suggestions, see 4.1.3.4.2 Estimation of Certificate Reconfiguration Impact .

 NOTE

If OM data flows are transmitted by the IPsec tunnel, the OM IP address cannot be used for data that is not protected by IPsec. If OM data flows are not transmitted by the IPsec tunnel, the OM IP address cannot be used for data that is protected by IPsec.

Table 4-9 lists the data to be prepared for a device certificate (corresponding to the **CERTMK** MO).

Table 4-9 Data to be prepared for a device certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT <i>(LTE eNodeB, 5G gNodeB)</i>	- If an operator-issued device certificate is used for identity authentication between the base station and SeGW, two CERTMK MOs must be configured to specify an operator-issued device certificate and a Huawei initial certificate, respectively. For Huawei initial certificates, this parameter must be set to appcert.pem. For operator-issued device certificates, this parameter must be set to OPKIDevCert.cer during base station deployment by PnP. - If a Huawei initial certificate is used for identity authentication between the base station and SeGW, only one CERTMK MO needs to be configured to specify a Huawei initial certificate. This parameter is set to appcert.pem accordingly. Users cannot modify or remove this MO.

Table 4-10 lists the data to be prepared for a device certificate in use.

Table 4-10 Data to be prepared for a device certificate in use

Parameter Name	Parameter ID	Setting Notes
Application Type	APPCERT.APPTYPE <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter must be set to IKE for IKE authentication. (The base station controllers do not support IKE authentication.) This parameter must be set to SSL for SSL authentication.
Certificate File Name	APPCERT.APPCERT <i>(5G gNodeB, LTE eNodeB)</i>	Base stations do not have special requirements for the setting of this parameter.

Table 4-11 lists the data to be prepared for a trust certificate (corresponding to the **TRUSTCERT** MO).

Table 4-11 Data to be prepared for a trust certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	TRUSTCERT.CERTNAME (LTE eNodeB, 5G gNodeB)	The base station must be configured with an operator's trust certificate and a Huawei trust certificate. For the Huawei trust certificate, this parameter is set to caroot.pem on the base station side. For an operator's trust certificate, this parameter is set to CN.cer when automatic certificate application is used. The value of <i>CN</i> must be the same as that in the Subject field of the trust certificate. If the operator's CA system has a multi-layer structure, a trust certificate chain needs to be configured. If the certificates at different layers of CAs in the certificate chain are separately stored, run the ADD TRUSTCERT command to add certificates at each layer.

Table 4-12 lists the data to be prepared for a trust group (corresponding to the **TRUSTGROUP MO**).

Table 4-12 Data to be prepared for a trust group

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTGROUP.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)	-
Trust Group Description	TRUSTGROUP.TRUSTGROUPDESC (LTE eNodeB, 5G gNodeB)	-

Table 4-13 lists the data to be prepared for the association between a trust group and a trust certificate (corresponding to the **TRUSTCERT2GROUP MO**).

Table 4-13 Data to be prepared for associating a trust group with a trust certificate

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTCERT2GROUP. TRUSTGROUPLD (LTE eNodeB, 5G gNodeB)	-
Certificate File Name	TRUSTCERT2GROUP. CERTNAME (LTE eNodeB, 5G gNodeB)	-

Table 4-14 lists the data to be prepared for a periodic certificate validity check task (corresponding to the **CERTCHKTSK** MO).

Table 4-14 Data to be prepared for a periodic certificate validity check task

Parameter Name	Parameter ID	Setting Notes
Checking Period	CERTCHKTSK.PERIOD (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Alarm Threshold	CERTCHKTSK.ALMRN G (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Update Method	CERTCHKTSK.UPDAT EMETHOD (LTE eNodeB, 5G gNodeB)	The value CMP is recommended.
Automatic Certificate Reapplication Switch	CERTCHKTSK.AUTOR EAPPLYSW (5G gNodeB, LTE eNodeB)	The value ON is recommended.
Initial Certificate Expire Threshold	CERTCHKTSK.VENDO RALMRNG (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Initial Certificate Verification Switch	CERTCHKTSK.VENDO RCERTCHKSW (LTE eNodeB, 5G gNodeB)	If the initial certificate is used, the value ON is recommended. Otherwise, the value OFF is recommended. After a third-party certificate is replaced, you are advised to set this parameter to OFF . Otherwise, related alarms may be reported after the initial certificate expires.

(Optional) Prepare CRL data if the base station needs to obtain CRL information from the CA. [Table 4-15](#) lists the data to be prepared for a CRL (corresponding to the **CRL MO**).

Table 4-15 Data to be prepared for a CRL

Parameter Name	Parameter ID	Setting Notes
CRL File Name	CRL.CERTNAME (5G gNodeB, LTE eNodeB)	-

(Optional) Prepare data related to CRL usage policies. [Table 4-16](#) lists the data to be prepared for these policies (corresponding to the **CRLPOLICY MO**).

Table 4-16 Data to be prepared for CRL usage policies

Parameter Name	Parameter ID	Setting Notes
CRL Using Policy	CRLPOLICY.CRLPOLICY (LTE eNodeB, 5G gNodeB)	Operators can set this parameter based on site requirements. During base station deployment through PnP, the base station does not support verifying peer certificate validity against the CRL.

(Optional) Prepare data related to a periodic CRL download task. [Table 4-17](#) lists the data to be prepared for the task (corresponding to the **CRLTSK MO**).

Table 4-17 Data to be prepared for a periodic CRL download task

Parameter Name	Parameter ID	Setting Notes
IP Address	CRLTSK.IP (5G gNodeB, LTE eNodeB)	Set this parameter to the IP address of the CRL server.
User Name	CRLTSK.USR (LTE eNodeB, 5G gNodeB)	-
Password	CRLTSK.PWD (LTE eNodeB, 5G gNodeB)	-
File Name	CRLTSK.FILENAME (LTE eNodeB, 5G gNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Using CRL's Next Update	CRLTSK.ISCRLTIME <i>(LTE eNodeB, 5G gNodeB)</i>	If this parameter is set to ENABLE , the base station downloads a CRL when the next update time arrives.
CRL Updating Period	CRLTSK.PERIOD <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter must be set when the CRLTSK.ISCRLTIME <i>(LTE eNodeB, 5G gNodeB)</i> parameter is set to DISABLE .
Access Method	CRLTSK.CRLGETMET HOD <i>(LTE eNodeB, 5G gNodeB)</i>	The value LDAP is recommended. Set this parameter to FTP or HTTP only when the peer device does not support LDAP.
Distinguish Name	CRLTSK.SEARCHDN <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter must be set when the CRLTSK.CRLGETMETHOD <i>(5G gNodeB, LTE eNodeB)</i> parameter is set to LDAP .
Port No.	CRLTSK.PORT <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter must be set when the CRLTSK.CRLGETMETHOD <i>(5G gNodeB, LTE eNodeB)</i> parameter is set to LDAP or HTTP .
Task ID	CRLTSK.TSKID <i>(LTE eNodeB, 5G gNodeB)</i>	-
Source IP	CRLTSK.SIP <i>(LTE eNodeB, 5G gNodeB)</i>	If this parameter is not set, the base station uses the OM IP address as the source IP address to update a CRL.
Connection Mode	CRLTSK.CONNMODE <i>(5G gNodeB, LTE eNodeB)</i>	This parameter is valid only when CRLTSK.CRLGETMETHOD <i>(5G gNodeB, LTE eNodeB)</i> is set to LDAP or HTTP .

Parameter Name	Parameter ID	Setting Notes
Authenticate Peer	CRLTSK.AUTHPEER (5G gNodeB, LTE eNodeB)	- This parameter is valid only when CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) is set to LDAP or HTTP and CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB) is set to SSL. - When CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) is set to HTTP, this parameter can only be set to ENABLE. - If this parameter is set to ENABLE, ensure that the NEs and CRL server have been correctly configured with the CA trust certificates and device digital certificates. - If this parameter is set to DISABLE, a base station cannot verify the identity of an LDAP server, posing security risks. - You can run SET SECPLY to adjust the configuration security level to prohibit the parameter from being set to DISABLE.
Certificate File Name	CRLTSK.CERTNAME (LTE eNodeB, 5G gNodeB)	This parameter specifies a device certificate name. This parameter specifies that the device certificate configured in the CERTMKN MO is used for authentication when establishing a TLS connection between the local end and an FTP, LDAP, or HTTP server. If this parameter is not specified, the system uses the preconfigured Huawei initial certificate for authentication.
Trust Group ID	CRLTSK.TRUSTGROU PID (LTE eNodeB, 5G gNodeB)	This parameter specifies the ID of a trust group. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust certificate group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

(Optional) Prepare data to manually download an operator's root certificate or CRL from an FTP server. [Table 4-18](#) lists the data to be prepared for downloading a certificate file (corresponding to the **DLD CERTFILE** command).

Table 4-18 Data to be prepared for downloading a certificate file

Parameter Name	Parameter ID	Setting Notes
FTP Server IP	<i>IP</i>	-
User Name	<i>USR</i>	-
Password	<i>PWD</i>	-
Source File Name	<i>SRCF</i>	-
Destination File Name	<i>DSTF</i>	It is recommended that this parameter be set to the same value as <i>SRCF</i> .
Guage Option	<i>GA</i>	This parameter determines whether to report the progress of file downloading. The value Y is recommended.
Certificate Type	<i>CT</i>	-

 NOTE

The file path of the FTP server connected to the base station must be in English. That is, the values of **SRCF** in **DLD CERTFILE** and **DSTF** in **ULD CERTFILE** must be in English and cannot be in Chinese.

(Optional) Prepare data to manually trigger a CMPv2-based certificate application. [Table 4-19](#) lists the data to be prepared for applying for a device certificate (corresponding to the **REQ DEVCERT** command) based on CMPv2.

Table 4-19 Data to be prepared for applying for a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	<i>CA.CANAME (LTE eNodeB, 5G gNodeB)</i>	-
Certificate File Name	<i>CERTMK.APPCERT (5G gNodeB, LTE eNodeB)</i>	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate update. [Table 4-20](#) lists the data to be prepared for updating a device certificate (corresponding to the **UPD DEVCERT** command) based on CMPv2.

Table 4-20 Data to be prepared for updating a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT <i>(LTE eNodeB, 5G gNodeB)</i>	This parameter specifies a certificate to be updated.
Renew Key	REKEY	The value Yes is recommended.

4.4.3.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.4.1 When to Use](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

The following is an MML command example of how to activate an operator-issued device certificate on the base station side.

```
//Setting the deployment position of a certificate (the base station needs to be reset for the modification to take effect)
SET CERTDEPLOY: DEPLOYTYPE=DEFAULT;
//Modifying configurations of a certificate request template
MOD CERTREQ: COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="xx", ORG="ITEF",
ORGUNIT="Hw", STATEPROVINCENAME="sc", LOCALITY="cd",
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,
SIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048, LOCALNAME="abcdefghijklmn.huawei.com",
LOCALIP="10.20.20.188";
//Setting the CA server information
//If the CA server is reachable through a public or internal network and OM data is protected by IPsec, you
are advised to set the source IP addresses for certificate application and update to an interface IP address
and an OM IP address (for example, 10.31.31.188), respectively. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR,
UPDSIP="10.31.31.188",INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",
TRUSTGROUPID=65535, OCSPURL="http://10.88.88.88:80/pkix/";
//If the CA server is reachable through a public or internal network and OM data is not protected by IPsec,
you are advised to set the source IP addresses for certificate application and update to an interface IP
address and an internal-network IP address (for example, 10.45.45.45), respectively. The following is an
example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR,
UPDSIP="10.45.45.45",INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",
VENDORCERT=APPCERT;
//If the CA server is reachable through only an external network, you are advised to set the source IP
```

addresses for both certificate application and update to interface IP addresses. The following is an example:

```
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR,
UPDSIP="10.20.20.188",INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",
VENDORCERT=APPCERT;
//Downloading an operator's root certificate from an FTP server (assume that the FTP server is on the MAE,
indicating that the IP address of the FTP server is the same as that of the MAE)
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="OperationCA.cer",DSTF="OperationCA.cer";
//Adding an operator's root certificate as the trust certificate
ADD TRUSTCERT: CERTNAME="OperationCA.cer";
//Adding a trust group
ADD TRUSTGROUP: TRUSTGROUPID=0;
//Adding a trust certificate to the trust group
ADD TRUSTCERT2GROUP: TRUSTGROUPID=0, CERTNAME="OperationCA.cer";
//Setting information required for the base station to apply for an operator-issued device certificate based
on CMPv2 when the application needs to be manually triggered (skip this step when the application is
automatically triggered)
REQ DEVCERT: CANAME="C=AU, S=Some-State, O=Internet Widgits Pty Ltd, CN=eca1",
APPCERT="OPKIDevCert.cer";
//Adding a device certificate (For an operator-issued device certificate, set APPCERT to OPKIDevCert.cer
during base station deployment by PnP and set APPCERT to the planned name of the operator-issued
device certificate in other deployment scenarios.)
ADD CERTMK: APPCERT="OPKIDevCert.cer", CASW=ON, CANAME="C=AU, S=Some-State, O=Internet
Widgits Pty Ltd, CN=eca1";
//Verifying whether the device certificate can be used for normal IKE and SSL connections to ensure that
secure channels can be successfully established between the base station and the peer end
TST APPCERT: APPTYPE=IKE, APPCERT="OPKIDevCert.cer";
//Setting information about the device certificate in use
MOD APPCERT: APPTYPE=IKE, APPCERT="OPKIDevCert.cer";
//Setting a periodic certificate validity check task
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDITEMETHOD=CMP;
//(Optional) Downloading the CRL from the FTP server (If the FTP server is on the MAE, the IP address of
the FTP server is the same as that of the MAE.)
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="eNodeB.crl",DSTF="eNodeB.crl";
//(Optional, required only when a manual certificate application procedure is used) Loading a CRL
ADD CRL: CERTNAME="eNodeB.crl";
//(Optional) Setting a CRL usage policy
SET CRLPOLICY:CRLPOLICY= NOVERIFY;
//(Optional) Adding a task of periodically downloading the CRL
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="eNodeB.crl", ISCRLTIME=DISABLE,
PERIOD=24, TSKID=0, CRLGETMETHOD=FTP, TRUSTGROUPID=65535;
```

NOTE

After the activated device certificate used by IKE is changed by running the **MOD APPCERT** command, if IKE authentication uses the new certificate and the current IKE SA is normal, the base station automatically initiates an IKE renegotiation.

In addition to the preceding steps, perform the following step to manually trigger a certificate update. When you run the **UPD DEVCERT** command to update a certificate, the certificate update will fail if the base station is performing an IKE or SSL negotiation. You need to run this command after the negotiation is completed.

```
UPD DEVCERT: APPCERT="OPKIDevCert.cer",REKEY=YES;
```

Deactivation

Step 1 Run the **RMV CA** command to remove the CA.

Step 2 (Optional) Run the **RMV CRLTSK** command to remove the task of periodically updating a CRL.

----End

4.4.3.3 Using the MAE-Deployment

MAE-Deployment Summary batch configuration and MAE-Deployment transport security wizard configuration are available for deploying the PKI feature for new base stations. You can use one of them based on site requirements.

MAE-Deployment Summary Batch Configuration

For detailed operations using an eNodeB as an example, see Feature Configuration Using the MAE-Deployment.

MAE-Deployment Transport Security Wizard Configuration

You can use the transport security wizard to configure the parameters for the PKI and IPsec features. The wizard will guide you through the selection of the key parameters for PKI and IPsec networking and achieve automatic configuration of most of the parameters related to PKI and IPsec features. After the wizard configuration is complete, the MAE-Deployment automatically imports the configured parameters to the Summary data file and prompts which parameters must be manually configured in the Summary data file (for example, the *UPDSIP* parameter in the **CA MO**).

Figure 4-21 shows the procedure for configuring data using the transport security wizard.

Figure 4-21 Procedure for configuring data using the transport security wizard

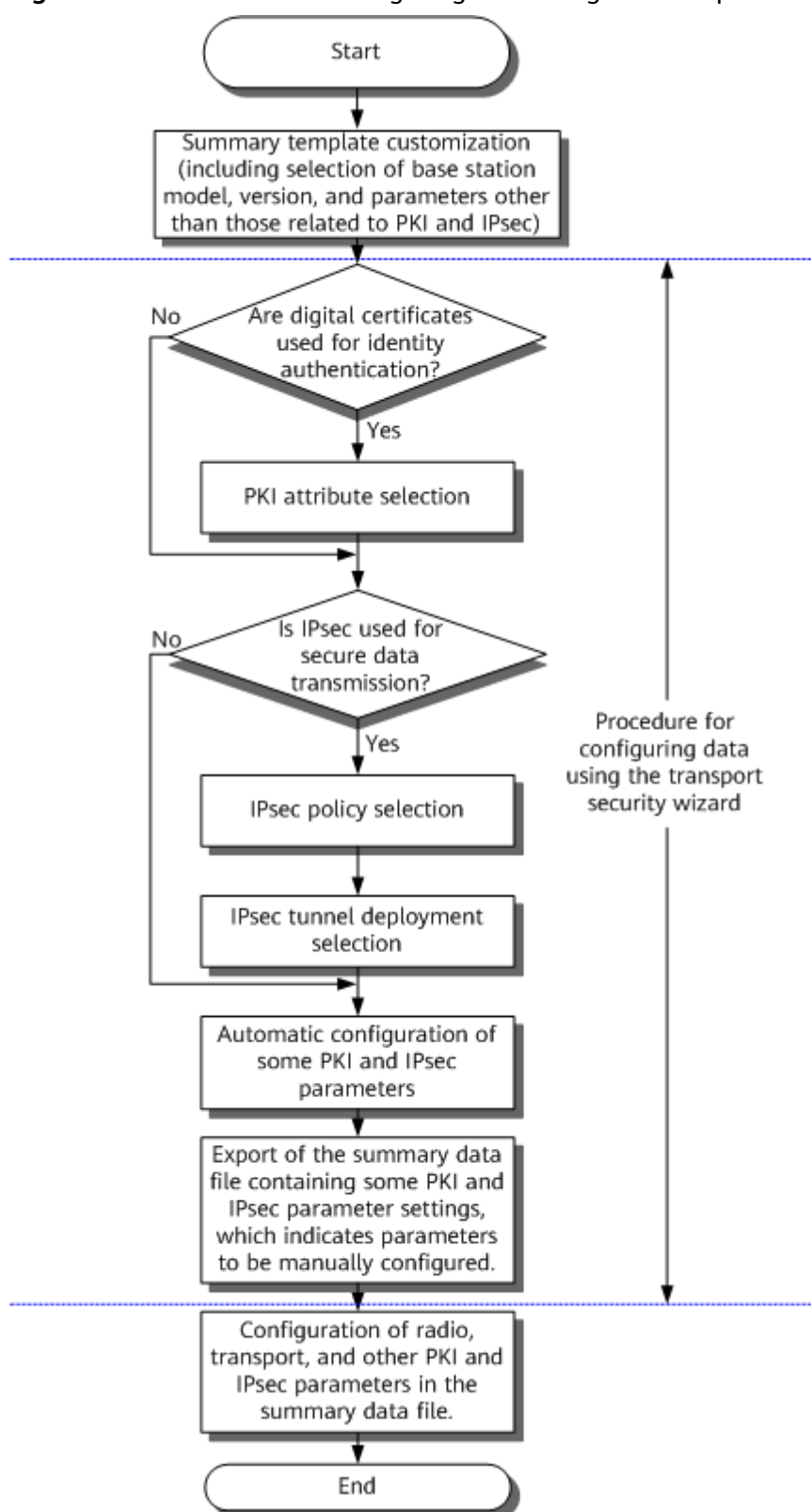
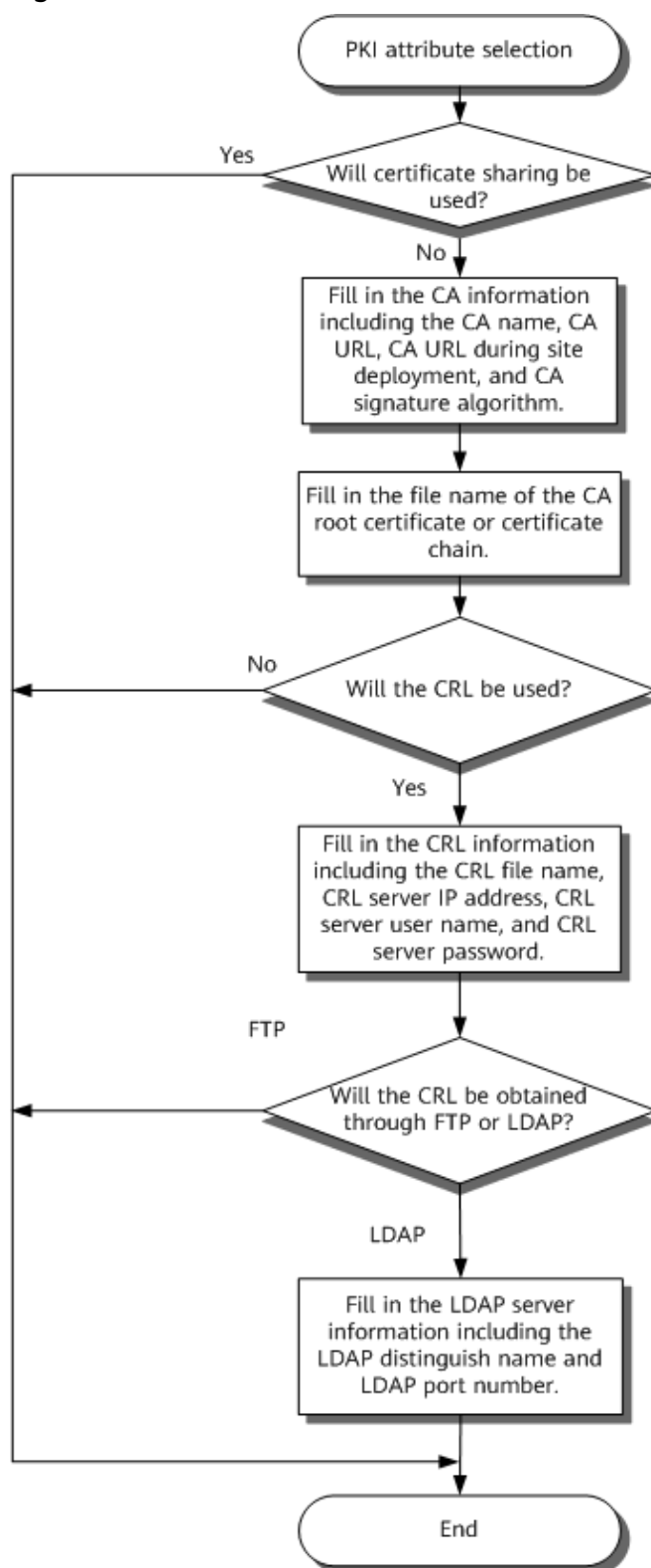


Figure 4-22 shows the PKI attribute selection in the transport security wizard.

Figure 4-22 PKI attribute selection



For details on how to configure PKI parameters in the MAE-Deployment transport security wizard, see [4.4.3.1 Data Preparation](#). For details on how to configure IPsec parameters, see the "Using the MAE-Deployment" section in *IPsec*.

After configurations on the transport security wizard are complete, the IPsec and PKI parameter setting tables are exported, displaying the IPsec and PKI parameters that have been configured and the parameters that need to be manually configured in the summary data file. You can adjust the configured parameters in the summary data file based on actual conditions.

For the configuration path and interface for the transport security wizard, see "Customizing a Summary Data File Using the Transmission Security Wizard" in *MAE Product Documentation*.

The MAE-Deployment transport security wizard has the following restrictions for configuring PKI:

- PKI redundancy cannot be configured.
- SSL transmission cannot be configured for obtaining the CRL.
- Base Station Supporting Multi-operator PKI cannot be configured.
- PKI parameters for the eGBTS, NodeB, eNodeB, and gNodeB can be configured. PKI parameters can be configured for the GBTS only when it is configured with GTMUC+UMPT_L.

4.4.3.4 Activation Verification

Observing the PKI Feature

Step 1 Run the **DSP APPCERT** command to check the status of device certificates.

If the values of **Certificate File Name**, **Issuer**, and **Common Name** are correct and the value of **Status** is **Normal** in the query result, the device certificate has been loaded to the base station.

Step 2 Run the **DSP TRUSTCERT** command to check the status of trust certificates.

If the value of **Status** is **Normal** in the query result, the trust certificate has been loaded to the base station.

Step 3 (Optional) Run the **DSP CRL** command to check the CRL status.

If the value of **Status** is **Normal** in the query result, the CRL has been loaded to the base station.

----End

Observing Offline Certificate Monitoring

Step 1 On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management > NE Certificate Management > Offline Certificate**. All device certificates in abnormal states are displayed on the MAE-Access.

Step 2 (Optional) To export information of device certificates in abnormal states, click **Export**. If the **On disconnected NE** or **On deleted NE** check box is selected, you also need to set the duration in which the device certificates remain in the state.

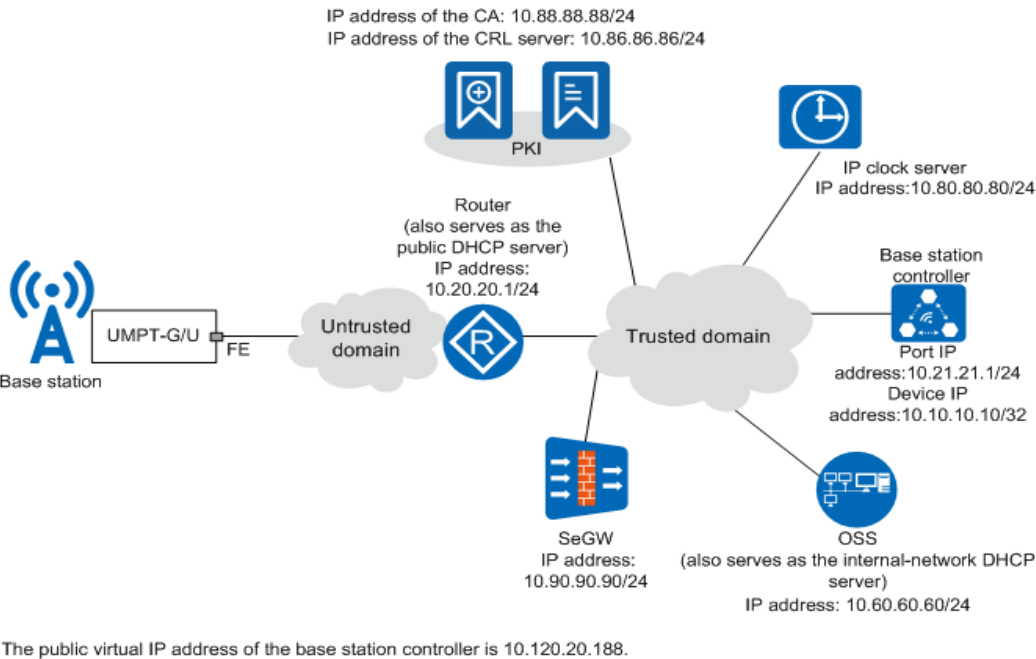
----End

4.4.4 Deployment of PKI on the BSC6900/BSC6910 (GSM/UMTS)

This section describes how to deploy the PKI feature by using MML commands. For details about how to deploy the PKI feature by using the MAE, see the MAE online help.

This section uses the networking illustrated in [Figure 4-23](#) as an example to describe how to deploy the PKI feature on the BSC6900/BSC6910.

Figure 4-23 Example of the secure networking for the BSC6900/BSC6910



4.4.4.1 Data Preparation

"-" in the following tables in this section indicates that there is no special requirement for parameter setting. Set the parameters according to network plans.

[Table 4-21](#) lists the data to be prepared for a certificate request template of the BSC6900/BSC6910 (corresponding to the **CERTREQ MO**).

Table 4-21 Data to be prepared for a certificate request template of the BSC6900/BSC6910

Parameter Name	Parameter ID	Setting Notes
Common Name	CERTREQ.COMMNAME	-
Common Name Additional Info.	CERTREQ.USERADDINFO	-

Parameter Name	Parameter ID	Setting Notes
Country	CERTREQ.COUNTRY	-
Organization	CERTREQ.ORG	-
Organizational Unit	CERTREQ.ORGUNIT	-
State or Province	CERTREQ.STATEPROVINCE	-
Locality	CERTREQ.LOCALITY	-
Key Usage	CERTREQ.KEYUSAGE	Selection of at least the DIGITAL_SIGNATURE and KEY_ENCIIPHERMENT options is recommended. If the DIGITAL_SIGNATURE option is selected, the key is used to verify the peer's digital signature during CMPv2-based certificate application or update and SSL authentication. If the KEY_ENCIIPHERMENT option is selected, the key is used to encrypt transmission data during SSL-based key exchange.
Signature Algorithm	CERTREQ.SIGNALG	SHA256 is recommended. Message digest algorithm 5 (MD5) is too weak and is not recommended from a security perspective.
Key Size	CERTREQ.KEYSIZE	-
Local Name	CERTREQ.LOCALNAME	If this parameter is not set, the value of the Common Name field in a certificate is used by default. If this parameter is set, the value of the Local Name field in a certificate must be the same as the value of this parameter.
Local IP	CERTREQ.LOCALIP	The recommended value of this parameter is the public-network virtual IP address of the OMU.

The BSC6900/BSC6910 must be configured with CA server information (as listed in [Table 4-22](#)) to apply for a certificate from the CA server. The data to be prepared for the CA server corresponds to the **CA MO**.

Table 4-22 Data to be prepared for the BSC6900/BSC6910 CA

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME	This parameter indicates the name of the CA on the operator's PKI server. During a CMPv2-based certificate procedure, the value of the Recipient field in a CMPv2 message sent to the CMP server equals the value of this parameter. For details about how to configure this parameter, see Figure 4-24. The following is an example: The Subject name of the certificate used for signing CMP messages on the CA/RA is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com, and the corresponding CANAME is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com. The character type for the C, S, L, O, OU, and CN fields is PRINTABLE. The character type for the E field is IA5. The other character types are not supported. The S field can be replaced with the ST field. For the BSC6900/BSC6910, the S and ST fields are regarded as the same field. For details about the character set of the PRINTABLE type, see RFC 3642.
Certificate Authority URL	CA.URL	Currently, the BSC6900/BSC6910 cannot translate domain names. Therefore, use an IP address instead of a domain name in the URL. By default, the CA uses TCP port 80 for HTTP services and TCP port 443 for HTTPS services. The TCP port number is determined by the CA server. For example, the URL domain name of the CA server can be set to http://10.88.88.88:80/pkix/.
Signature Algorithm	CA.SIGNALG	-

Parameter Name	Parameter ID	Setting Notes
Certificate Fetch Mode	CA.MODE	When this parameter is set to DEFAULT_MODE, the BSC6900/BSC6910 applies for and updates a certificate by using the default IP address 0.0.0.0 as the source IP address for certificate updates. When this parameter is set to CFG_UPD_SIP, the BSC6900/BSC6910 applies for and updates a certificate by using the external virtual IP address as the source IP address for certificate updates.

Figure 4-24 CANAME configuration

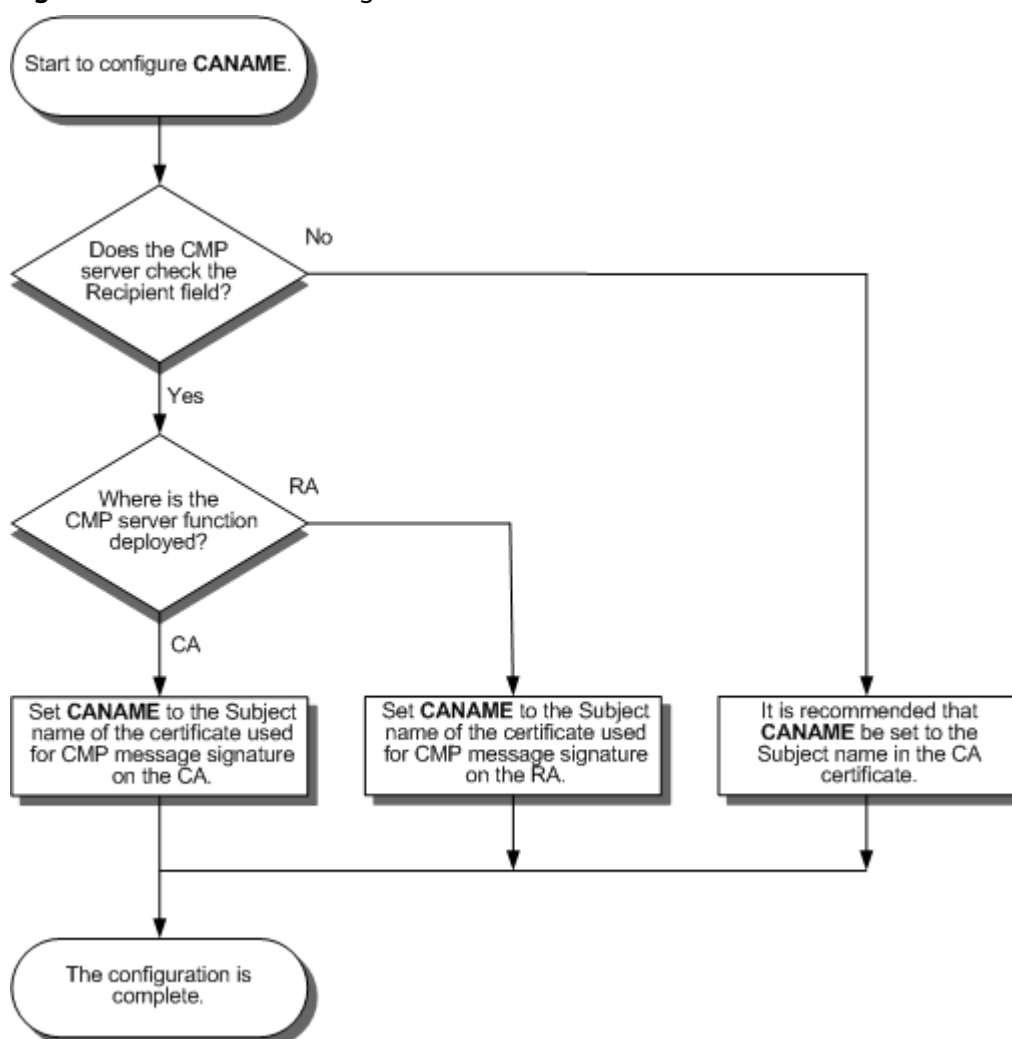


Table 4-23 lists the data to be prepared for a device certificate (corresponding to the **CERTMK MO**).

Table 4-23 Data to be prepared for a device certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT	• If an operator-issued device certificate is used for identity authentication between the BSC6900/BSC6910 and a device such as MAE, two CERTMK MOs must be configured to specify an operator-issued device certificate and a Huawei initial certificate, respectively. This parameter is set to OPKIDevCert.cer and usercert.pem or hwusercert.pem for the two certificates, respectively. • If a Huawei initial certificate is used for identity authentication between the BSC6900/BSC6910 and a device such as MAE, only one CERTMK MO needs to be configured to specify a Huawei initial certificate. This parameter is set to usercert.pem or hwusercert.pem accordingly. Users cannot modify or remove this MO.

Table 4-24 lists the data to be prepared for an application's certificate (corresponding to the **APPCERT** MO). An application's certificate specifies a device certificate that is currently used by a BSC6900/BSC6910.

Table 4-24 Data to be prepared for an application's certificate

Parameter Name	Parameter ID	Setting Notes
Application Type	APPCERT.APPTYPE	This parameter must be set to SSL .

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	APPCERT.APPCERT	• If an operator-issued device certificate is used for identity authentication between the BSC6900/BSC6910 and a device such as MAE, this parameter must be set to OPKIDevCert.cer. • If a Huawei initial certificate is used for identity authentication between the BSC6900/BSC6910 and a device such as MAE, this parameter must be set to usercert.pem or hwusercert.pem.

Table 4-25 lists the data to be prepared for a trust certificate (corresponding to the **TRUSTCERT** MO).

Table 4-25 Data to be prepared for a trust certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	TRUSTCERT.CERTNAME	An operator's trust certificate and a Huawei trust certificate must be configured. For a Huawei trust certificate, set this parameter to rootca.pem on the BSC6900/BSC6910. For an operator's trust certificate, set this parameter to CN.pem when automatic certificate application is used. The value of <i>CN</i> must be the same as that in the Subject field of the trust certificate. If the operator's CA system has a multi-layer structure, a trust certificate chain needs to be configured.

Table 4-26 lists the data to be prepared for a periodic certificate validity check task (corresponding to the **CERTCHKTSK** MO).

Table 4-26 Data to be prepared for a periodic certificate validity check task

Parameter Name	Parameter ID	Setting Notes
Checking Period	CERTCHKTSK.PERIOD	The default value is recommended.
Alarm Threshold	CERTCHKTSK.ALMRNG	The default value is recommended.
Update Method	CERTCHKTSK.UPDATEMETHOD	The value CMP is recommended.
Initial Certificate Expire Threshold	CERTCHKTSK.VENDORALMRNG	The default value is recommended.
Initial Certificate Check Switch	CERTCHKTSK.VENDORCERTCHKSW	If the initial certificate is used, the value ON is recommended. Otherwise, the value OFF is recommended. After a third-party certificate is replaced, you are advised to set this parameter to OFF . Otherwise, related alarms may be reported after the initial certificate expires.

(Optional) Prepare CRL data if the BSC6900/BSC6910 needs to obtain CRL information from the CA. [Table 4-27](#) lists the data to be prepared for a CRL (corresponding to the **CRL MO**).

Table 4-27 Data to be prepared for a CRL

Parameter Name	Parameter ID	Setting Notes
CRL File Name	CRL.CERTNAME	-

(Optional) Prepare data related to CRL usage policies. [Table 4-28](#) lists the data to be prepared for these policies (corresponding to the **CRLPOLICY MO**).

Table 4-28 Data to be prepared for CRL usage policies

Parameter Name	Parameter ID	Setting Notes
CRL Using Policy	CRLPOLICY.CRLPOLICY	Operators can set this parameter based on site requirements. If this parameter is set to NOVERIFY , the revocation of peer certificates cannot be detected in a timely manner. As a result, the revoked certificates may be used by unauthorized users. Therefore, it is recommended that this parameter not be set to NOVERIFY . The value ALARM or DISCONNECT is recommended.

(Optional) Prepare data related to a periodic CRL download task. [Table 4-29](#) lists the data to be prepared for the task (corresponding to the **CRLTSK MO**).

Table 4-29 Data to be prepared for a periodic CRL download task

Parameter Name	Parameter ID	Setting Notes
Task ID	CRLTSK.TSKID	-
IP Address	CRLTSK.IP	Set this parameter to the IP address of the digital certificate and CRL database.
Access Method	CRLTSK.CRLGETMETHOD	The value FTP is recommended.
Port No.	CRLTSK.PORT	-
User Name	CRLTSK.USR	-
Password	CRLTSK.PWD	-
File Name	CRLTSK.FILENAME	-
Using CRL's Next Update	CRLTSK.ISCRLTIME	If this parameter is set to ENABLE , the BSC6900/BSC6910 downloads a CRL when the next update time arrives.
CRL Updating Period	CRLTSK.PERIOD	This parameter must be set when the CRLTSK.ISCRLTIME parameter is set to DISABLE .

Parameter Name	Parameter ID	Setting Notes
Source IP	CRLTSK.SIP	This parameter indicates the source IP address to download a CRL. When the IP address is set to 0.0.0.0, the system automatically uses the OM IP address as the source IP address to obtain the updated CRL from the CRL server.
Distinguish Name	CRLTSK.SEARCHDN	This parameter must be set when the CRLTSK.CRLGETMETHOD parameter is set to LDAP .
Connection Mode	CRLTSK.CONNMODE	This parameter indicates whether to use SSL connections. This parameter takes effect only when the CRLTSK.CRLGETMETHOD parameter is set to LDAP .
Authenticate Peer	CRLTSK.AUTHPEER	This parameter indicates whether to authenticate the peer certificate when the SSL connections are used. This parameter takes effect only when the CRLTSK.CRLGETMETHOD parameter is set to LDAP. If this parameter is set to authenticate the peer certificate, ensure that the NEs and CRL server have been correctly configured with the CA trust certificates and device certificates.

(Optional) Prepare data to manually download an operator's root certificate or CRL from an FTP server. [Table 4-30](#) lists the data to be prepared for downloading a certificate file (corresponding to the **DLD CERTFILE** command).

Table 4-30 Data to be prepared for downloading a certificate file

Parameter Name	Parameter ID	Setting Notes
FTP Server IP	IP	-
User Name	USR	-
Password	PWD	-
Source File Name	SRCF	-
Destination File Name	DSTF	It is recommended that this parameter be set to the same value as SRCF .

Parameter Name	Parameter ID	Setting Notes
Guage Option	GA	This parameter determines whether to report the progress of file downloading. The value Yes is recommended.
Certificate Type	CT	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate application. [Table 4-31](#) lists the data to be prepared for applying for a device certificate (corresponding to the **REQ DEVCERT** command) based on CMPv2.

Table 4-31 Data to be prepared for applying for a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME	-
Certificate File Name	CERTMK.APPCERT	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate update. [Table 4-32](#) lists the data to be prepared for updating a device certificate (corresponding to the **UPD DEVCERT** command) based on CMPv2.

Table 4-32 Data to be prepared for updating a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT	This parameter specifies a certificate to be updated.
Renew Key	REKEY	The value YES is recommended.

4.4.4.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

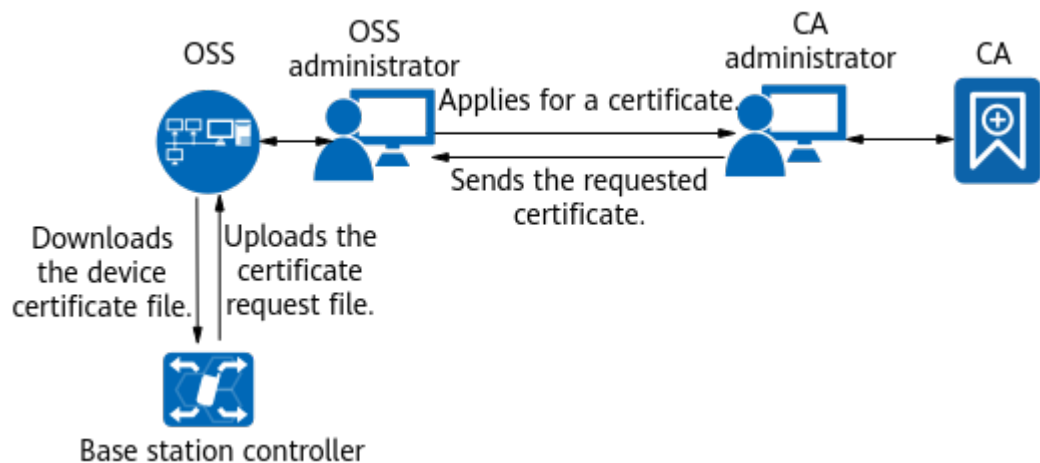
Prerequisites

- The operator has deployed a PKI system.
- The route from the operator's PKI system to the external IP address of the BSC6900/BSC6910 is reachable.
- Huawei root certificate is configured as the trust certificate on the operator's CA server. If the operator's CA servers work in the active/standby mode, Huawei root certificate is configured as the trust certificates on both the active and standby CA servers.
- The BSC6900/BSC6910 has been connected to the MAE.
- The OMU board of the BSC6900/BSC6910 has a unique device certificate that can identify its identity.
- The operator has been granted system administrator privileges.

Activation

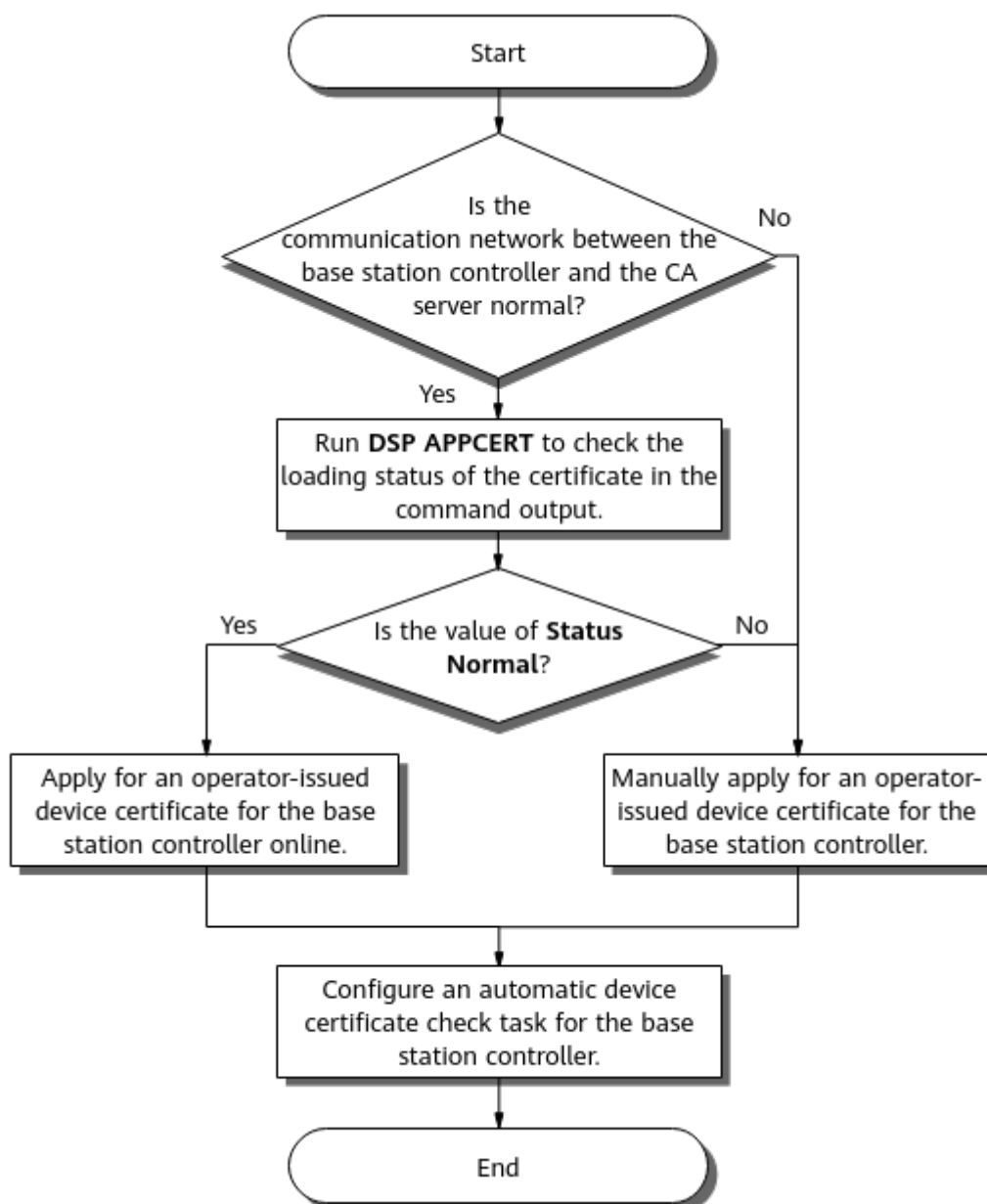
[Figure 4-25](#) shows the procedure for manually applying for a device certificate on the BSC6900/BSC6910.

Figure 4-25 Manual certificate application for the BSC6900/BSC6910



[Figure 4-26](#) shows the steps to activate an operator-issued device certificate on the BSC6900/BSC6910.

Figure 4-26 Procedure for connecting the BSC6900/BSC6910 to a PKI system



Step 1 Apply for a device certificate issued by the operator's CA for the BSC6900/BSC6910.

- Manually apply for an operator-issued device certificate for the BSC6900/BSC6910.
 - a. Run the **MOD CERTREQ** command to set parameters related to the certificate request template. The SHA1 algorithm has security risks. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.
 - b. Run the **CRE CERTREQFILE** command to generate a certificate request file.
 - c. Obtain a certificate request file from the BSC6900/BSC6910 and send the file to the MAE to apply for a digital certificate. You can obtain the

certificate request file from the BSC6900/BSC6910 using either of the following methods:

- Use the file manager of the LMT to download the certificate request file from the **ftp/cert/CertApplyFile** directory of the OMU active workspace.
 - Run the **ULD CERTFILE** command to upload the certificate request file to the FTP server, and then download the file from the FTP server to a local PC.
- d. The OSS administrator applies for a certificate from the CA administrator. The OSS administrator manually submits the certificate request file to the CA administrator for an operator-issued device certificate. Then, the CA administrator manually returns the operator-issued device certificate to the OSS administrator.
- e. Transfer the operator's CA certificate or certificate chain file to the BSC6900/BSC6910. You can use either of the following methods:
- Use the file manager of the LMT to upload the CA certificate or certificate chain file to the **ftp/cert/CACertFile** directory of the OMU active workspace.
 - Upload the CA certificate or certificate chain file to the FTP server. Then, run the **DLD CERTFILE** command to download the CA certificate or certificate chain file from the FTP server to the BSC6900/BSC6910.
- f. Run the **ADD TRUSTCERT** command to add the operator's CA certificate or certificate chain file.
- g. Transfer the device certificate file to the BSC6900/BSC6910. You can use either of the following methods:
- Use the file manager of the LMT to upload the device certificate file to the **ftp/cert/CertFile** directory of the OMU active workspace.
 - Upload the device certificate file to the FTP server. Then, run the **DLD CERTFILE** command to download the device certificate file from the FTP server to the BSC6900/BSC6910.
- h. Run the **ADD CERTMK** command to add the device certificate to the device.
- i. On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management**. In the certificate management window, select the NE from the navigation tree on the left. On the displayed device certificate management page, select the assigned operator's certificate and click **Test** to check whether an SSL connection to the MAE can be established.
- Bidirectional authentication is used for SSL certificate testing. That is, the BSC6900/BSC6910 and MAE authenticate the device certificates of each other. The SSL certificate testing result reflects whether the certificates can be used.
- j. Run the **MOD APPCERT** command to set the device certificate file for an application.

- Apply for an operator-issued device certificate for the BSC6900/BSC6910 online.
 - a. Run the **MOD CERTREQ** command to set parameters related to the certificate request template. The SHA1 algorithm has security risks. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.
 - b. Run the **ADD CA** command to add a CA server. The SHA1 algorithm has security risks. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.

The **CA.CANAME** parameter needs to be configured according to [Figure 4-24](#). The CA URL is the URL used by the operator to provide services and needs to be provided by the CA administrator.
 - c. Transfer the operator's CA certificate or certificate chain file to the BSC6900/BSC6910. You can use either of the following methods:
 - Use the file manager of the LMT to upload the CA certificate or certificate chain file to the **ftp/cert/CACertFile** directory of the OMU active workspace.
 - Upload the CA certificate or certificate chain file to the FTP server. Then, run the **DLD CERTFILE** command to download the CA certificate or certificate chain file from the FTP server to the BSC6900/BSC6910.
 - d. Run the **ADD TRUSTCERT** command to add the operator's CA certificate or certificate chain file.
 - e. Run the **REQ DEVCERT** command with information required for applying for an operator-issued device certificate specified to trigger a certificate application procedure.

If the certificate application succeeds, the output of the **REQ DEVCERT** command will indicate successful execution. In this case, the **DSP CERTMK** command can be executed to query whether a certificate has been successfully applied for.

The CA name can be queried by running the **LST CA** command.
 - f. On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management**. In the certificate management window, select the NE from the navigation tree on the left. On the displayed device certificate management page, select the assigned operator's certificate and click **Test** to check whether an SSL connection to the MAE can be established.

Bidirectional authentication is used for SSL certificate testing. That is, the BSC6900/BSC6910 and MAE authenticate the device certificates of each other. The SSL certificate testing result reflects whether the certificates can be used.
 - g. Run the **MOD APPCERT** command to set the device certificate file for an application.

Step 2 Configure an automatic device certificate check task for the BSC6900/BSC6910.

1. Run the **SET CERTCHKTSK** command to set a periodic certificate validity check task.

2. (Optional) Transfer the CRL to the BSC6900/BSC6910. You can use either of the following methods:
 - Use the file manager of the LMT to upload the CRL to the **ftp/cert/CRL** directory of the OMU active workspace.
 - Upload the CRL to the FTP server. Then, run the **DLD CERTFILE** command to download the CRL file from the FTP server to the BSC6900/BSC6910.
3. (Optional) Run the **ADD CRL** command to add a CRL.
4. (Optional) Run the **SET CRLPOLICY** command to set a CRL usage policy.

 NOTE

If **CRLPOLICY.CRLPOLICY** is set to **NOVERIFY**, the revocation of peer certificates cannot be detected in a timely manner. As a result, the revoked certificates may be used by unauthorized users. Therefore, it is not recommended that this parameter be set to **NOVERIFY**. The value **ALARM** or **DISCONNECT** is recommended.

5. (Optional) Run the **ADD CRLTSK** command to add a periodic CRL download task.

----End

In addition to the preceding steps, perform the following step to manually trigger a certificate update:

- Step 1** (Optional) Obtain a new subordinate CA certificate from the CA server.
- Step 2** (Optional) Transfer the new subordinate CA certificate to the BSC6900/BSC6910. You can use either of the following methods:
- Use the file manager of the LMT to upload the subordinate CA certificate file to the **ftp/cert/CACertFile** directory of the OMU active workspace.
 - Upload the subordinate CA certificate file to the FTP server. Then, run the **DLD CERTFILE** command to download the subordinate CA certificate file from the FTP server to the BSC6900/BSC6910.
- Step 3** (Optional) Run the **ADD TRUSTCERT** command to add the new subordinate CA certificate of the BSC6900/BSC6910.
- Step 4** Run the **UPD DEVCERT** command to set information about a certificate update. After the setting takes effect, a CMPv2-based certificate update procedure is triggered.
- Step 5** (Optional) Run the **RMV TRUSTCERT** command to remove the old subordinate CA certificate of the BSC6900/BSC6910.

----End

Activation Command Examples

The following is an MML command example of how to activate an operator-issued device certificate on the BSC6900/BSC6910.

```
//Modifying configurations of a certificate request template
MOD CERTREQ: COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="cn", ORG="ITEF",
ORGUNIT="Hw", STATEPROVINCENAME="sc", LOCALITY="cd",
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,
SIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048, LOCALNAME="abcdefghijklmn.huawei.com",
LOCALIP="10.120.20.188";
//Setting the CA server information
```

```
//If the CA server is reachable only through a public network, you are advised to set the public-network
virtual IP address of the BSC6900/BSC6910 for certificate application and update. The following is an
example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=CFG_UPD_SIP;
//Setting information required for the BSC6900/BSC6910 to apply for an operator-issued device certificate
based on CMPv2 when the application needs to be manually triggered
REQ DEVCERT: CANAME="C=AU, S=Some-State, O=Internet Widgits Pty Ltd, CN=eca1",
APPCERT="OPKIDevCert.cer";
//Modifying configurations of a certificate in use
MOD APPCERT: APPTYPE=SSL, APPCERT="OPKIDevCert.cer";
//Setting a periodic certificate validity check task with Initial Certificate Verification Switch set to OFF
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDATEMETHOD=CMP, VENDORALMRNG=180,
VENDORCERTCHKSW=OFF;
//(Optional) Downloading the CRL from the FTP server (If the FTP server is deployed on the MAE, the IP
address of the FTP server is the same as that of the MAE.)
DLD CERTFILE: CT=CRL, SRCF="bsc.crl", DSTF="bsc.crl", IP="10.120.86.86", USR="admin";
//(Optional) Loading the CRL
ADD CRL: CERTNAME="bsc.crl";
//(Optional) Setting a CRL usage policy
SET CRLPOLICY: CRLPOLICY= ALARM;
//(Optional) Adding a task of periodically downloading the CRL
ADD CRTSK: TSKID=0, IP="10.120.86.86", CRLGETMETHOD=LDAP, USR="admin", PWD="*****",
FILENAME="bsc.crl", ISCRITIME=DISABLE, PERIOD=24;
//Adding the following configuration in the case of a manual certificate update
UPD DEVCERT: APPCERT="OPKIDevCert.cer", REKEY=YES;
```

Deactivation

- Step 1** Run the **RMV CA** command to remove the CA.
- Step 2** (Optional) Run the **RMV CRTSK** command to remove the task of periodically updating a CRL.
- End

4.4.4.3 Activation Verification

- Step 1** Check the status of device certificates.

Run the MML command **DSP APPCERT** and check the values of the **Certificate File Name**, **Issuer**, **Common Name**, and **Status** parameters in the query result. If the values of **Certificate File Name**, **Issuer**, and **Common Name** are correct and the value of **Status** is **Normal**, the device certificate has been loaded to the BSC6900/BSC6910.

The following is an example.

```

+++   GUAN_JI      2013-09-06 16:02:03
O&M   #27
%%DSP APCERT:;%
RETCODE = 0   Execution succeeded.

Display Appcert
-----
      Application Type   = SSL
Certificate File Name   = usercert.pem
      Issuer             = Huawei Wireless Network Product CA
      Common Name        = MBSC Certificate
      Serial Number      = 76:41:2d:00:1e:25:60:8f:a6:d6:c2:e5:d1:33:81:75:55
      Effective Date     = 2012-03-21 14:55:48
      Expiration Date    = 2027-03-18 14:55:48
      Status             = Normal
      (Number of results = 1)

---   END

```

Step 2 Check the status of trust certificates.

Run the MML command **DSP TRUSTCERT** and check the value of **Status** in the query result. If **Normal** is displayed, the trust certificate has been loaded to the BSC6900/BSC6910.

The following is an example.

```

+++   GUAN_JI      2013-09-06 16:05:21
O&M   #48
%%DSP TRUSTCERT:;%
RETCODE = 0   Execution succeeded.

Display CA Certificates or Chain's Details
-----
Certificate File Name   = rootca.pem
      Serial Number      = 76:25:60:02:2a:47:5e:ec:a6:2b:7c:68:ac:9b:3f:29:66
      Common Name        = Huawei Equipment CA
      Issuer Name         = Huawei Equipment CA
      Effective Date      = 2011-12-06 15:34:23
      Expiring Date       = 2041-11-28 15:34:23
      Status              = Normal

Certificate File Name   = rootca.pem
      Serial Number      = 76:20:c1:80:f9:d2:0b:a2:a1:12:99:07:38:38:ff:79:62
      Common Name        = Huawei Wireless Network Product CA
      Issuer Name         = Huawei Equipment CA
      Effective Date      = 2011-12-06 16:09:59
      Expiring Date       = 2036-11-29 16:09:59
      Status              = Normal
      (Number of results = 2)

---   END

```

Step 3 (Optional) Check the CRL status.

Run the MML command **DSP CRL** and check the value of **Status** in the query result. If **Normal** is displayed, the CRL has been loaded to the BSC6900/BSC6910.

The following is an example.

```
+++ huawei_207 2013-11-20 19:17:02
OAM #20359
%%DSP CRL:%%
RETCODE = 0 Execution succeeded.

Display CRL's Details
-----
CRL File Name Issuer This Update Next Update Status CRL Server IP Address Last Used Last Update Time of CRL
-----
12q.crl HWSUBCA 2013-11-18 01:44:18 2013-12-03 14:25:00 Normal 10.144.242.154 2013-11-19 19:43:17
aa.crl HWSUBCA 2013-11-18 01:44:18 2013-12-03 14:25:00 Normal 10.144.242.154 2013-11-20 10:06:39
jkl <NULL> <NULL> <NULL> File does not exist <NULL> <NULL>
(Number of results = 3)

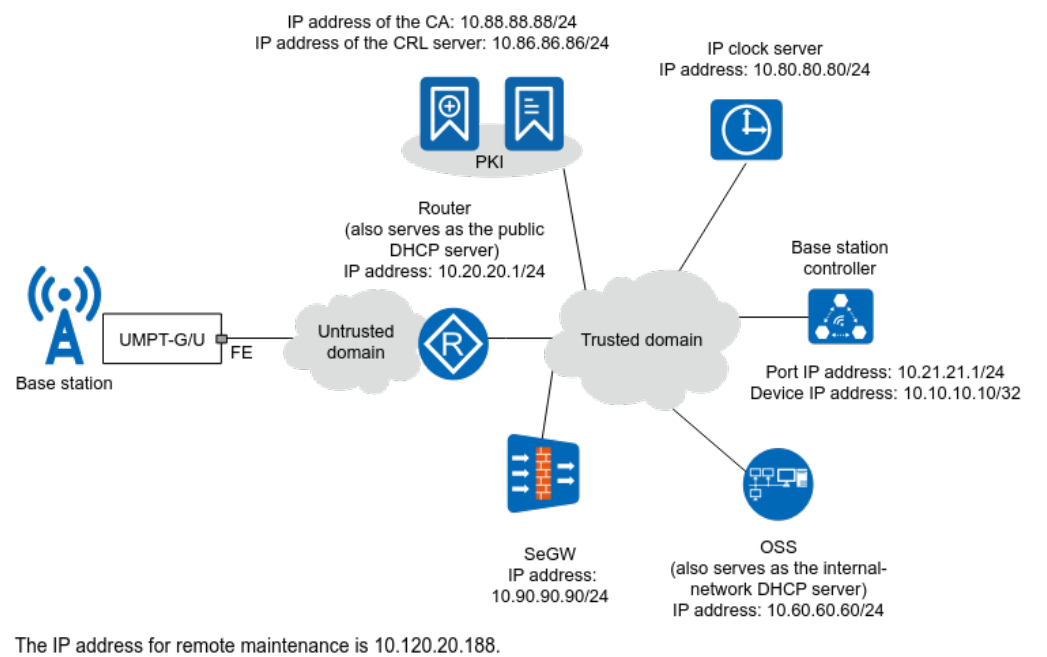
END
```

----End

4.4.5 Deployment of PKI on the BSC6960 (GSM/UMTS)

This section uses the networking illustrated in [Figure 4-27](#) as an example to describe how to deploy the PKI feature on the BSC6960.

Figure 4-27 Example of the secure networking for the BSC6960



4.4.5.1 Data Preparation

"-" in the following tables in this section indicates that there is no special requirement for parameter setting. Set the parameters according to network plans.

[Table 4-33](#) lists the data to be prepared for a certificate request template of the BSC6960 (corresponding to the CERTREQ MO).

Table 4-33 Data to be prepared for a certificate request template

Parameter Name	Parameter ID	Setting Notes
Common Name Additional Info.	CERTREQ.USERADDINFO	-
Country	CERTREQ.COUNTRY	-
Organization	CERTREQ.ORG	-
Organizational Unit	CERTREQ.ORGUNIT	-
State or Province	CERTREQ.STATEPROVINCENAME	-
Locality	CERTREQ.LOCALITY	-
Key Usage	CERTREQ.KEYUSAGE	It is recommended that at least the DIGITAL_SIGNATURE option be selected. A digital signature is used to verify the peer's digital signature during CMPv2-based certificate application and update and SSL authentication.
Signature Algorithm	CERTREQ.SIGNALG	The value SHA256 is recommended.
Key Size	CERTREQ.KEYSIZE	This parameter is valid only when the CERTREQ.KEYTYPE parameter is set to RSA . You are advised to set this parameter to KEYSIZE2048 .
Key Type	CERTREQ.KEYTYPE	An ECDSA digital certificate features fast computing, a short key length, and a small space for storage. If all the operator's devices support ECDSA digital certificates, it is recommended that this parameter be set to ECDSA .
ECC Curve	CERTREQ.ECCCURVE	If this parameter is set to P256 , the security strength meets the security requirements. If the operator's devices support the P384 and ED25519 algorithms, P384 or ED25519 can also be used.

The BSC6960 must be configured with CA server information to apply for a certificate from the CA server. [Table 4-34](#) lists the data to be prepared for the CA server (corresponding to the **CA MO**).

Table 4-34 Data to be prepared for the CA

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME	This parameter indicates the name of the CA. The Common Name field of the CA name cannot contain any of the following characters: backslash (\), slash (/), colon (:), asterisk (*), question mark (?), double quotation marks ("), left angle bracket (<), right angle bracket (>), and bar (). Otherwise, an error occurs when you run the REQ DEVCERT command to apply for a device certificate. For example: The Subject name of the certificate used for signing CMP messages on the CA/RA is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com, and the corresponding CANAME is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com.
Certificate Authority URL	CA.URL	Currently, the BSC6960 cannot translate domain names. Therefore, use an IP address instead of a domain name in the URL. The TCP port number is determined by the CA server. For example, the URL domain name of the CA server can be set to <code>http://10.88.88.88:80/pkix/</code>. For an IPv6 URL address, the format "[<i>IP address</i>]:<i>Port Number</i>" must be used to separate the IP address from the port number, for example, <code>https://[2050::20:23:123]:34481/pkix/</code>.
Signature Algorithm	CA.SIGNALG	The value SHA256 is recommended.

Parameter Name	Parameter ID	Setting Notes
Certificate Registration Name	CA.RANAME	Set this parameter if the customer's certificate registration server requires that the value of the recipient field in the received CMPv2-based certificate message be the name of the certificate registration server.
Certificate Request Switch	CA.CERTREQSW	Set this parameter to DEFAULT .
Key Type	CA.KEYTYPE	An ECDSA digital certificate features fast computing, a short key length, and a small space for storage. If all the operator's devices support ECDSA digital certificates, it is recommended that this parameter be set to ECDSA .
ECC Curve	CA.ECCCURVE	If this parameter is set to P256 , the security strength meets the security requirements. The recommended value is therefore P256 . If the operator's devices support the P384 and ED25519 algorithms, P384 or ED25519 can also be used.
Certificate Request Signature Algorithm	CA.CERTREQSIGNALG	The value SHA256 is recommended.
Key Size	CA.KEYSIZE	This parameter is valid only when the CA.KEYTYPE parameter is set to RSA .
Certificate File Name	CA.CERTNAME	After an alarm indicating that a Huawei initial certificate is about to expire is reported, an operator-issued device certificate must be configured before the certificate expires.
Trust Group ID	CA.TRUSTGROUPID	This parameter specifies the ID of a trust group. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust certificate group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

Figure 4-28 CANAME configuration

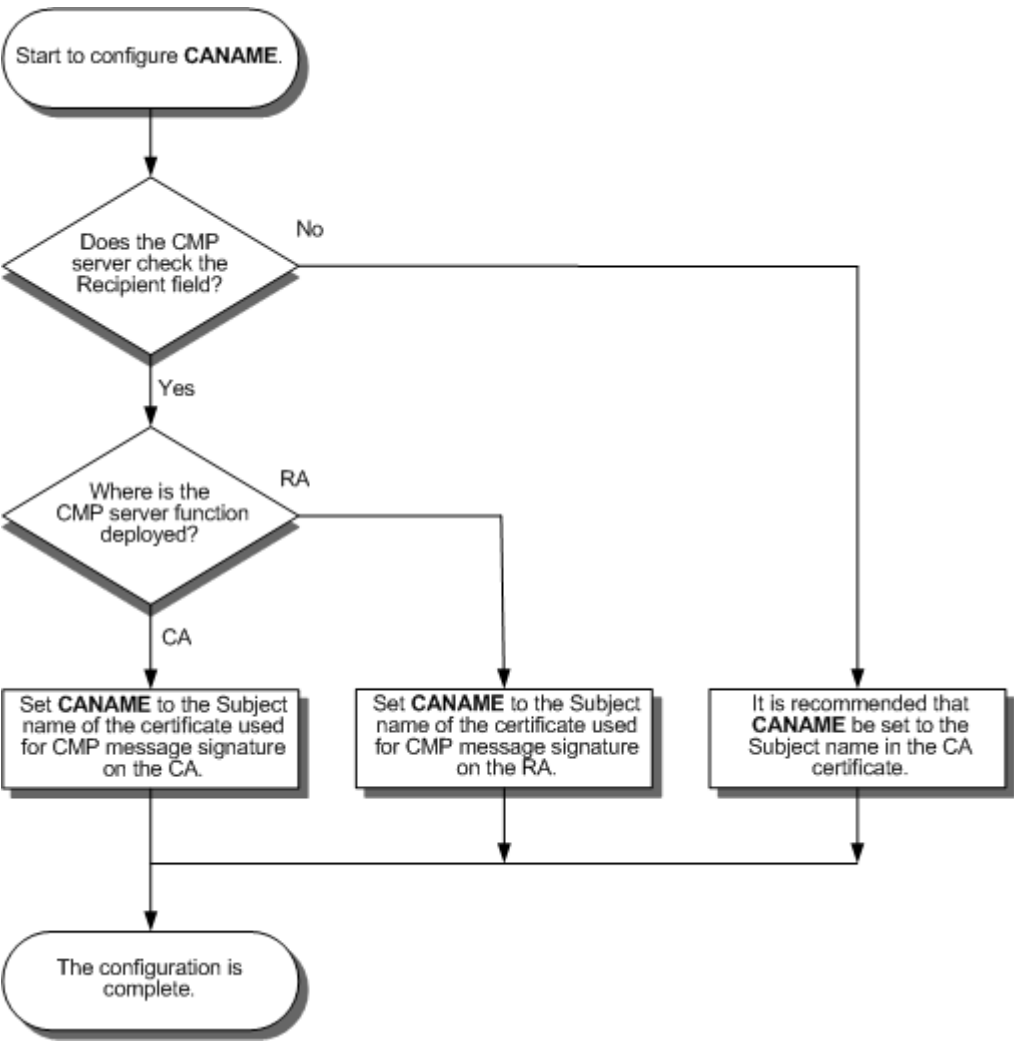


Table 4-35 lists the data to be prepared for a device certificate (corresponding to the CERTMK MO).

Table 4-35 Data to be prepared for a device certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT	• If an operator-issued device certificate is used for identity authentication between the BSC6960 and a device such as MAE, two CERTMK MOs must be configured to specify an operator-issued device certificate and a Huawei initial certificate, respectively. This parameter is set to OPKIDevCert.cer and appcert.pem for the two certificates, respectively. • If a Huawei initial certificate is used for identity authentication between the BSC6960 and a device such as MAE, only one CERTMK MO needs to be configured to specify a Huawei initial certificate. This parameter is set to appcert.pem accordingly. Users cannot modify or remove this MO.

Table 4-36 lists the data to be prepared for a device certificate in use.

Table 4-36 Data to be prepared for a device certificate in use

Parameter Name	Parameter ID	Setting Notes
Application Type	APPCERT.APPTYPE	This parameter must be set to SSL .
Certificate File Name	APPCERT.APPCERT	The BSC6960 has no special configuration requirements for this parameter.

Table 4-37 lists the data to be prepared for a trust certificate (corresponding to the **TRUSTCERT** MO).

Table 4-37 Data to be prepared for a trust certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	TRUSTCERT.CERTNAME	An operator's trust certificate and a Huawei trust certificate must be configured. For a Huawei trust certificate, set this parameter to caroot.pem for the BSC6960. For an operator's trust certificate, this parameter is set to CN.cer when automatic certificate application is used. The value of <i>CN</i> must be the same as that in the Subject field of the trust certificate. If the operator's CA system has a multi-layer structure, a trust certificate chain needs to be configured. If the certificates at different layers of CAs in the certificate chain are separately stored, run the ADD TRUSTCERT command to add certificates at each layer.

(Optional) [Table 4-38](#) lists the data to be prepared for a trust group (corresponding to the **TRUSTGROUP MO**).

Table 4-38 Data to be prepared for a trust group

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTGROUP.TRUSTGROUPID	-
Trust Group Description	TRUSTGROUP.TRUSTGROUPDESC	-

(Optional) [Table 4-39](#) lists the data to be prepared for the association between a trust group and a trust certificate (corresponding to the **TRUSTCERT2GROUP MO**).

Table 4-39 Data to be prepared for associating a trust group with a trust certificate

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTCERT2GROUP. TRUSTGROUPID	-
Certificate File Name	TRUSTCERT2GROUP. CERTNAME	-

Table 4-40 lists the data to be prepared for a periodic certificate validity check task (corresponding to the **CERTCHKTSK** MO).

Table 4-40 Data to be prepared for a periodic certificate validity check task

Parameter Name	Parameter ID	Setting Notes
Checking Period	CERTCHKTSK.PERIOD	The default value is recommended.
Alarm Threshold	CERTCHKTSK.ALMRNG	The default value is recommended.
Update Method	CERTCHKTSK.UPDATEMETHOD	The value CMP is recommended.
Automatic Certificate Reapplication Switch	CERTCHKTSK.AUTOREAPPLYSW	The value ON is recommended.
Initial Certificate Expire Threshold	CERTCHKTSK.VENDORALMRNG	The default value is recommended.
Initial Certificate Verification Switch	CERTCHKTSK.VENDORCERTCHKSW	If the initial certificate is used, the value ON is recommended. Otherwise, the value OFF is recommended. After a third-party certificate is replaced, you are advised to set this parameter to OFF . Otherwise, related alarms may be reported after the initial certificate expires.

(Optional) Prepare CRL data if the BSC6960 needs to obtain CRL information from the CA. **Table 4-41** lists the data to be prepared for a CRL (corresponding to the **CRL** MO).

Table 4-41 Data to be prepared for a CRL

Parameter Name	Parameter ID	Setting Notes
CRL File Name	CRL.CERTNAME	-

(Optional) Prepare data related to CRL usage policies. [Table 4-42](#) lists the data to be prepared for these policies (corresponding to the **CRLPOLICY** MO).

Table 4-42 Data to be prepared for CRL usage policies

Parameter Name	Parameter ID	Setting Notes
CRL Using Policy	CRLPOLICY.CRLPOLICY	-

(Optional) Prepare data related to a periodic CRL download task. [Table 4-43](#) lists the data to be prepared for the task (corresponding to the **CRLTSK** MO).

Table 4-43 Data to be prepared for a periodic CRL download task

Parameter Name	Parameter ID	Setting Notes
IP Address	CRLTSK.IP	Set this parameter to the IP address of the CRL server.
User Name	CRLTSK.USR	-
Password	CRLTSK.PWD	-
File Name	CRLTSK.FILENAME	-
Using CRL's Next Update	CRLTSK.ISCRITIME	If this parameter is set to ENABLE , the BSC6960 downloads a CRL when the next update time arrives.
CRL Updating Period	CRLTSK.PERIOD	This parameter must be set when the CRLTSK.ISCRITIME parameter is set to DISABLE .
Access Method	CRLTSK.CRLGETMETHOD	The value LDAP is recommended. Set this parameter to FTP only when the peer device does not support LDAP.
Distinguish Name	CRLTSK.SEARCHDN	This parameter must be set when the CRLTSK.CRLGETMETHOD parameter is set to LDAP .

Parameter Name	Parameter ID	Setting Notes
Port No.	CRLTSK.PORT	This parameter must be set when the CRLTSK.CRLGETMETHOD parameter is set to LDAP .
Task ID	CRLTSK.TSKID	-
Source IP	CRLTSK.SIP	If this parameter is not set, the BSC6960 uses the OM IP address as the source IP address to update a CRL.
Certificate File Name	CRLTSK.CERTNAME	This parameter specifies a device certificate name. This parameter specifies that the device certificate configured in the CERTMK MO is used for authentication when a TLS connection is set up between the local end and the FTP or LDAP server. If this parameter is not specified, the system uses the preconfigured Huawei initial certificate for authentication.
Trust Group ID	CRLTSK.TRUSTGROU PID	This parameter specifies the ID of a trust group. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust certificate group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

(Optional) Prepare data to manually download an operator's root certificate or CRL from an FTP server. [Table 4-44](#) lists the data to be prepared for downloading a certificate file (corresponding to the **DLD CERTFILE** command).

Table 4-44 Data to be prepared for downloading a certificate file

Parameter Name	Parameter ID	Setting Notes
FTP Server IP	IP	-
User Name	USR	-
Password	PWD	-

Parameter Name	Parameter ID	Setting Notes
Source File Name	<i>SRCF</i>	-
Destination File Name	<i>DSTF</i>	It is recommended that this parameter be set to the same value as <i>SRCF</i> .
Guage Option	<i>GA</i>	This parameter determines whether to report the progress of file downloading. The value Y is recommended.
Certificate Type	<i>CT</i>	-

 NOTE

The file path of the FTP server connected to the BSC6960 must be in English. That is, the values of **SRCF** in **DLD CERTFILE** and **DSTF** in **ULD CERTFILE** must be in English.

(Optional) Prepare data to manually trigger a CMPv2-based certificate application. [Table 4-45](#) lists the data to be prepared for applying for a device certificate (corresponding to the **REQ DEVCERT** command) based on CMPv2.

Table 4-45 Data to be prepared for applying for a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	<i>CA.CANAME</i>	-
Certificate File Name	<i>CERTMK.APPCERT</i>	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate update. [Table 4-46](#) lists the data to be prepared for updating a device certificate (corresponding to the **UPD DEVCERT** command) based on CMPv2.

Table 4-46 Data to be prepared for updating a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	<i>CERTMK.APPCERT</i>	This parameter specifies a certificate to be updated.

Parameter Name	Parameter ID	Setting Notes
Renew Key	<i>REKEY</i>	The value Yes is recommended.

4.4.5.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.4.1 When to Use](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Prerequisites

- The operator has deployed a PKI system.
- The route from the operator's PKI system to the external IP address of the BSC6960 is reachable.
- Huawei root certificate is configured as the trust certificate on the operator's CA server. If the operator's CA servers work in the active/standby mode, Huawei root certificate is configured as the trust certificates on both the active and standby CA servers.
- The BSC6960 has been connected to the MAE.
- The CPMU board of the BSC6960 has a unique device certificate that can identify its identity.
- The operator has been granted system administrator privileges.

Activation

[Figure 4-29](#) shows the procedure for manually applying for a device certificate on the BSC6960.

Figure 4-29 Manual certificate application for the BSC6960

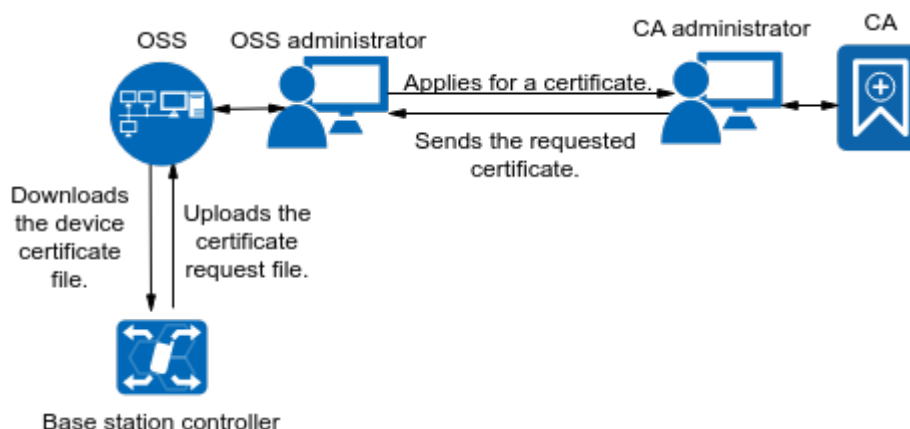
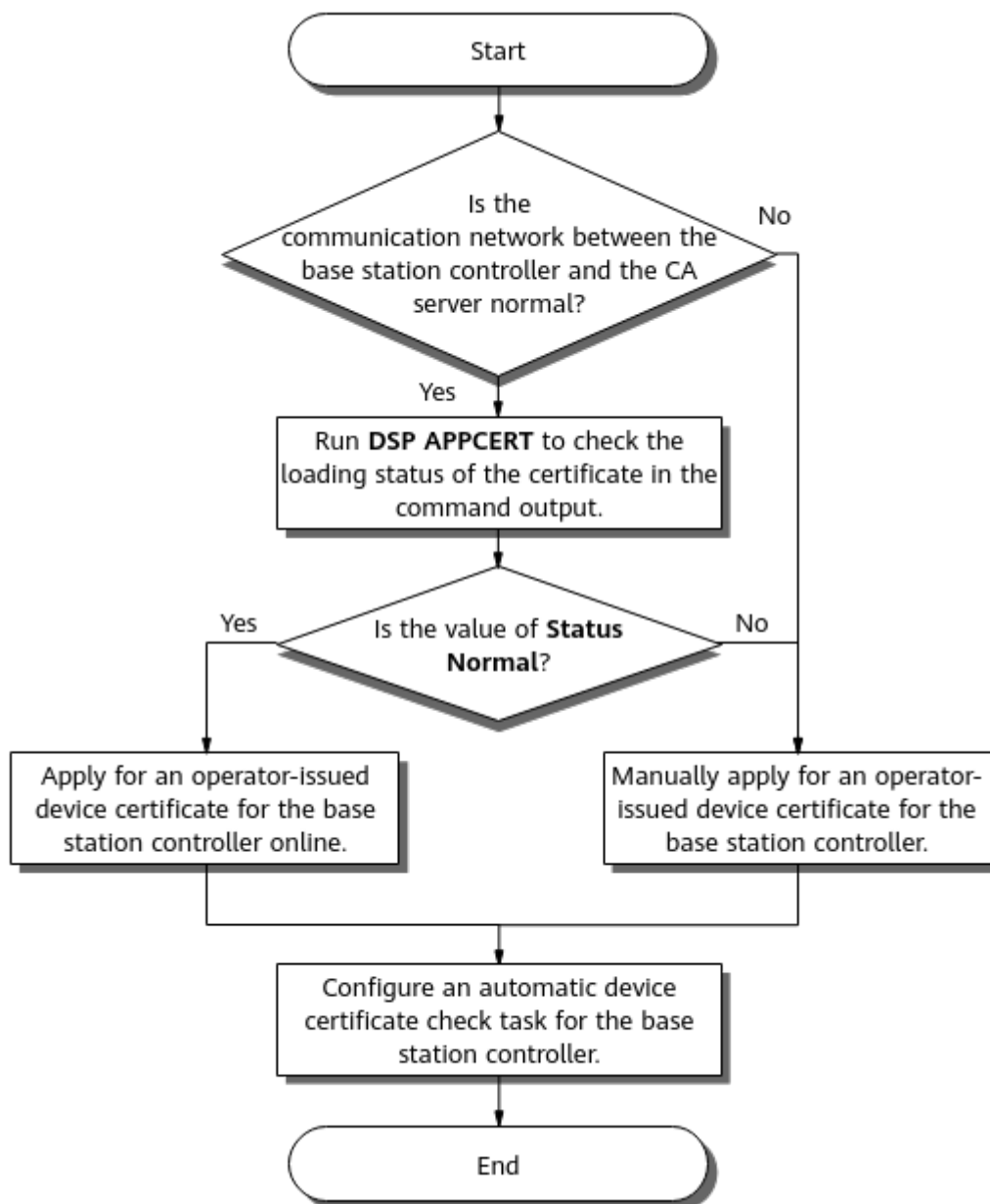


Figure 4-30 shows the steps to activate an operator-issued device certificate on the BSC6960.

Figure 4-30 Procedure for connecting the BSC6960 to a PKI system



Step 1 Apply for a device certificate issued by the operator's CA for the BSC6960.

- Manually apply for an operator-issued device certificate for the BSC6960.
 - a. Transfer the operator's CA certificate or certificate chain file to the BSC6960. You can use the following method:
Upload the CA certificate or certificate chain file to the FTP server. Then, run the **DLD CERTFILE** command to download the CA certificate or certificate chain file from the FTP server to the BSC6960.
 - b. Run the **ADD TRUSTCERT** command to add the operator's CA certificate or certificate chain file.

- c. Run the **MOD CERTREQ** command to set parameters related to the certificate request template. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.
- d. Run the **CRE CERTREQFILE** command to generate a certificate request file.
- e. Obtain a certificate request file from the BSC6960 and send the file to the CA server to apply for a digital certificate. You can obtain the certificate request file from the BSC6960 using the following method:
Run the **ULD CERTFILE** command to upload the certificate request file to the FTP server, and then download the file from the FTP server to a local PC.
- f. Transfer the device certificate file to the BSC6960. You can use the following method:
Upload the device certificate file to the FTP server. Then, run the **DLD CERTFILE** command to download the device certificate file from the FTP server to the BSC6960.

 CAUTION

Names of device certificate files uploaded to the active and standby boards must be the same.

- g. Run the **ADD CERTMK** command to add the device certificate to the device.
- h. On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management**. In the certificate management window, select the NE from the navigation tree on the left. On the displayed device certificate management page, select the assigned operator's certificate and click **Test** to check whether an SSL connection to the MAE can be established.
Bidirectional authentication is used for SSL certificate testing. That is, the BSC6960 and MAE authenticate the device certificates of each other. The SSL certificate testing result reflects whether the certificates can be used.
- i. After the operations on the active board are finished, configure the standby board. Once you have confirmed that the services are not impacted, run the **SWP BRD** command with **Forced Switchover Flag** set to **NO** to switch over the boards. When the switchover is successfully completed, perform step [Step 1.d](#) to step [Step 1.f](#) on the new active board.
- j. Run the **MOD CERTMK** command on the new active board to make the uploaded **CERTMK** take effect. Set **CERTMK.APPCERT** to the name of the uploaded device file and retain the values of other parameters.
- k. Perform step [Step 1.h](#).
- l. Once you have confirmed that the services are not impacted, run the **SWP BRD** command with **Forced Switchover Flag** set to **NO** to switch over the boards.
- m. After the board switchover is successful and services are normal, perform step [Step 1.h](#) to test the certificate validity.

- n. After the certificate validity test is successful, run the **MOD APPCERT** command to set the certificate information.
- Apply for an operator-issued device certificate for the BSC6960 online.
 - a. Run the **MOD CERTREQ** command to set parameters related to the certificate request template. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.
 - b. Run the **ADD CA** command to add a CA server. If the CA supports the SHA256 algorithm, you are advised to use the SHA256 algorithm as the signature algorithm.

The **CA.CANAME** parameter needs to be configured according to [Figure 4-28](#). The CA URL is the URL used by the operator to provide services and needs to be provided by the CA administrator.

- c. Transfer the operator's CA certificate or certificate chain file to the BSC6960. You can use the following method:

Upload the CA certificate or certificate chain file to the FTP server. Then, run the **DLD CERTFILE** command to download the CA certificate or certificate chain file from the FTP server to the BSC6960.

- d. Run the **ADD TRUSTCERT** command to add the operator's CA certificate or certificate chain file.
- e. Run the **REQ DEVCERT** command with information required for applying for an operator-issued device certificate specified to trigger a certificate application procedure.

If the certificate application succeeds, the output of the **REQ DEVCERT** command will indicate successful execution. In this case, the **DSP CERTMK** command can be executed to query whether a certificate has been successfully applied for.

The CA name can be queried by running the **LST CA** command.

 CAUTION

The certificate file name specified by **CERTMK.APPCERT** in the **REQ DEVCERT** command must be the same for the active and standby boards.

- f. On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management**. In the certificate management window, select the NE from the navigation tree on the left. On the displayed device certificate management page, select the assigned operator's certificate and click **Test** to check whether an SSL connection to the MAE can be established.

Bidirectional authentication is used for SSL certificate testing. That is, the BSC6960 and MAE authenticate the device certificates of each other. The SSL certificate testing result reflects whether the certificates can be used.
- g. After the operations on the active board are finished, configure the standby board. Then, run the **SWP BRD** command with **Forced Switchover Flag** set to **NO** to switch over the active and standby boards.
- h. After the switchover is successful and the new active board is running properly, run the **DSP CERTMK** command to check whether the

automatic certificate application is successful. If it is not, perform step [Step 1.e](#).

- i. Perform step [Step 1.f](#).
- j. Run the **SWP BRD** command with **Forced Switchover Flag** set to **NO** to switch over the active and standby boards.
- k. After the board switchover is successful, perform step [Step 1.f](#) to test the certificate validity.
- l. After the certificate validity test is successful, run the **MOD APPCERT** command to set the certificate information.

Step 2 Configure an automatic device certificate check task for the BSC6960.

1. Run the **SET CERTCHKTSK** command to set a periodic certificate validity check task.
2. (Optional) Transfer the CRL to the BSC6960. You can use the following method:
Upload the CRL to the FTP server. Then, run the **DLD CERTFILE** command to download the CRL from the FTP server to the BSC6960.
3. (Optional) Run the **ADD CRL** command to add a CRL.
4. (Optional) Run the **SET CRLPOLICY** command to set a CRL usage policy.
5. (Optional) Run the **ADD CRLTSK** command to add a periodic CRL download task.

----End

In addition to the preceding steps, perform the following step to manually trigger a certificate update:

Step 1 (Optional) Obtain a new subordinate CA certificate from the CA server.

Step 2 (Optional) Transfer the new subordinate CA certificate to the BSC6960. You can use either of the following methods:

Upload the subordinate CA certificate file to the FTP server. Then, run the **DLD CERTFILE** command to download the subordinate CA certificate file from the FTP server to the BSC6960.

Step 3 (Optional) Run the **ADD TRUSTCERT** command to add the new subordinate CA certificate of the BSC6960.

Step 4 Run the **UPD DEVCERT** command to set information about a certificate update. After the setting takes effect, a CMPv2-based certificate update procedure is triggered.

Step 5 (Optional) Run the **RMV TRUSTCERT** command to remove the old subordinate CA certificate of the BSC6960.

----End

Activation Command Examples

The following is an MML command example of how to activate an operator-issued device certificate on the BSC6960.

```
//Modifying configurations of a certificate request template  
MOD CERTREQ: COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="xx", ORG="ITEF",
```

```
ORGUNIT="Hw", STATEPROVINCENAME="sc", LOCALITY="cd",  
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,  
SIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048;  
//Setting the CA server information  
//If the CA server is reachable through a public or internal network, you are advised to set the source IP  
address for certificate application to the interface IP address. The following is an example:  
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Wdights Pty Ltd, CN = eca1", URL="http://  
10.88.88.88:80/pkix/", SIGNALG=SHA256, TRUSTGROUPID=65535;  
//If the CA server is reachable through only an external network, you are advised to set the source IP  
addresses for both certificate application and update to the interface IP address. The following is an  
example:  
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Wdights Pty Ltd, CN = eca1", URL="http://  
10.88.88.88:80/pkix/", SIGNALG=SHA256;  
//Downloading an operator's root certificate from an FTP server (assume that the FTP server is on the MAE,  
indicating that the IP address of the FTP server is the same as that of the MAE)  
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="OperationCA.cer",DSTF="OperationCA.cer";  
//Adding an operator's root certificate as the trust certificate  
ADD TRUSTCERT: CERTNAME="OperationCA.cer";  
//(Optional) Adding a trust group  
ADD TRUSTGROUP: TRUSTGROUPID=0;  
//(Optional) Adding a trust certificate to the trust group  
ADD TRUSTCERT2GROUP: TRUSTGROUPID=0, CERTNAME="OperationCA.cer";  
//Setting information required for the BSC6960 to apply for an operator-issued device certificate based on  
CMPv2 when the application needs to be manually triggered (skip this step when the application is  
automatically triggered)  
REQ DEVCERT: CANAME="C=AU, S=Some-State, O=Internet Wdights Pty Ltd, CN=eca1",  
APPCERT="OPKIDevCert.cer";  
//Adding a device certificate (For an operator-issued device certificate, set APPCERT to the planned name.)  
ADD CERTMK: APPCERT="OPKIDevCert.cer", CASW=ON, CANAME="C=AU, S=Some-State, O=Internet  
Wdights Pty Ltd, CN=eca1";  
//Verifying whether the device certificate can be used for normal SSL connections to ensure that secure  
channels can be successfully established between the BSC6960 and the peer end  
TST APPCERT: APPTYPE=SSL, APPCERT="OPKIDevCert.cer";  
//Setting information about the device certificate in use  
MOD APPCERT: APPTYPE=SSL, APPCERT="OPKIDevCert.cer";  
//Setting a periodic certificate validity check task with Initial Certificate Verification Switch set to OFF  
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDTEMETHOD=CMF, VENDORALMRNG=180,  
VENDORCERTCHKSW=OFF;  
//(Optional) Downloading the CRL from the FTP server (If the FTP server is on the MAE, the IP address of  
the FTP server is the same as that of the MAE.)  
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="eNodeB.crl",DSTF="eNodeB.crl";  
//(Optional, required only when a manual certificate application procedure is used) Loading a CRL  
ADD CRL: CERTNAME="eNodeB.crl";  
//(Optional) Setting a CRL usage policy  
SET CRLPOLICY:CRLPOLICY= NOVERIFY;  
//(Optional) Adding a task of periodically downloading the CRL  
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="eNodeB.crl", ISCRLTIME=DISABLE,  
PERIOD=24, TSKID=0, CRLGETMETHOD=FTP, TRUSTGROUPID=65535;
```

In addition to the preceding steps, perform the following step to manually trigger a certificate update. Running the **UPD DEVCERT** command to update a certificate will fail if the BSC6960 is performing an SSL negotiation. You need to run this command after the negotiation is completed.

```
UPD DEVCERT: APPCERT="OPKIDevCert.cer",REKEY=YES;
```

Deactivation

- Step 1** Run the **RMV CA** command to remove the CA.
- Step 2** (Optional) Run the **RMV CRLTSK** command to remove the task of periodically updating a CRL.
- End

4.4.5.3 Activation Verification

Step 1 Run the **DSP APPCERT** command to check the status of device certificates.

If the values of **Certificate File Name**, **Issuer**, and **Common Name** are correct and the value of **Status** is **Normal** in the query result, the device certificate has been loaded to the BSC6960.

Step 2 Run the **DSP TRUSTCERT** command to check the status of trust certificates.

If the value of **Status** is **Normal** in the query result, the trust certificate has been loaded to the BSC6960.

Step 3 (Optional) Run the **DSP CRL** command to check the CRL status.

If the value of **Status** is **Normal** in the query result, the CRL has been loaded to the BSC6960.

----End

4.4.6 Deployment of PKI on a Board with eCPRI Ports

4.4.6.1 Data Preparation

Before this function is enabled, engineering personnel must obtain the requirements for applying for certificates from the CA by the BBU proxy from the CA maintenance engineer.

To deploy the PKI feature on a board with an eCPRI port, prepare the following data on the base station side.

Table 4-47 Parameters in the **MOD APPCERT** command

Parameter Name	Parameter ID	Setting Notes
RRU Certificate Request Switch	APPCERT.RRUCERTREQSW (5G gNodeB, LTE eNodeB)	Set this parameter to ON(On) .

4.4.6.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
//Turning on the switch for the AAU to apply for an operator-issued device certificate  
MOD APPCERT: APPTYPE=SSL, APPCERT="eNodeBCert.pem", RRUCERTREQSW=ON;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for the AAU to apply for an operator-issued device certificate  
MOD APPCERT: APPTYPE=SSL, APPCERT="eNodeBCert.pem", RRUCERTREQSW=OFF;
```

4.4.6.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.6.4 Activation Verification

Perform the following operations to check whether an operator-issued device certificate has been successfully applied for:

- Step 1** Run the **SET RRUSECPOLICY** command to disable Huawei certificate authentication.
- Step 2** Check whether the eCPRI connection is normal.
- Initiate a voice service and a data service and then check whether the two services are running normally.
 - Check whether the corresponding base station is online on the topology view of the MAE.

----End

4.4.7 Deployment of PKI for Transmission Link Backup of Main Control Boards (LTE/NR)

4.4.7.1 Data Preparation

"-" in the following tables in this section indicates that there is no special requirement for parameter setting. Set the parameters according to network plans.

Table 4-48 lists the data to be prepared for the certificate deployment position on the base station (corresponding to the **CERTDEPLOY MO**).

Table 4-48 Data to be prepared for the deployment position of a certificate

Parameter Name	Parameter ID	Setting Notes
Certification Deploy Position Type	CERTDEPLOY.DEPLOY TYPE (LTE eNodeB, 5G gNodeB)	Set this parameter to DEFAULT .

Table 4-49 lists the data to be prepared for a certificate request template of the base station (corresponding to the **CERTREQ MO**).

Table 4-49 Data to be prepared for a certificate request template

Parameter Name	Parameter ID	Setting Notes
Common Name Additional Info.	CERTREQ.USERADDI NFO (LTE eNodeB, 5G gNodeB)	Retain the default value.
Country	CERTREQ.COUNTRY (LTE eNodeB, 5G gNodeB)	-
Organization	CERTREQ.ORG (LTE eNodeB, 5G gNodeB)	-
Organizational Unit	CERTREQ.ORGUNIT (LTE eNodeB, 5G gNodeB)	-
State or Province	CERTREQ.STATEPRO VINCENAME (LTE eNodeB, 5G gNodeB)	-
Locality	CERTREQ.LOCALITY (LTE eNodeB, 5G gNodeB)	-
Key Usage	CERTREQ.KEYUSAGE (LTE eNodeB, 5G gNodeB)	Retain the default value.
Signature Algorithm	CERTREQ.SIGNALG (LTE eNodeB, 5G gNodeB)	-
Local Name	CERTREQ.LOCALNAME (LTE eNodeB, 5G gNodeB)	Retain the default value.

Parameter Name	Parameter ID	Setting Notes
Local IP	CERTREQ.LOCALIP (LTE eNodeB, 5G gNodeB)	Retain the default value.
Local IPv6	CERTREQ.LOCALIP6 (5G gNodeB, LTE eNodeB)	Retain the default value.
Key Type	CERTREQ.KEYTYPE (5G gNodeB, LTE eNodeB)	-
ECC Curve	CERTREQ.ECCCURVE (5G gNodeB, LTE eNodeB)	-

The base station must be configured with CA server information to apply for a certificate from the CA server. [Table 4-50](#) lists the data to be prepared for the CA server (corresponding to the **CA MO**).

Table 4-50 Data to be prepared for the CA

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME (LTE eNodeB, 5G gNodeB)	This parameter indicates the name of the CA. The fields of the Certificate Authority Name parameter except Organization must not contain the following invalid characters: backslash (\), slash (/), colon (:), asterisk (*), question mark (?), double quotation marks ("), left angle bracket (<), right angle bracket (>), and bar (). Otherwise, an error occurs when you run the REQ DEVCERT command to apply for a device certificate. For example: The Subject name of the certificate used for signing CMP messages on the CA/RA is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com, and the corresponding CANAME is C = AU, S = Some-State, L = cd, O = Internet Widgits Pty Ltd, OU = Wireless, CN = eca1, E = rosa@huawei.com.

Parameter Name	Parameter ID	Setting Notes
IP Version	CA.IPVERSION (LTE eNodeB, 5G gNodeB)	-
Certificate Authority URL	CA.URL (LTE eNodeB, 5G gNodeB)	Currently, base stations cannot translate domain names. Therefore, use an IP address instead of a domain name in the URL. The TCP port number is determined by the CA server. For example, the URL domain name of the CA server can be set to http://10.88.88.88:80/pkix/. For an IPv6 URL address, the format "[IP address]:Port Number" must be used to separate the IP address from the port number, for example, https://[2050::20:23:123]:34481/pkix/.
Signature Algorithm	CA.SIGNALG (LTE eNodeB, 5G gNodeB)	-
Certificate Registration Name	CA.RANAME (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Certificate Fetch Mode	CA.MODE (LTE eNodeB, 5G gNodeB)	- If the parameter is set to DEFAULT_MODE, the base station uses the effective local OM IP address as the source address and the URL as the destination address for initial certificate application during base station deployment, manual certificate application, automatic certificate update, manual certificate update, and OCSP status acquisition. - If the parameter is set to CFG_INIT_UPD_ADDR, the base station uses the values of the INITREQSIP/INITREQSIP6, INITREQURL, and SLVINITREQSIP/SLVINITREQSIP6 parameters as the source, destination, and slave source addresses, respectively, for initial certificate application during base station deployment. In addition, it uses the values of the UPDSIP/UPDSIP6, URL, and SLVUPDSIP/SLVUPDSIP6 parameters as the source, destination, and slave source addresses, respectively, for automatic certificate update, manual certificate update, and manual certificate application. - If the parameter is set to CFG_UPD_SIP, the base station uses the values of the UPDSIP/UPDSIP6, URL, and SLVUPDSIP/SLVUPDSIP6 parameters as the source, destination, and slave source addresses, respectively, for initial certificate application during base station deployment, manual certificate application, automatic certificate update, and manual certificate update. - If the parameter is set to CFG_INIT_UPD_ADDR or CFG_UPD_SIP, the values of the UPDSIP/UPDSIP6 and SLVUPDSIP/SLVUPDSIP6 parameters are used as the source and slave source

Parameter Name	Parameter ID	Setting Notes
		addresses, respectively, for OSCP status acquisition.
Certificate Update Source IP	CA.UPDSIP (LTE eNodeB, 5G gNodeB)	-
CA URL During Site Deployment	CA.INITREQURL (LTE eNodeB, 5G gNodeB)	-
Source IP for Applying for a Certificate During Site Deployment	CA.INITREQSIP (LTE eNodeB, 5G gNodeB)	-
Certificate Update Source IPv6 Address	CA.UPDSIP6 (5G gNodeB, LTE eNodeB)	-
Source IPv6 Address for Applying for a Certificate During Site Deployment	CA.INITREQSIP6 (5G gNodeB, LTE eNodeB)	-
Slave Certificate Authority URL	CA.SLVURL (5G gNodeB, LTE eNodeB)	-
Slave CA URL During Site Deployment	CA.SLVINITREQURL (5G gNodeB, LTE eNodeB)	-
Certificate Request Switch	CA.CERTREQSW (5G gNodeB, LTE eNodeB)	Retain the default value.
Local Name	CA.LOCALNAME (5G gNodeB, LTE eNodeB)	-
Local IP	CA.LOCALIP (5G gNodeB, LTE eNodeB)	-
Local IPv6 Address	CA.LOCALIP6 (5G gNodeB, LTE eNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Key Type	CA.KEYTYPE (5G gNodeB, LTE eNodeB)	-
ECC Curve	CA.ECCCURVE (5G gNodeB, LTE eNodeB)	-
Certificate Request Signature Algorithm	CA.CERTREQSIGNALG (LTE eNodeB, 5G gNodeB)	-
Certificate File Name	CA.CERTNAME (LTE eNodeB, 5G gNodeB)	After an alarm indicating that a Huawei initial certificate is about to expire is reported, an operator-issued device certificate must be configured before the certificate expires.
Initial Certificate File Name	CA.VENDORCERT (5G gNodeB, LTE eNodeB)	-
Certificate Update Slave Source IP	CA.SLVUPDSIP (LTE eNodeB, 5G gNodeB)	-
Initial Certificate Apply Slave Source IP	CA.SLVINITREQSIP (LTE eNodeB, 5G gNodeB)	-
Certificate Update Slave Source IPv6 Address	CA.SLVUPDSIP6 (LTE eNodeB, 5G gNodeB)	-
Initial Certificate Apply Slave Source IPv6	CA.SLVINITREQSIP6 (LTE eNodeB, 5G gNodeB)	-
Support Forcible Execution	CA.FORCEEXECUTE (5G gNodeB, LTE eNodeB)	For setting suggestions, see 4.1.3.4.2 Estimation of Certificate Reconfiguration Impact .

 NOTE

If OM data flows are transmitted by the IPsec tunnel, the OM IP address cannot be used for data that is not protected by IPsec. If OM data flows are not transmitted by the IPsec tunnel, the OM IP address cannot be used for data that is protected by IPsec.

Table 4-51 lists the data to be prepared for a device certificate (corresponding to the **CERTMK** MO).

Table 4-51 Data to be prepared for a device certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT <i>(LTE eNodeB, 5G gNodeB)</i>	- For Huawei initial certificates, this parameter must be set to appcert.pem. - For operator-issued device certificates, this parameter must be set to OPKIDevCert.cer during base station deployment by PnP.

Table 4-52 lists the data to be prepared for a trust certificate (corresponding to the **TRUSTCERT** MO).

Table 4-52 Data to be prepared for a trust certificate

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	TRUSTCERT.CERTNAME <i>(LTE eNodeB, 5G gNodeB)</i>	- The base station must be configured with an operator's trust certificate and a Huawei trust certificate. For the Huawei trust certificate, this parameter is set to caroot.pem on the base station side. For an operator's trust certificate, this parameter is set to CN.cer when automatic certificate application is used. The value of CN must be the same as that in the Subject field of the trust certificate. - If the operator's CA system has a multi-layer structure, a trust certificate chain needs to be configured. If the certificates at different layers of CAs in the certificate chain are separately stored, run the ADD TRUSTCERT command to add certificates at each layer.

Table 4-53 lists the data to be prepared for a trust group (corresponding to the **TRUSTGROUP** MO).

Table 4-53 Data to be prepared for a trust group

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTGROUP.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)	-
Trust Group Description	TRUSTGROUP.TRUSTGROUPDESC (LTE eNodeB, 5G gNodeB)	-

Table 4-54 lists the data to be prepared for the association between a trust group and a trust certificate (corresponding to the **TRUSTCERT2GROUP** MO).

Table 4-54 Data to be prepared for associating a trust group with a trust certificate

Parameter Name	Parameter ID	Setting Notes
Trust Group ID	TRUSTCERT2GROUP.TRUSTGROUPID (LTE eNodeB, 5G gNodeB)	-
Certificate File Name	TRUSTCERT2GROUP.CERTNAME (LTE eNodeB, 5G gNodeB)	-

Table 4-55 lists the data to be prepared for a periodic certificate validity check task (corresponding to the **CERTCHKTSK** MO).

Table 4-55 Data to be prepared for a periodic certificate validity check task

Parameter Name	Parameter ID	Setting Notes
Checking Period	CERTCHKTSK.PERIOD (LTE eNodeB, 5G gNodeB)	Retain the default value.
Alarm Threshold	CERTCHKTSK.ALMRNG (LTE eNodeB, 5G gNodeB)	Retain the default value.
Update Method	CERTCHKTSK.UPDATEMETHOD (LTE eNodeB, 5G gNodeB)	Set this parameter to CMP .

Parameter Name	Parameter ID	Setting Notes
Automatic Certificate Reapplication Switch	CERTCHKTSK.AUTOR EAPPLYSW (5G gNodeB, LTE eNodeB)	Set this parameter to ON .

(Optional) Prepare CRL data if the base station needs to obtain CRL information from the CA. [Table 4-56](#) lists the data to be prepared for a CRL (corresponding to the **CRL MO**).

Table 4-56 Data to be prepared for a CRL

Parameter Name	Parameter ID	Setting Notes
CRL File Name	CRL.CERTNAME (5G gNodeB, LTE eNodeB)	-

(Optional) Prepare data related to CRL usage policies. [Table 4-57](#) lists the data to be prepared for these policies (corresponding to the **CRLPOLICY MO**).

Table 4-57 Data to be prepared for CRL usage policies

Parameter Name	Parameter ID	Setting Notes
CRL Using Policy	CRLPOLICY.CRLPOLICY (LTE eNodeB, 5G gNodeB)	Operators can set this parameter based on site requirements. During base station deployment through PnP, the base station does not support verifying peer certificate validity against the CRL.

(Optional) Prepare data related to a periodic CRL download task. [Table 4-58](#) lists the data to be prepared for the task (corresponding to the **CRLTSK MO**).

Table 4-58 Data to be prepared for a periodic CRL download task

Parameter Name	Parameter ID	Setting Notes
IP Version	CRLTSK.IPVERSION (LTE eNodeB, 5G gNodeB)	-

Parameter Name	Parameter ID	Setting Notes
IP Address	CRLTSK.IP (5G gNodeB, LTE eNodeB)	Set this parameter to the IPv4 address of the CRL server.
IPv6 Address	CRLTSK.IP6 (LTE eNodeB, 5G gNodeB)	Set this parameter to the IPv6 address of the CRL server.
User Name	CRLTSK.USR (LTE eNodeB, 5G gNodeB)	-
Password	CRLTSK.PWD (LTE eNodeB, 5G gNodeB)	-
File Name	CRLTSK.FILENAME (LTE eNodeB, 5G gNodeB)	-
Using CRL's Next Update	CRLTSK.ISCRLTIME (LTE eNodeB, 5G gNodeB)	If this parameter is set to ENABLE , the base station downloads a CRL when the next update time arrives.
CRL Updating Period	CRLTSK.PERIOD (LTE eNodeB, 5G gNodeB)	This parameter must be set when the CRLTSK.ISCRLTIME (LTE eNodeB, 5G gNodeB) parameter is set to DISABLE .
Access Method	CRLTSK.CRLGETMET HOD (LTE eNodeB, 5G gNodeB)	The value FTP is recommended.
Distinguish Name	CRLTSK.SEARCHDN (LTE eNodeB, 5G gNodeB)	This parameter must be set when the CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) parameter is set to LDAP .
Port No.	CRLTSK.PORT (LTE eNodeB, 5G gNodeB)	This parameter must be set when the CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) parameter is set to LDAP or HTTP .
Task ID	CRLTSK.TSKID (LTE eNodeB, 5G gNodeB)	-
Source IP	CRLTSK.SIP (LTE eNodeB, 5G gNodeB)	If this parameter is set to the default value (all 0s), the base station uses the OM IP address as the source IP address for CRL updates.
IPv6 Source Address	CRLTSK.SIP6 (LTE eNodeB, 5G gNodeB)	If this parameter is set to the default value (all 0s), the base station uses the OM IP address as the source IP address for CRL updates.

Parameter Name	Parameter ID	Setting Notes
Connection Mode	CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB)	This parameter is valid only when CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) is set to LDAP or HTTP .
Authenticate Peer	CRLTSK.AUTHPEER (5G gNodeB, LTE eNodeB)	- This parameter is valid only when CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) is set to LDAP or HTTP and CRLTSK.CONNMODE (5G gNodeB, LTE eNodeB) is set to SSL. - When CRLTSK.CRLGETMETHOD (5G gNodeB, LTE eNodeB) is set to HTTP, this parameter can only be set to ENABLE. - If this parameter is set to ENABLE, ensure that the NEs and CRL server have been correctly configured with the CA trust certificates and device digital certificates. - If this parameter is set to DISABLE, a base station cannot verify the identity of an LDAP server, posing security risks.
Certificate File Name	CRLTSK.CERTNAME (LTE eNodeB, 5G gNodeB)	This parameter specifies a device certificate name. This parameter specifies that the device certificate configured in the CERTMK MO is used for authentication when establishing a TLS connection between the local end and an FTP, LDAP, or HTTP server. If this parameter is not specified, the system uses the preconfigured Huawei initial certificate for authentication.
Trust Group ID	CRLTSK.TRUSTGROU PID (LTE eNodeB, 5G gNodeB)	This parameter specifies the ID of a trust group. When certificate authentication is used for link negotiation or establishment, the trust certificates in the trust group configured in the TRUSTGROUP MO are used to authenticate the peer end. If this parameter is set to 65535 , no trust certificate group is associated, and all loaded trust certificates in the system are used to authenticate the device certificates of the peer end.

Parameter Name	Parameter ID	Setting Notes
Slave Source IP	CRLTSK.SLVSIP (LTE eNodeB, 5G gNodeB)	-
Slave Source IPv6 Address	CRLTSK.SLVSIP6 (LTE eNodeB, 5G gNodeB)	-

(Optional) Prepare data to manually download an operator's root certificate or CRL from an FTP server. [Table 4-59](#) lists the data to be prepared for downloading a certificate file (corresponding to the **DLD CERTFILE** command).

Table 4-59 Data to be prepared for downloading a certificate file

Parameter Name	Parameter ID	Setting Notes
FTP Server IP	IP	-
User Name	USR	-
Password	PWD	-
Source File Name	SRCF	-
Destination File Name	DSTF	It is recommended that this parameter be set to the same value as SRCF .
Guage Option	GA	This parameter determines whether to report the progress of file downloading. The value Y is recommended.
Certificate Type	CT	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate application. [Table 4-60](#) lists the data to be prepared for applying for a device certificate (corresponding to the **REQ DEVCERT** command) based on CMPv2.

Table 4-60 Data to be prepared for applying for a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME (LTE eNodeB, 5G gNodeB)	-

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT (5G gNodeB, LTE eNodeB)	-

(Optional) Prepare data to manually trigger a CMPv2-based certificate update. [Table 4-61](#) lists the data to be prepared for updating a device certificate (corresponding to the **UPD DEVCERT** command) based on CMPv2.

Table 4-61 Data to be prepared for updating a device certificate based on CMPv2

Parameter Name	Parameter ID	Setting Notes
Certificate File Name	CERTMK.APPCERT (LTE eNodeB, 5G gNodeB)	This parameter specifies a certificate to be updated.
Renew Key	REKEY	Use the recommended value.

4.4.7.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

The following is an MML command example of how to activate an operator-issued device certificate on the base station side.

```
//Setting the deployment position of a certificate (the base station needs to be reset for the modification to take effect)
SET CERTDEPLOY: DEPLOYTYPE=DEFAULT;
//Modifying configurations of a certificate request template
MOD CERTREQ: COMMNAME=ESN, USERADDINFO=".huawei.com", COUNTRY="xx", ORG="ITEF",
ORGUNIT="Hw", STATEPROVINCENAME="sc", LOCALITY="cd",
KEYUSAGE=DATA_ENCIPHERMENT-1&DIGITAL_SIGNATURE-1&KEY_AGREEMENT-1&KEY_ENCIPHERMENT-1,
SIGNALG=SHA256, KEYTYPE=RSA, KEYSIZE=KEYSIZE2048, LOCALNAME="abcdefghijklmn.huawei.com",
LOCALIP="10.20.20.188";
//Setting the CA server information
//If the CA server is reachable through a public or internal network and OM data is protected by IPsec, you
are advised to set the source IP addresses for certificate application and update to an interface IP address
and an OM IP address (for example, 10.31.31.188), respectively, and set slave source IP addresses for
certificate application and update. The following is an example:
```

```
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", IPVERSION=IPV4,
URL="http://10.88.88.88:80/pkix/", MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",
INITREQURL="http://10.89.89.89:80/pkix/", INITREQSIP="10.20.20.188", CERTREQSW=DEFAULT,
OCSURL="http://10.88.88.88:80/pkix/", SLVUPDSIP="10.31.32.188";
//If the CA server is reachable through a public or internal network and OM data is not protected by IPsec,
you are advised to set the source IP addresses for certificate application and update to an interface IP
address and an internal-network IP address (for example, 10.45.45.45), respectively, and set slave source IP
addresses for certificate application and update. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", IPVERSION=IPV4,
URL="http://10.88.88.88:80/pkix/", MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.45.45.45",
INITREQURL="http://10.89.89.89:80/pkix/", INITREQSIP="10.20.20.188", CERTREQSW=DEFAULT,
SLVUPDSIP="10.45.46.45", SLVINITREQSIP="10.20.21.188", VENDORCERT=APPCERT;
//If the CA server is reachable only through a public network, you are advised to set the IP address for
certificate application and update to an interface IP address, and set slave IP addresses for certificate
application and update to the interface IP address of another board. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", IPVERSION=IPV4,
URL="http://10.88.88.88:80/pkix/", MODE=CFG_INIT_UPD_ADDR, UPDSIP="10.20.20.188",
INITREQURL="http://10.89.89.89:80/pkix/", INITREQSIP="10.20.20.188", CERTREQSW=DEFAULT,
SLVUPDSIP="10.20.21.188", SLVINITREQSIP="10.20.21.188", VENDORCERT=APPCERT;
//Downloading an operator's root certificate from an FTP server (assume that the FTP server is deployed on
the MAE, indicating that the IP address of the FTP server is the same as that of the MAE)
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="OperationCA.cer",DSTF="OperationCA.cer";
//Adding an operator's root certificate as the trust certificate
ADD TRUSTCERT: CERTNAME="OperationCA.cer";
//Adding a trust group
ADD TRUSTGROUP: TRUSTGROUPID=0;
//Adding a trust certificate to the trust group
ADD TRUSTCERT2GROUP: TRUSTGROUPID=0, CERTNAME="OperationCA.cer";
//Setting information required for the base station to apply for an operator-issued device certificate based
on CMPv2 when the application needs to be manually triggered (skip this step when the application is
automatically triggered)
REQ DEVCERT: CANAME="C=AU, S=Some-State, O=Internet Widgits Pty Ltd, CN=eca1",
APPCERT="OPKIDevCert.cer";
//Adding a device certificate (For an operator-issued device certificate, set APPCERT to OPKIDevCert.cer
during base station deployment by PnP and set APPCERT to the planned name of the operator-issued
device certificate in other deployment scenarios.)
ADD CERTMK: APPCERT="OPKIDevCert.cer", CASW=ON, CANAME="C=AU, S=Some-State, O=Internet
Widgits Pty Ltd, CN=eca1";
//Verifying whether the device certificate can be used for normal IKE and SSL connections to ensure that
secure channels can be successfully established between the base station and the peer end
TST APPCERT: APPTYPE=IKE, APPCERT="OPKIDevCert.cer";
//Setting information about the device certificate in use
MOD APPCERT: APPTYPE=IKE, APPCERT="OPKIDevCert.cer";
//Setting a periodic certificate validity check task
SET CERTCHKTSK: PERIOD=1, ALMRNG=30, UPDATEMETHOD=CMP;
//Optional) Downloading the CRL from the FTP server (If the FTP server is on the MAE, the IP address of
the FTP server is the same as that of the MAE.)
DLD CERTFILE:IP="10.60.60.60",USR="admin",PWD="*****",SRCF="eNodeB.crl",DSTF="eNodeB.crl";
//Optional, required only when a manual certificate application procedure is used) Loading a CRL
ADD CRL: CERTNAME="eNodeB.crl";
//Optional) Setting a CRL usage policy
SET CRLPOLICY:CRLPOLICY= NOVERIFY;
//Optional) Adding a task of periodically downloading the CRL and setting the slave source IP address
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="eNodeB.crl", ISCRITIME=DISABLE,
PERIOD=24, SIP="10.20.20.188", TSKID=0, CRLGETMETHOD=FTP, TRUSTGROUPID=65535,
SLVSIP="10.20.21.188";
```

Deactivation

- Step 1** Run the **RMV CA** command to remove the CA.
- Step 2** (Optional) Run the **RMV CRLTSK** command to remove the task of periodically updating a CRL.
- End

4.4.7.3 Activation Verification

Observing the PKI Feature

Step 1 Run the **DSP APPCERT** command to check the status of device certificates.

If the values of **Certificate File Name**, **Issuer**, and **Common Name** are correct and the value of **Status** is **Normal** in the query result, the device certificate has been loaded to the base station.

Step 2 Run the **DSP TRUSTCERT** command to check the status of trust certificates.

If the value of **Status** is **Normal** in the query result, the trust certificate has been loaded to the base station.

Step 3 (Optional) Run the **DSP CRL** command to check the CRL status.

If the value of **Status** is **Normal** in the query result, the CRL has been loaded to the base station.

----End

Observing Offline Certificate Monitoring

Step 1 On the MAE-Access, choose **Security > Certificate Authentication Management > Certificate Management > NE Certificate Management > Offline Certificate**. All device certificates in abnormal states are displayed on the MAE-Access.

Step 2 (Optional) To export information of device certificates in abnormal states, click **Export**. If the **On disconnected NE** or **On deleted NE** check box is selected, you also need to set the duration in which the device certificates remain in the state.

----End

4.4.8 Replacing the Initial Certificate

A CA has been deployed on the operator's network, or the LiteCA service of the MAE has been enabled. After the LiteCA is enabled, it can be considered as a CA server deployed on the network. Then, Huawei base stations, base station controllers, and OSSs can apply for new device certificates from the LiteCA.

Before enabling the LiteCA service, perform the following steps on the MAE-Access:

Step 1 Configure certificate templates, including a root CA template and a device certificate template. For details, see "Configuring a Certificate Profile" in *MAE Product Documentation*.

Step 2 Add a CA. For details, see "Configuring a CA" in *MAE Product Documentation*.

----End

The certificate replacement process is the same as the PKI feature deployment process. For details, see the following sections:

- [4.4.3 Deployment of PKI on the GBTS/eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station](#)

- [4.4.4 Deployment of PKI on the BSC6900/BSC6910 \(GSM/UMTS\)](#)
- [4.4.5 Deployment of PKI on the BSC6960 \(GSM/UMTS\)](#)
- [4.4.6 Deployment of PKI on a Board with eCPRI Ports](#)

The following parameter settings are different:

Parameter Name	Parameter ID	Setting Notes
Certificate Authority Name	CA.CANAME (LTE eNodeB, 5G gNodeB)	On the MAE-Common, choose Security > Certificate Authority Service . On the displayed tab page, select CA , view the created CA information, and set this parameter to the value of Subject .
Certificate Authority URL	CA.URL (LTE eNodeB, 5G gNodeB)	Set this parameter to http://OSS IP address:Port number/cmp/CA name?certprofile:Device certificate template name . For example, if the IP address of the OSS is 10.181.21.44, the port number is 10080, the CA name is templateCA, and the name of the device certificate template is rsa2048, then the URL address is http://10.181.21.44:10080/cmp/templateCA?certprofile:rsa2048.

4.4.9 Reconfiguration

Reconfiguration of CA Name

In **Certificate Authority Name**, the S and ST fields are regarded as the same field. Services can be properly provided if the S field is used at one end but the ST field is used at the other end.

To change the S field to the ST field, perform the following steps:

```
//Removing the original CA configuration
RMV CA:CANAME="C=AU, S=Some-State, O=Internet Widgits Pty Ltd, CN=eca1";
//Adding a CA
ADD CA:CANAME="C=AU, ST=Some-State, O=Internet Widgits Pty Ltd, CN=eca1", URL="http://10.88.88.88:80/pkix/";
```

Estimation of Certificate Reconfiguration Impact

If the following commands are involved during reconfiguration, the base station estimates whether services will be affected by the reconfiguration. For details, see [4.1.3.4.2 Estimation of Certificate Reconfiguration Impact](#).

- **MOD CERTREQ**

- MOD CA
- MOD APPCERT
- MOD CERTMK

Activation of Automatic Certificate Application After a CA Change (in Base Station Deployment/IKE Negotiation Failure Scenarios)

If the RA name is specified by the **CA.CANAME (5G gNodeB, LTE eNodeB)** parameter, remove this CA record and then reconfigure a correct one.

If a CA record with the following parameter settings exists:

CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2",
URL="http://10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE=
CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",INITREQURL="http://10.89.89.89:80/
pkix/",INITREQSIP="10.20.20.188";

Then, it is recommended that **RANAME** be set to **C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca2** and **CANAME** be set to **C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1** through the following operations:

```
//Adding a CA
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",RANAME="C = AU, S
= Some-State, O = Internet Widgits Pty Ltd, CN = eca2", URL="http://10.88.88.88:80/pkix/",
SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",INITREQURL="http://
10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188";
//Turning on the CA switch
MOD CERTMK:APPCERT="opki1.cer",CASW=ON,CANAME="C = AU, S = Some-State, O = Internet Widgits
Pty Ltd, CN = eca1";
//Removing the original CA configuration
RMV CA:CANAME="C=AU, ST=Some-State, O=Internet Widgits Pty Ltd, CN=eca2";
//Turning off the CA switch
MOD CERTMK:APPCERT="opki1.cer",CASW=OFF;
//Turning on the automatic certificate reapplication switch
SET CERTCHKTSK:PERIOD=1, ALMRNG=30, UPDTEMETHOD=CMP, AUTOREAPPLYSW = ON;
```

4.4.10 Network Monitoring

None

5 NE Supporting PKI Redundancy

5.1 Principles

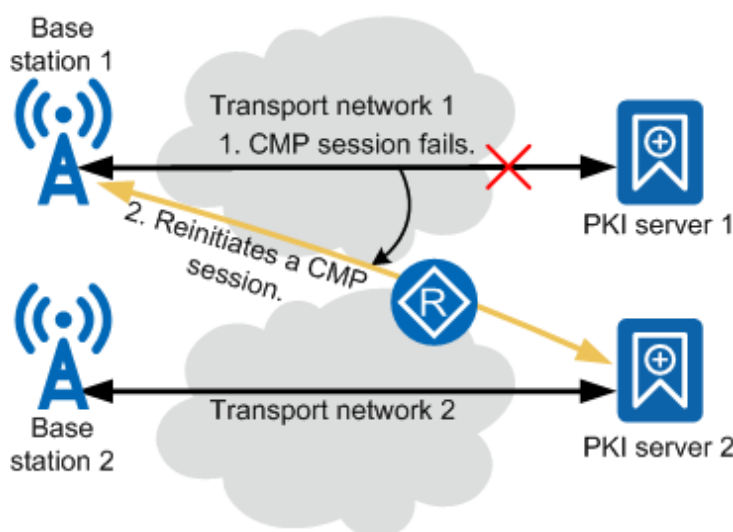
PKI redundancy is introduced to improve the reliability of PKI-based networking.

To achieve PKI redundancy, two PKI servers must be deployed on the network. There should be reachable routes between the base station/base station controller and the two PKI servers. In addition, the following conditions should be met:

- The two PKI servers have the same **CANAME** and root certificate or certificate chain and synchronize certificate management databases between them.
- The two CAs must have different IP addresses, and so do active and standby CRL servers.

Every time before certificate application, certificate update, and CRL acquisition, the base station or base station controller first initiates a session with the active PKI server. If the session fails, the base station or base station controller reinitiates a session with the standby PKI server. This mechanism ensures successful certificate applications and updates as well as CRL acquisitions.

Figure 5-1 Working principles of PKI redundancy



The following parameters specify the URL of the standby CA on the base station and base station controller sides:

- **CA.SLVURL** (5G gNodeB, LTE eNodeB)
- **CA.SLVINITREQURL** (5G gNodeB, LTE eNodeB)

The following parameters specify the login information of the standby CRL server on the base station and base station controller sides:

- **CRLTSK.SLVIP** (5G gNodeB, LTE eNodeB)
- **CRLTSK.SLVPORT** (5G gNodeB, LTE eNodeB)
- **CRLTSK.SLVUSR** (5G gNodeB, LTE eNodeB)
- **CRLTSK.SLVPWD** (5G gNodeB, LTE eNodeB)

During certificate updates or CRL acquisitions, the base station/base station controller reports ALM-26842 Automatic Certificate Update Failed only when the sessions between the base station/base station controller and both the active and standby PKI servers fail.

PKI redundancy has the following application limitations: PKI redundancy is not supported during base station deployment by PnP. The operator must ensure that the active PKI server works properly during base station deployment by PnP.

5.2 Network Analysis

5.2.1 Benefits

This feature improves networking reliability.

5.2.2 Impacts

None

5.3 Requirements

5.3.1 Licenses

None

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 GBFD-160210 BTS Supporting PKI Redundancy

5.3.2.1.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
GS M	BTS Supporting PKI	None	<i>PKI</i>

5.3.2.1.2 Mutually Exclusive Functions

None

5.3.2.1.3 Function Impacts

None

5.3.2.2 GBFD-160208 BSC Supporting PKI Redundancy

5.3.2.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
GS M	BSC Supporting PKI	None	<i>PKI</i>

5.3.2.2.2 Mutually Exclusive Functions

None

5.3.2.2.3 Function Impacts

None

5.3.2.3 WRFD-160275 NodeB Supporting PKI Redundancy

5.3.2.3.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
UM TS	NodeB PKI Support	None	<i>PKI</i>

5.3.2.3.2 Mutually Exclusive Functions

None

5.3.2.3.3 Function Impacts

None

5.3.2.4 WRFD-160277 RNC Supporting PKI Redundancy

5.3.2.4.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
UM TS	RNC Supporting PKI	None	<i>PKI</i>

5.3.2.4.2 Mutually Exclusive Functions

None

5.3.2.4.3 Function Impacts

None

5.3.2.5 LOFD-070212 eNodeB Supporting PKI Redundancy

5.3.2.5.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
LTE FD D	Public Key Infrastructur e (PKI)	None	<i>PKI</i>

5.3.2.5.2 Mutually Exclusive Functions

None

5.3.2.5.3 Function Impacts

None

5.3.2.6 MLOFD-070212 eNodeB Supporting PKI Redundancy

5.3.2.6.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
NB- IoT	Public Key Infrastructur e (PKI)	None	<i>PKI</i>

5.3.2.6.2 Mutually Exclusive Functions

None

5.3.2.6.3 Function Impacts

None

5.3.2.7 TDLOFD-070212 eNodeB Supporting PKI Redundancy

5.3.2.7.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference
LTE TD D	Public Key Infrastructur e (PKI)	None	<i>PKI</i>

5.3.2.7.2 Mutually Exclusive Functions

None

5.3.2.7.3 Function Impacts

None

5.3.2.8 FBFD-010023 Security Mechanism (gNodeB Supporting PKI Redundancy)

5.3.2.8.1 Prerequisite Functions

None

5.3.2.8.2 Mutually Exclusive Functions

None

5.3.2.8.3 Function Impacts

None

5.3.3 Hardware

For details, see [4.3.3 Hardware](#).

5.3.4 Networking

- Two PKI servers are deployed on the operator's network. For the requirements for PKI servers, see [4.3.4 Others](#).
- The two PKI servers have the same CA name and root certificate or certificate chain and synchronize certificate management databases between them.
- There are reachable routes between the base station/base station controller and the two PKI servers.

5.3.5 Others

None

5.4 Operation and Maintenance

5.4.1 Precautions

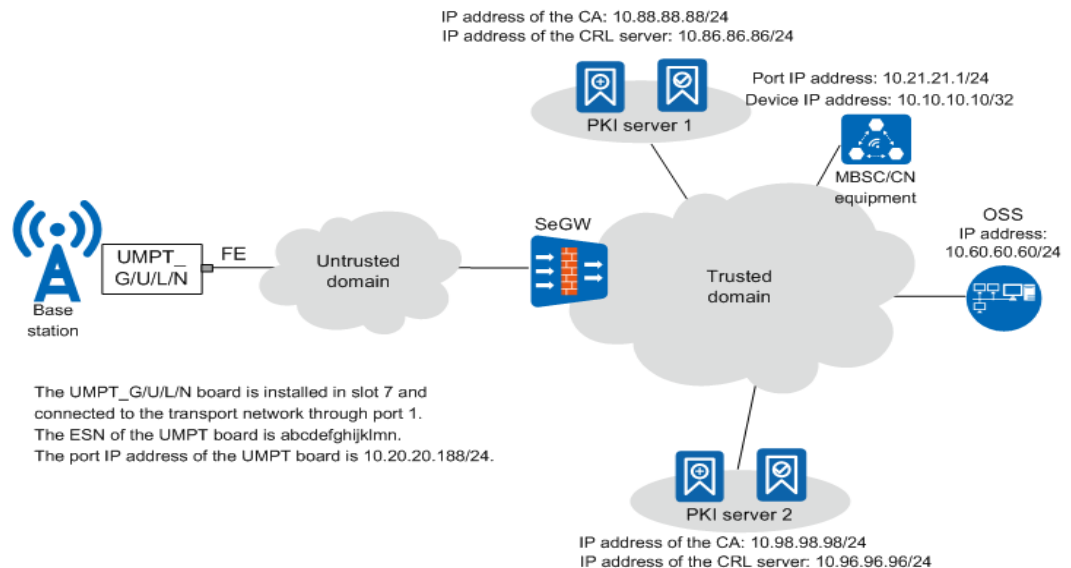
Before deploying the PKI redundancy feature, engineering personnel need to collect the following information besides that listed in [4.4.2 Precautions](#).

Items to Be Collected	Required Parameter on the Base Station or Base Station Controller Side
URL of the standby CA	CA.SLVURL (5G gNodeB, LTE eNodeB) (Optional) CA.SLVINITREQURL (5G gNodeB, LTE eNodeB)
(Optional) IP address of the standby CRL server	CRLTSK.SLVIP (5G gNodeB, LTE eNodeB)
(Optional) user name for logging in to the standby CRL server	CRLTSK.SLVUSR (5G gNodeB, LTE eNodeB)
(Optional) password for logging in to the standby CRL server	CRLTSK.SLVPWD (5G gNodeB, LTE eNodeB)
(Optional) port number of the standby CRL server	CRLTSK.SLVPORT (5G gNodeB, LTE eNodeB)

5.4.2 Deployment of PKI Redundancy on the eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station

This section uses the networking illustrated in [Figure 5-2](#) as an example to describe how to deploy the PKI redundancy feature on the GBTS/eGBTS/NodeB/

Figure 5-2 Example of the secure networking for the eGBTS/NodeB/eNodeB/gNodeB/multimode base station



Compared with the data to be prepared described in section [4.4.3.1 Data Preparation](#), the following tables list the additional data for preparation.

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.
Slave CA URL During Site Deployment	CA.SLVINITREQURL (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.

152

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP (5G gNodeB, LTE eNodeB)	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.
Slave Port No.	CRLTSK.SLVPORT (5G gNodeB, LTE eNodeB)	This parameter can be set only when PKI redundancy is enabled.

5.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

For details, see [4.4.3.2 Using MML Commands](#). The following describes the differences in configurations.

In the PKI redundancy scenario, the configurations of the CA server and periodic CRL download task are as follows:

```
//Setting the CA server information
//If the CA server is reachable through a public or internal network and OM data is protected by IPsec, you
are advised to set the source IP addresses for certificate application and update to an interface IP address
and an OM IP address (for example, 10.31.31.188), respectively. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR, UPDSIP="10.31.31.188",
INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",SLVURL="http://10.98.98.98:80/
pkix/",SLVINITREQURL="http://10.99.99.99:80/pkix/";
//If the CA server is reachable through a public or internal network and OM data is not protected by IPsec,
you are advised to set the source IP addresses for certificate application and update to an interface IP
address and an internal-network IP address (for example, 10.45.45.45), respectively. The following is an
example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_INIT_UPD_ADDR, UPDSIP="10.45.45.45",
INITREQURL="http://10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",SLVURL="http://10.98.98.98:80/
pkix/",SLVINITREQURL="http://10.99.99.99:80/pkix/";
//If the CA server is reachable only through an external network, you are advised to set the source IP
addresses for both certificate application and update to interface IP addresses. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",URL="http://
```

```
10.88.88.88:80/  
pkix/",SIGNALG=SHA256,MODE=CFG_INIT_UPD_ADDR,UPDSIP="10.20.20.188",INITREQURL="http://  
10.89.89.89:80/pkix/",INITREQSIP="10.20.20.188",SLVURL="http://10.98.98.98:80/  
pkix/",SLVINITREQURL="http://10.99.99.99:80/pkix/",CERTREQSW=DEFAULT;  
//(Optional) Adding a task of periodically downloading the CRL  
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="eNodeB.crl", ISCRLTIME=DISABLE,  
PERIOD=24, TSKID=0, CRLGETMETHOD=FTP, SLVIP="10.96.96.96", SLVUSR="admin2", SLVPWD="*****";
```

5.4.2.3 Activation Verification

The certificate update and CRL file obtaining succeed when the active PKI server is faulty and the standby PKI server is normal. After MML commands are executed to query the status of the device certificates and CRL files, the results shown in the following figures indicate that PKI redundancy functions properly.

Step 1 Check the status of device certificates.

Run the MML command **DSP CERTMK**. In the command output, **CA URL Last Used** indicates the URL of the standby CA, and **Last Update Time of Certificate** indicates the time of the latest certificate update.

```
DSP CERTMK;  
The result is shown as follows:  
+++ HUAWEI 2013-11-08 12:59:22  
O&M #91  
%%DSP CERTMK:%%  
RETCODE = 0 Operation succeeded.  
  
Display Device Certificate  
-----  
Certificate File Name Issuer Common Name Serial Number Effective Date Expiration Date Status CA URL Last Used Last Update Time of Certificate  
OPKIDevCertcer embca 210305394710C2000064.huawei.com 76:61:1e:45:0b:57:4f:27:a8:e7:e9:c9:cc:c6:e2:17:72 2013-09-27 12:44:30 2021-05-03 19:53:44 Normal http://98.98.98.80/pkix/ 2013-09-08 13:05:08  
NULL  
(Number of results = 1)
```

Step 2 Check the status of CRL files.

Run the MML command **DSP CRL**. In the command output, **CRL Server IP Address Last Used** indicates the IP address of the standby CRL server, and **Last Update Time of CRL** indicates the time of the latest CRL obtaining.

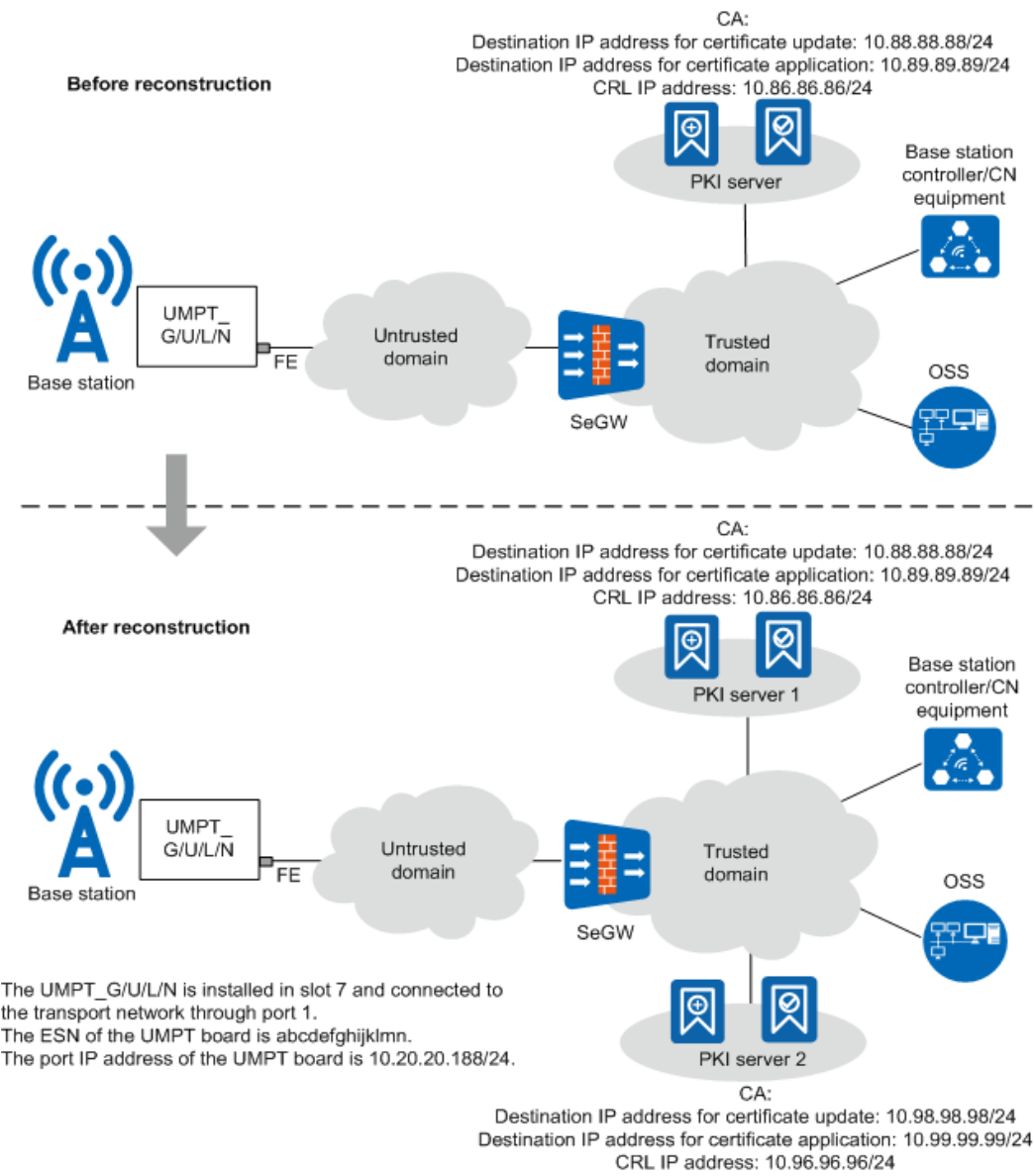
```
DSP CRL;  
+++ HUAWEI 2011-11-08 13:06:52  
O&M #4915  
%%DSP CRL:%%  
RETCODE = 0 Operation succeeded.  
  
Display CRL's Details  
-----  
CRL File Name Issuer This Update Next Update CRL Server IP Address Last Used Last Update Time of CRL Status  
eNodeB.crl ecal 2013-09-08 16:50:24 2013-09-16 05:10:24 10.166.158.241 2009-08-10 14:12:06 Normal  
(Number of results = 1)
```

----End

5.4.3 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the eGBTS/NodeB/eNodeB/gNodeB/Multimode Base Station

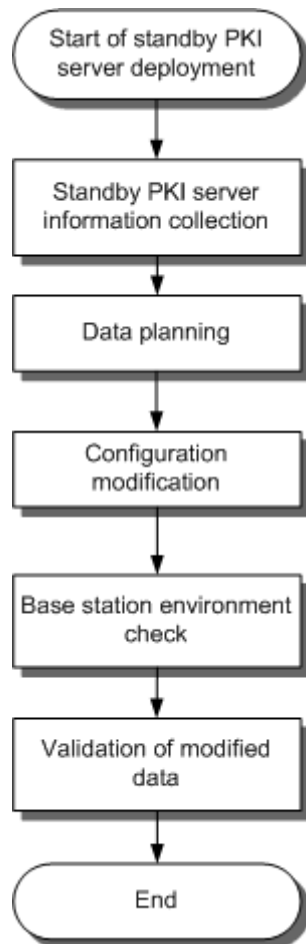
This section uses the networking illustrated in [Figure 5-3](#) as an example to describe the reconstruction requirements and reconfiguration procedure when a PKI-based secure network is reconstructed into a PKI redundancy network on the eGBTS/NodeB/eNodeB/gNodeB/multimode base station. A UMDU can also be used in a co-MPT multimode base station in the secure networking shown in this figure. However, a UMDU/MDUC cannot be used in a separate-MPT multimode base station.

Figure 5-3 Example of reconstructing a PKI-based secure network into a PKI redundancy network on the eGBTS/NodeB/eNodeB/gNodeB/multimode base station



5.4.3.1 Data Preparation

General Procedure



Network Deployment and Information Collection

- A standby PKI server has been deployed on the network.
- The active and standby PKI servers have the same CA name and root certificate or certificate chain and synchronize certificate management databases between them. There should be reachable routes between the base station and the two PKI servers.
- Engineering personnel collect information about the standby PKI server, including the URL of the standby CA, URL of the CA which issues the certificate during deployment, and IP address, user name, password, and port number of the standby CRL server.

Checking the Base Station Environment

- The base station meets the hardware requirements described in section [5.3.3 Hardware](#).
- The license for the PKI redundancy feature has been activated on the base station.

Data Planning

In addition to the original configuration data, you need to configure data for the standby CA and standby CRL server.

The following table lists the data to be prepared for the standby CA.

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.
Slave CA URL During Site Deployment	CA.SLVINITREQURL (5G gNodeB, LTE eNodeB)	

(Optional) The following table lists the data to be prepared for a periodic CRL download task.

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP (5G gNodeB, LTE eNodeB)	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR (5G gNodeB, LTE eNodeB)	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD (5G gNodeB, LTE eNodeB)	
Slave Port No.	CRLTSK.SLVPORT (5G gNodeB, LTE eNodeB)	

5.4.3.2 Data Configuration

Preparing the Incremental Script

An incremental script is generated based on data of existing base stations and includes configuration modifications.

For details about how to modify PKI redundancy configurations, see [4.4.3.2 Using MML Commands](#).

Downloading the Modified Data

1. From the MAE-Deployment main menu, choose **Area Management > Planned Area > Export Incremental Scripts**.
2. In the **Export Incremental Scripts** dialog box, choose a specific NE to which the script is exported, specify **Output Path** and **Script Executor Operation**, and click **OK**.
3. On the displayed **Script Executor** page, observe the progress.

5.4.3.3 Activation Verification

For details, see [5.4.2.3 Activation Verification](#).

5.4.4 Deployment of PKI Redundancy on the BSC6900/BSC6910 (GSM/UMTS)

PKI redundancy on the BSC6900/BSC6910 helps improve the reliability of updating the device certificate and CRL. Therefore, the standby CA and CRL servers must be configured before PKI redundancy is enabled.

5.4.4.1 Data Preparation

Compared with the data to be prepared described in section [4.4.4.1 Data Preparation](#), the following tables list the additional data for preparation.

The following table lists the additional data to be prepared for the CA (the **CA MO**).

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL	This parameter needs to be set when PKI redundancy is enabled.

(Optional) The following table lists the additional data to be prepared for a periodic CRL download task (the **CRLTSK MO**).

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD	This parameter needs to be set when PKI redundancy is enabled.

Parameter Name	Parameter ID	Setting Notes
Slave Port No.	CRLTSK.SLVPORT	This parameter can be set only when PKI redundancy is enabled.

5.4.4.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

For details, see [4.4.4.2 Using MML Commands](#). The following describes the differences in configurations.

```
//Setting the CA server information
//If the BSC6900/BSC6910 can access the CA only through an external network, you are advised to use the
external virtual IP address of the BSC6900/BSC6910 for certificate update. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256, MODE= CFG_UPD_SIP, UPDSIP="10.120.20.188",SLVURL="http://
10.98.98.98:80/pkix/";
//Adding a task of periodically downloading the CRL
ADD CRLTSK: TSKID=1, IP="10.86.86.86", CRLGETMETHOD=LDAP, PORT=389, USR="admin", PWD="*****",
FILENAME="bsc.crl", ISCRLTIME=ENABLE, SIP="10.120.20.188", SLVIP="10.86.86.90", SLVPORT=389,
SLVUSR="test", SLVPWD="*****", SEARCHDN="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN =
eca1";
```

5.4.4.3 Activation Verification

The certificate update and CRL file obtaining succeed when the active PKI server is faulty and the standby PKI server is normal. After MML commands are executed to query the status of the device certificates and CRL files, the results shown in the following figures indicate that PKI redundancy functions properly.

Step 1 Check the status of device certificates.

Run the MML command **DSP CERTMK**. In the command output, **CA URL Last Used** indicates the URL of the standby CA, and **Last Update Time of Certificate** indicates the time of the latest certificate update.

```
*** RJG-17 2014-10-23 17:36:15
DWM 46213
WDMSP CERTMK: OK
RETCODE = 0 Execution succeeded.
Display Device Certificate Details
Certificate File Name Issuer Common Name Serial Number Effective Date Expiration Date Status CA URL Last Used Last Update Time of Certificate
aa.cer Huawei RadioShack SRA1 WISC Certificate 00:e4:e0:33:e4:5c:3d:db:bf:e4:1e:e3:2d:94:8b:ca:ca 2013-10-23 15:17:22 2015-10-23 15:37:22 Normal http://10.145.24.100:8081/pkix/ 2014-10-23 15:17:17
uncert.pem Huawei Wireless Network Product CA WISC Certificate 7b:41:24:00:1a:25:00:0f:a6:a6:c2:a5:41:33:01:75:95 2012-07-21 14:55:40 2027-07-10 14:55:40 Normal (URL) (URL)
(Number of results = 2)
--- END
```

Step 2 Check the status of CRL files.

Run the MML command **DSP CRL**. In the command output, **CRL Server IP Address Last Used** indicates the IP address of the standby CRL server, and **Last Update Time of CRL** indicates the time of the latest CRL obtaining.

```
+++ RJG-17      2014-10-23 17:42:17
O&M #2620
%%DSP CRL:%%
RETCODE = 0 Execution succeeded.

Display CRL's Details
-----
CRL File Name      Issuer              This Update      Next Update      Status      CRL Server IP Address Last Used      Last Update Time of CRL
-----
/123/2.crl         <NULL>              <NULL>           <NULL>           File does not exist <NULL>                               <NULL>
/ban/aaa_crl.crl   <NULL>              <NULL>           <NULL>           File does not exist <NULL>                               <NULL>
1.crl              <NULL>              <NULL>           <NULL>           File does not exist <NULL>                               <NULL>
aaa_crl.crl        Huawei RadioSubCA SHA1 2013-08-15 18:19:25 2013-08-16 18:19:26 Expired <NULL>                               <NULL>
test.crl           <NULL>              <NULL>           <NULL>           File does not exist <NULL>                               <NULL>
(Number of results = 5)

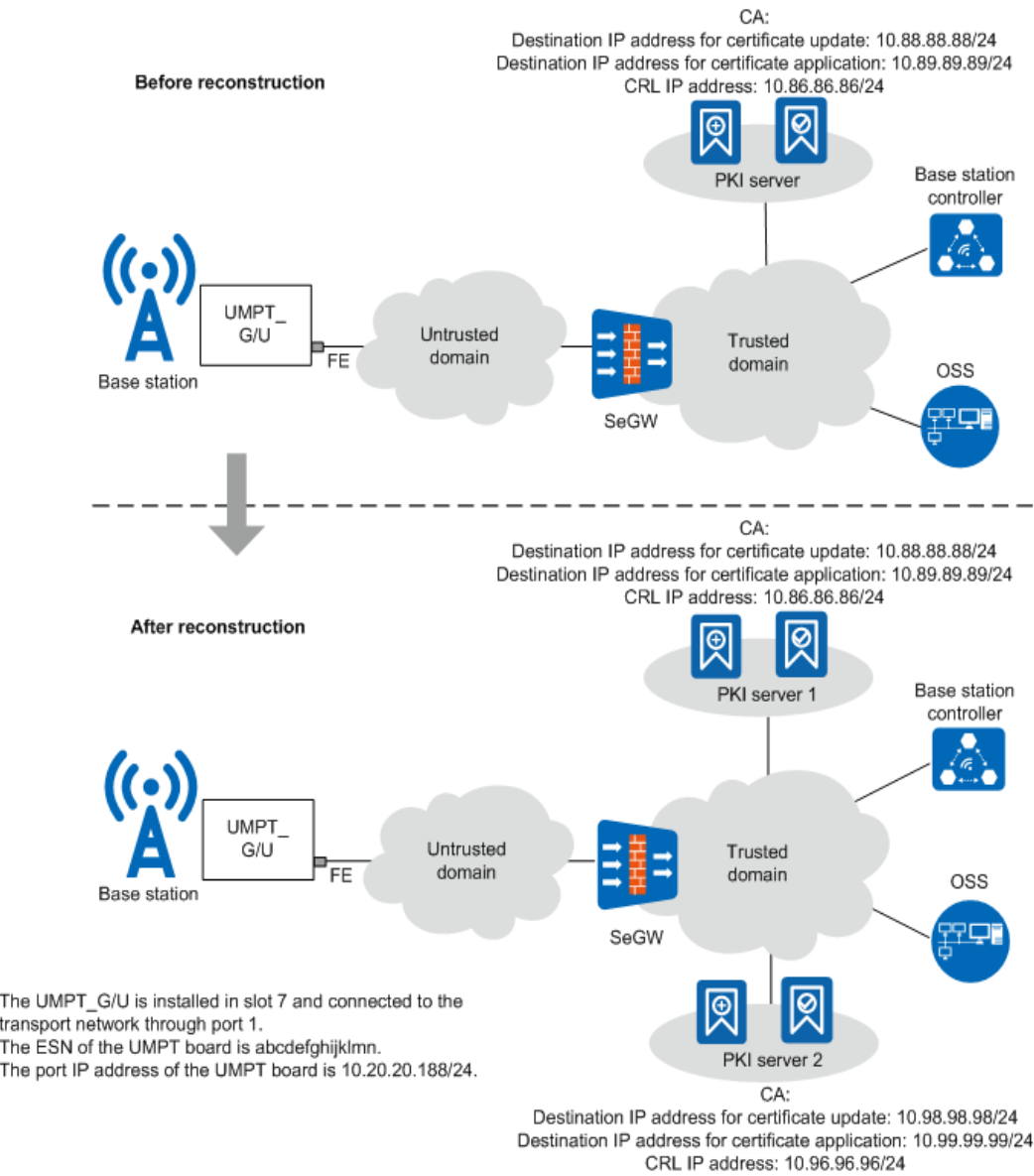
--- END
```

----End

5.4.5 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the BSC6900/BSC6910 (GSM/UMTS)

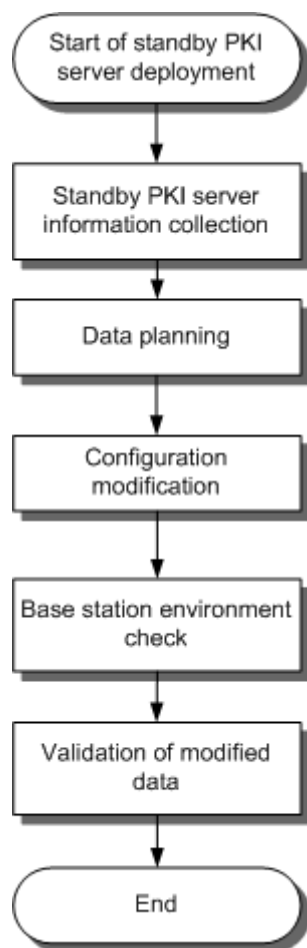
This section uses the networking illustrated in [Figure 5-4](#) as an example to describe the reconstruction requirements and reconfiguration procedure when a PKI-based secure network is reconstructed into a PKI redundancy network on the BSC6900/BSC6910.

Figure 5-4 Example of reconstructing a PKI-based secure network into a PKI redundancy network on the BSC6900/BSC6910



5.4.5.1 Data Preparation

General Procedure



Network Deployment and Information Collection

- A standby PKI server has been deployed on the network.
- The active and standby PKI servers have the same CA name and root certificate or certificate chain and synchronize certificate management databases between them. There are reachable routes between the BSC6900/BSC6910 and the two PKI servers.
- Engineering personnel collect information about the standby PKI server, including the URL of the standby CA, and IP address, user name, password, and port number of the standby CRL server.

Data Planning

In addition to the original configuration data, you need to configure data for the standby CA and standby CRL server.

The following table lists the data to be prepared for the standby CA.

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL	This parameter needs to be set when PKI redundancy is enabled.

(Optional) The following table lists the additional data to be prepared for a periodic CRL download task.

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD	This parameter needs to be set when PKI redundancy is enabled.
Slave Port No.	CRLTSK.SLVPORT	This parameter can be set only when PKI redundancy is enabled.

Checking the BSC6900/BSC6910 Environment

The license for the PKI redundancy feature has been activated on the BSC6900/BSC6910.

5.4.5.2 Data Configuration

Preparing the Incremental Script

An incremental script is generated based on data of existing sites and includes configuration modifications.

For details about how to modify PKI redundancy configurations, see [4.4.4.2 Using MML Commands](#).

Downloading and Activating the Modified Data

- From the MAE-Deployment main menu, choose **Area Management > Planned Area > Export Incremental Scripts**.
- In the **Export Incremental Scripts** dialog box, choose a specific NE to which the script is exported, specify **Output Path** and **Script Executor Operation**, and click **OK**.
- On the displayed **Script Executor** page, observe the progress.

- Restart the BSC6900/BSC6910 to make the script take effect.

5.4.5.3 Activation Verification

For details, see [5.4.4.3 Activation Verification](#).

5.4.6 Deployment of PKI Redundancy on the BSC6960 (GSM/UMTS)

PKI redundancy on the BSC6960 helps improve the reliability of updating the device certificate and CRL. Therefore, the standby CA and CRL servers must be configured before PKI redundancy is enabled.

5.4.6.1 Data Preparation

Compared with the data to be prepared described in [4.4.5.1 Data Preparation](#), the following tables list the additional data for preparation.

The following table lists the additional data to be prepared for the CA (the **CA MO**).

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL	This parameter needs to be set when PKI redundancy is enabled.

(Optional) The following table lists the additional data to be prepared for a periodic CRL download task.

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD	This parameter needs to be set when PKI redundancy is enabled.
Slave Port No.	CRLTSK.SLVPORT	This parameter can be set only when PKI redundancy is enabled.

5.4.6.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

For details, see [4.4.5.2 Using MML Commands](#). The following describes the differences in configurations.

In the PKI redundancy scenario, the configurations of the CA server and periodic CRL download task are as follows:

```
//Setting the CA server information
//If the CA server is accessible through either a public network or an internal network, set the source IP
addresses for certificate application to the interface IP address. The following is an example:
ADD CA: CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1", URL="http://
10.88.88.88:80/pkix/", SIGNALG=SHA256;
//If the CA server is accessible only through an external network, you are advised to set the source IP
addresses for both certificate application and update to the interface IP address. The following is an
example:
ADD CA:CANAME="C = AU, S = Some-State, O = Internet Widgits Pty Ltd, CN = eca1",URL="http://
10.88.88.88:80/pkix/",SIGNALG=SHA256,CERTREQSW=DEFAULT;
//Optional Adding a task of periodically downloading the CRL
ADD CRLTSK: IP="10.86.86.86", USR="admin", PWD="*****", FILENAME="bsc.crl", ISCRLTIME=DISABLE,
PERIOD=24, TSKID=0, CRLGETMETHOD=FTP, SLVIP="10.96.96.96", SLVUSR="admin2", SLVPWD="*****";
```

5.4.6.3 Activation Verification

The certificate update and CRL file obtaining succeed when the active PKI server is faulty and the standby PKI server is normal. After MML commands are executed to query the status of the device certificates and CRL files, the results shown in the following figures indicate that PKI redundancy functions properly.

Step 1 Check the status of device certificates.

Run the MML command **DSP CERTMK**. In the command output, **CA URL Last Used** indicates the URL of the standby CA server, and **Last Update Time of Certificate** indicates the time of the latest certificate update.

```
DSP CERTMK;
The result is shown as follows:
+++ HUAWEI 2013-11-08 12:59:22
O&M #91
%%DSP CERTMK:%%
RETCODE = 0 Operation succeeded.

Display Device Certificate
-----
Certificate File Name Issuer Common Name Serial Number Effective Date Expiration Date Status CA URL Last Used Last Update Time of Certificate
OPKIDevCert.cer embca 210305394710C2000064.huawei.com 76:61:1e:45:0b:57:4f:27:a8:e7:e9:e9:cc:cf:e2:17:72 2013-09-27 12:44:30 2021-05-03 19:53:44 Normal http://98.98.98.80/pkix/ 2013-09-08 13:05:08
NULL
(Number of results = 1)
```

Step 2 Check the status of CRL files.

Run the MML command **DSP CRL**. In the command output, **CRL Server IP Address Last Used** indicates the IP address of the standby CRL server, and **Last Update Time of CRL** indicates the time of the latest CRL obtaining.

```
DSP CRL::
+++ HUAWEI 2011-11-08 13:06:52
O&M #4915
%%DSP CRL::%%
RETCODE = 0 Operation succeeded.

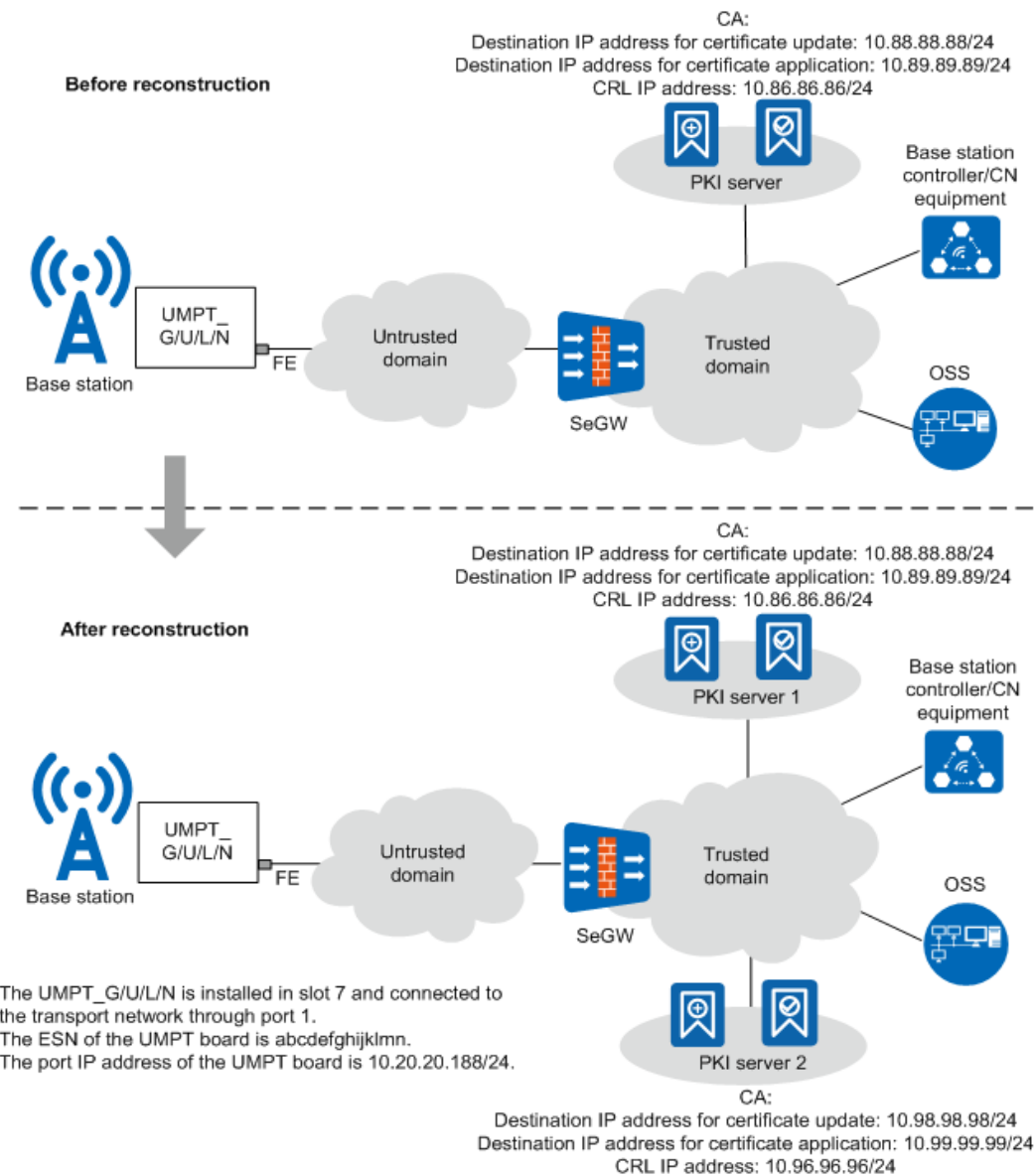
Display CRL's Details
-----
CRL File Name Issuer This Update Next Update CRL Server IP Address Last Used Last Update Time of CRL Status
eNodeB.crl ecal 2013-09-08 16:50:24 2013-09-16 05:10:24 10.166.158.241 2009-08-10 14:12:06 Normal
(Number of results = 1)
```

----End

5.4.7 Reconstruction from a PKI-based Secure Network to a PKI Redundancy Network on the BSC6960 (GSM/UMTS)

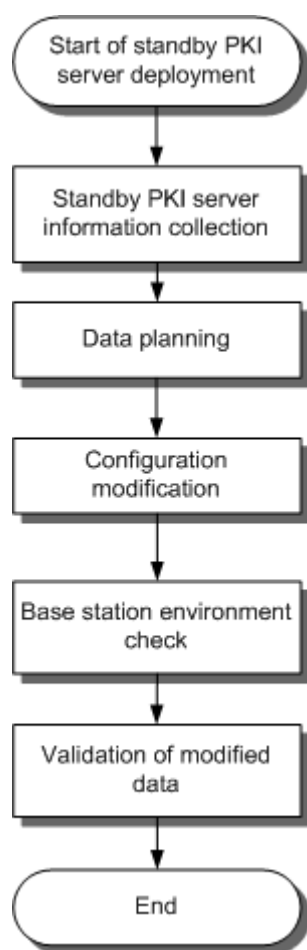
This section uses the networking illustrated in [Figure 5-5](#) as an example to describe the reconstruction requirements and reconfiguration procedure when a PKI-based secure network is reconstructed into a PKI redundancy network on the BSC6960.

Figure 5-5 Example of reconstructing a PKI-based secure network into a PKI redundancy network on the BSC6960



5.4.7.1 Data Preparation

General Procedure



Network Deployment and Information Collection

- A standby PKI server has been deployed on the network.
- The active and standby PKI servers have the same CA name and root certificate or certificate chain and synchronize certificate management databases between them. There are reachable routes between the BSC6960 and the two PKI servers.
- Engineering personnel collect information about the standby PKI server, including the URL of the standby CA, URL of the CA which issues the certificate during deployment, and IP address, user name, password, and port number of the standby CRL server.

Checking the BSC6960 Environment

The BSC6960 meets the hardware requirements described in [5.3.3 Hardware](#).

Data Planning

In addition to the original configuration data, you need to configure data for the standby CA and standby CRL server.

The following table lists the data to be prepared for the standby CA.

Parameter Name	Parameter ID	Setting Notes
Slave Certificate Authority URL	CA.SLVURL	This parameter needs to be set when PKI redundancy is enabled.

(Optional) The following table lists the additional data to be prepared for a periodic CRL download task.

Parameter Name	Parameter ID	Setting Notes
Slave IP Address	CRLTSK.SLVIP	This parameter needs to be set to the IP address of the standby CRL server when PKI redundancy is enabled.
Slave User Name	CRLTSK.SLVUSR	This parameter needs to be set when PKI redundancy is enabled.
Slave Password	CRLTSK.SLVPWD	
Slave Port No.	CRLTSK.SLVPORT	

5.4.7.2 Data Configuration

Preparing the Incremental Script

An incremental script is generated based on data of existing sites and includes configuration modifications.

For details about how to modify PKI redundancy configurations, see [4.4.5.2 Using MML Commands](#).

Downloading and Activating the Modified Data

1. From the MAE-Deployment main menu, choose **Area Management > Planned Area > Export Incremental Scripts**.
2. In the **Export Incremental Scripts** dialog box, choose a specific NE to which the script is exported, specify **Output Path** and **Script Executor Operation**, and click **OK**.
3. On the displayed **Script Executor** page, observe the progress.

5.4.7.3 Activation Verification

For details, see [5.4.6.3 Activation Verification](#).

5.4.8 Network Monitoring

None

6 NE Supporting Digital Certificate Whitelist Management

6.1 Principles

If digital certificates are required when IPsec is used, the customer should preferentially deploy a PKI system and use the operator-issued device certificate to replace the Huawei initial certificate.

If no PKI system is deployed on an operator's network, the base station cannot use an operator-issued device certificate to access the operator's network. In this case, the base station can use the digital certificate whitelist management function to access the operator's network. To support the digital certificate whitelist management, both the base station and SeGW must be Huawei equipment and be preconfigured with Huawei initial certificates. A digital certificate whitelist is a list of Common Name in the Huawei initial certificates preconfigured on the base station and SeGW.

NOTICE

The digital certificate whitelist management function takes effect only when the customer requires a Huawei initial certificate for routine O&M. Using Huawei initial certificates for routine O&M is not recommended. Therefore, this feature is not recommended. The customer should preferentially deploy a PKI system and use the operator-issued device certificate to replace the Huawei initial certificate.

A digital certificate whitelist is configured on the MAE and then loaded onto the base station. During IKE negotiation for IPsec tunnel establishment, the base station uses the digital certificate whitelist to authenticate each piece of equipment that expects to establish an IPsec tunnel with the base station. IPsec tunnels can be set up only between the base station and devices in the whitelist through IKE negotiation. The digital certificate whitelist is used for authentication between base stations only when there are links (for example, the X2 interface) between them or base stations are cascaded.

Digital certificate whitelist management has the following application limitations:

- Only Huawei initial certificates can be used for authentication. Security risks exist if Huawei initial certificates are always used for authentication. For details, see [4.1.2.1 Device Certificate](#).
- The digital certificate whitelist management function supports only IKE/IPsec negotiation and does not support other types of security channels (such as SSL).

6.2 Network Analysis

6.2.1 Benefits

The base station can use the digital certificate whitelist management function to access the operator's network.

6.2.2 Impacts

None

6.3 Requirements

6.3.1 Licenses

RA T	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
GS M	GBFD-1 81202	BTS Supporting Digital Certificate Whitelist Managemen t	LGB3BSD CWM01	BTS Supporting Digital Certificate Whitelist Managemen t(per BTS)	BTS	Per BTS
UM TS	WRFD-1 81220	NodeB Supporting Digital Certificate Whitelist Managemen t	LQW9SD CWLM01	NodeB Supporting Digital Certificate Whitelist Managemen t	NodeB	Per NodeB
LTE FD D	LOFD-1 11203	eNodeB Supporting Digital Certificate Whitelist Managemen t	LT1SDIG WHI00	eNodeB Supporting Digital Certificate Whitelist Managemen t	eNodeB	Per eNodeB

RA T	Feature ID	Feature Name	Model	License Control Item Name	NE	Sales Unit
NB- IoT	MLOFD- 111203	eNodeB Supporting Digital Certificate Whitelist Managemen t	ML1SDI GWHI00	eNodeB Supporting Digital Certificate Whitelist Managemen t	eNodeB	Per eNodeB
NR	FOFD-0 10080	IPsec (gNodeB Supporting Digital Certificate Whitelist Managemen t)	NR0S0IP SEC00	IPsec	gNodeB	Per gNodeB

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 GBFD-181202 BTS Supporting Digital Certificate Whitelist Management

6.3.2.1.1 Prerequisite Functions

Function Name	Function Switch	Reference
BTS Integrated IPSec	None	<i>IPsec</i>

6.3.2.1.2 Mutually Exclusive Functions

None

6.3.2.1.3 Function Impacts

None

6.3.2.2 WRFD-181220 NodeB Supporting Digital Certificate Whitelist Management

6.3.2.2.1 Prerequisite Functions

Function Name	Function Switch	Reference
NodeB Integrated IPsec	None	<i>IPsec</i>

6.3.2.2.2 Mutually Exclusive Functions

None

6.3.2.2.3 Function Impacts

None

6.3.2.3 LOFD-111203 eNodeB Supporting Digital Certificate Whitelist Management

6.3.2.3.1 Prerequisite Functions

Function Name	Function Switch	Reference
IPsec	None	<i>IPsec</i>

6.3.2.3.2 Mutually Exclusive Functions

None

6.3.2.3.3 Function Impacts

None

6.3.2.4 MLOFD-111203 eNodeB Supporting Digital Certificate Whitelist Management

6.3.2.4.1 Prerequisite Functions

Function Name	Function Switch	Reference
IPsec	None	<i>IPsec</i>

6.3.2.4.2 Mutually Exclusive Functions

None

6.3.2.4.3 Function Impacts

None

6.3.2.5 FOFD-010080 IPsec (gNodeB Supporting Digital Certificate Whitelist Management)

6.3.2.5.1 Prerequisite Functions

None

6.3.2.5.2 Mutually Exclusive Functions

None

6.3.2.5.3 Function Impacts

None

6.3.3 Hardware

Base Station Models

RAT	Base Station Model
GSM	3900 and 5900 series base stations (macro base stations)
UMTS	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
LTE	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
NR	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- For an eGBTS, this feature is supported by the Ethernet ports only on the UMPT and UTRPc.
- For a NodeB, this feature is supported only by the UMPT and UTRPc.
- For an LTE FDD eNodeB, this feature is supported only by the UMPT and UTRPc.
- For a gNodeB, this feature is supported only by the UMPT.
- This feature is supported by the integrated BBU3910A/BBU3910C/BBU5900C/BookBBU5901/BookBBU5901a.

RF Modules

This function does not depend on RF modules.

6.3.4 Others

SeGWs must be Huawei devices and support Digital Certificate Whitelist Management.

6.4 Operation and Maintenance

6.4.1 When to Use

This feature is recommended when no PKI system is deployed on the network and the Huawei initial certificates will be used for IKE/IPsec authentication.

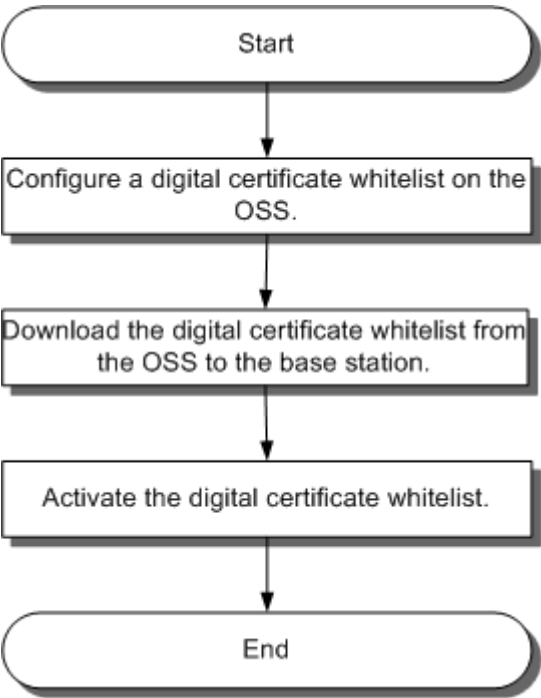
6.4.2 Data Configuration

6.4.2.1 Data Preparation

Collect the common names in the Huawei initial certificates of all base stations and SeGWs to be connected to the network.

Figure 6-1 shows the process of deploying Digital Certificate Whitelist Management.

Figure 6-1 Process of deploying Digital Certificate Whitelist Management



The following data must be prepared on the base station side:

Table 6-1 Parameters in the DLD CERTFILE command

Parameter Name	Parameter ID	Setting Notes
FTP Server IP	<i>IP</i>	Set these parameters based on the site conditions.
User Name	<i>USR</i>	
Password	<i>PWD</i>	
Source File Name	<i>SRCF</i>	
Destination File Name	<i>DSTF</i>	
Guage Option	<i>GA</i>	Set this parameter to CERTWHITELST(CERTWHITELST) .
Certificate Type	<i>CT</i>	

Table 6-2 Parameters in the **ACT CERTWHITELSTFILE** command

Parameter Name	Parameter ID	Setting Notes
Digital Certificate Whitelist File Name	CERTCFG.CERTWHITE LSTFILENAME (5G gNodeB, LTE eNodeB)	Set this parameter based on the site conditions.
Integrity Check Switch	CERTCFG.INTEGRITYC HECKSW (5G gNodeB, LTE eNodeB)	Set this parameter to ON(On) .
Integrity Check Password	CERTCFG.FILEPWD (5G gNodeB, LTE eNodeB)	Set this parameter based on the site conditions.

Table 6-3 Parameters in the **SET CERTCFG** command

Parameter Name	Parameter ID	Setting Notes
IKE Check Switch	CERTCFG.IKECHECKS W (5G gNodeB, LTE eNodeB)	Set this parameter to ON(On) .

6.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

You can configure a digital certificate whitelist on the MAE, download it from the MAE to a base station, and then activate it. For details, see descriptions in *MAE Product Documentation*. This section describes how to activate a digital certificate whitelist using MML commands on the base station.

```
//Downloading a digital certificate whitelist from the MAE to a base station
DLD CERTFILE: IP="192.168.1.1", USR="admin", PWD="*****", SRCF="certwhitelist.gz",
DSTF="certwhitelist.gz",CT=CERTWHITELST;
//Activating the digital certificate whitelist
ACT CERTWHITELSTFILE: CERTWHITELSTFILENAME="certwhitelist.gz", INTEGRITYCHECKSW=ON,
```

```
FILEPWD="*****",  
//Turning on the IKE check switch  
SET CERTCFG: IKECHECKSW=ON;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the IKE check switch  
SET CERTCFG: IKECHECKSW=OFF;
```

6.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.3 Activation Verification

After Digital Certificate Whitelist Management is deployed, observe the status of IPsec tunnels using the digital certificate whitelist to determine whether this feature has been successfully enabled.

Step 1 Run the **DSP IPSECSA** command to check the IPsec SA status.

Step 2 Check whether services protected by the IPsec tunnel are normal.

- Initiate a voice service and a data service and then check whether the two services are running normally.
- Check whether the corresponding base station is online on the topology view of the MAE.

-----End

6.4.4 Network Monitoring

None

7 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 33.310 V10.5.0: "Network Domain Security (NDS); Authentication Framework (AF)"
- IETF RFC 4210, "Internet X.509 Public Key Infrastructure Certificate Management Protocol (CMP)"
- IETF RFC 4211, "Internet X.509 Public Key Infrastructure Certificate Request Message Format (CRMF)"
- IETF RFC 5280, "Internet X.509 Public Key Infrastructure Certificate and CRL Profile"
- IETF RFC 2585, "Internet X.509 Public Key Infrastructure Operational Protocols: FTP and HTTP"
- *SRAN Networking and Evolution Overview*
- *IPsec*
- *OM Security*
- *SSL*
- *Equipment Security*
- *Access Control based on 802.1x*
- *Base Station Supporting Multi-operator PKI*
- *IPv4 Transmission*
- *Base Station Equipment Reliability*
- *RF Unit and Topology Management*
- Section "Creating USUs in Batches" in *USU3910 Initial Configuration Guide*.
- *eCPRI*
- *License Control Item Lists (FDD), License Control Item Lists (CIoT), and License Control Item Lists (TDD) in eRAN Feature Documentation*
- *License Control Item Lists in 5G RAN Feature Documentation*
- *5G RANDTLS*